The spectrum of a compact internal space. II. Effective couplings and mass scales

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Abstract: This paper studies the numerical consequences of a fixed compact spectral datum, taken here to be the Laplace and Dirac spectra of RP^3, together with a specified scale-flow prescription. The structural content supported by this datum, namely the gauge algebra, the fermion representation content, and the mass-gap structure, is recorded in the companion paper, and the present paper evaluates its numerical consequences. The calculation closes the dimensionless window capacity and evaluates a set of effective couplings, mass ratios, and low-energy scales. After the structural datum and the closure prescriptions are fixed, no continuously adjusted phenomenological parameter is introduced in the numerical chain. The resulting quantities are compared with representative Standard Model and cosmological reference values, and the deviations are reported together with the sensitivity to the main closure choices. The results are stated as scheme-conditioned consequences of the spectral construction. Several downstream applications are explicitly marked as exploratory and are not included in the core comparison.

Keywords: spectral datum, compact internal space, effective couplings, scale flow, mass gap, numerical evaluation

Contents

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1 Introduction

The companion paper [1] develops a spectral approach to gauge structure and to a mass gap. Its basic object is a discrete spectrum with multiplicities, and physical quantities are defined as convergent spectral sums over that spectrum rather than as expectation values with respect to a path-integral measure. From one principle, the disorder-dominated coarse-graining principle, together with a set of stated conventions, that paper obtains the internal-space geometry, the gauge algebra, the spacetime dimension, the fermion representation content, and the mass-gap structure, recorded as structural results. The present paper does not re-derive that structure. It takes the spectral datum fixed by the companion paper and records the numerical output of one specified scale-flow prescription built on it. The output is reported as a consistency check of the prescription rather than as a physical prediction.

The datum is the Laplace and Dirac spectrum of a compact three-dimensional internal space of constant positive curvature, the real projective space $RP^3 = S^3/\mathbb{Z}_2$. The core inputs of the computation are therefore two-fold. They are this spectrum, inherited from the companion paper, and the closure prescriptions recorded in Section 2, namely the window profile, the pole criterion, the transverse-traceless kernel, and the normalisation conventions. The core outputs are the dimensionless window capacity kL that closes the fixed point, the generation count, the effective gauge-coupling normalisations, the primary fermion mass ratios, and the electroweak scale. These are the quantities whose status is most direct within the construction.

Beyond these core outputs, the same chain yields a number of downstream quantities. These are the QCD scale, the proton and glueball mass estimates, the CKM and CP relations, and the cosmological fractions. They are reported as further tests of the extended construction rather than as independent predictions on the same footing as the core outputs. Several of them, in particular the baryon asymmetry, the dark-energy density, the acceleration scale, and the primordial helium fraction, rest on additional effective relations whose status is stated where they are introduced. They are treated as exploratory phenomenological implications.

The construction can be compared with other spectral and algebraic approaches. Connes noncommutative geometry takes a spectral triple as the primitive object and derives the action from the spectral action principle [2–4]. The present construction shares the use of spectral data as carriers of geometric and physical information, but defines physical quantities directly as convergent spectral sums without passing through an action functional, so the two approaches are complementary. The algebraic classification of the finite geometry underlying the Standard Model [5, 6] provides a natural consistency condition, and the heat-kernel expansion of the spectral action [4, 7] is structurally the same object that the spectral sums evaluate, here obtained from the spectrum rather than from the action. Algebraic quantum field theory likewise avoids a path-integral measure [8, 9], but reconstructs observables from nets of local algebras [8, 10], whereas here the algebraic structure is induced from the spectrum. The isomorphism between the spectral-scale flow and the standard renormalisation group equations, recorded in Appendix D.2, provides a bridge between the two [11, 12]. These references locate the construction in relation to existing spectral and algebraic formalisms; the numerical outputs themselves are computed in Sections 5–9 and compared with reference values in Section 10.

The computation is organised around a single dimensionful anchor. The Newton constant G_{N} is the only dimensionful quantity taken from observation, used solely to convert the dimensionless output of the spectral sums into physical units. Every mass scale descends from G_{N} through the spectral geometry. Within the adopted spectral datum and closure prescription, the dimensionless quantities considered here are computed without continuous phenomenological adjustment. After the structural inputs and closure prescriptions are specified, the numerical chain contains no continuously adjusted phenomenological parameter. This does not remove the scheme dependence of the closure prescription. The result nevertheless depends on discrete and functional choices, in particular the window profile, the pole criterion, and the normalisation conventions, and on the truncations and effective relations of the downstream applications. The sensitivity to these choices is quantified in Appendix D and summarised in Section 10.5. The explicit numerical inputs are limited to the stated structural constants and closure conventions.

The gravitational coupling carries two roles. In its first role G_{N} is the single observed anchor, and the identity $G_{\mathrm{N}} = 1/(8\pi M_{\mathrm{P}}^2)$ defines the reduced Planck mass. In its second role G_{N} belongs to the gravitational sector of Section 8, the transverse-traceless zero mode, whose normalisation is the residue $Z_{\mathrm{phys}} = \lambda/(\lambda + \sigma)$. The matter self-energy σ is tiny compared with the transverse-traceless eigenvalue λ, so $Z_{\mathrm{phys}} = 1$ to the displayed precision, and the two roles coincide.

The plan of the paper follows the chain of dependencies among the quantities. Section 2 states the computation scheme, the anchor, and the reproducibility convention. Section 3 closes the window capacity that fixes the internal scale and the generation count. Section 4 records the content symmetries and the torsion parameter. Section 5 computes the gauge couplings. Section 6 computes the fermion masses. Section 7 computes the electroweak order parameters. Section 8 computes the cosmological quantities. Section 9 computes the QCD scales. Section 10 collects the full parameter table and compares it with observation, closing with a statement of the theoretical sensitivity whose technical details are in Appendix D. The neutrino sector, the CKM and CP content, and the primordial helium abundance are recorded in Appendices E and F as extensions of this core chain. Section 11 discusses the robustness of the central values, the connection to the Standard Model, the scope of the acceleration-scale implication, and the open questions.

The scope is fixed at the outset. This paper computes numbers from a fixed spectral datum. The companion paper states precisely which of its ingredients are theorems, which are definitions, and which are conventions or assumptions, and the present paper inherits that classification. The numerical results below are computed values attached to a specific spectral datum. They are compared with observed values only after the calculation is complete, and no correction is applied to improve the comparison. The window profile, the pole criterion, the truncation order of the renormalisation-group running, the threshold matching, and the effective relations of the downstream applications are stated explicitly, and their consequences are assessed in Appendix D.

The derivations that produce the numbers of this paper are self-contained at the level of the numerical chain. The key closures, the window-capacity fixed point, the torsion modulus, the coupling conservation law, and the mass ladder, are derived in full in Appendix C. The companion paper is needed for the structural definitions of the spectral datum and the gauge and fermion content that this paper takes as its starting point. This paper is a numerical evaluation of a fixed spectral construction, not an independent claim that the closure principle uniquely determines nature.

2 The computation scheme

This section fixes the rules of the computation. It states the single dimensionful anchor, recalls the spectral-sum representation, records the closure convention, and specifies how reproducibility is documented.

2.1 The single dimensionful anchor

The only dimensionful quantity taken from observation is the Newton constant G_N. In natural units it is

$$G_\mathrm{N} = 6.70883 \times 10^{-39}\ \mathrm{GeV}^{-2},$$

the value compiled by the Particle Data Group [13]. From it one forms the reduced Planck mass

$$M_\mathrm{P} = \frac{1}{\sqrt{8\pi G_\mathrm{N}}} = 2.43532 \times 10^{18}\ \mathrm{GeV}.$$

The Newton constant is used only as the observed dimensionful anchor setting the overall mass scale. Every other mass scale in this paper is a dimensionless number multiplying M_{P}, with the dimensionless number computed from the spectral datum. The full Planck mass $M_{\mathrm{Pl}} = \sqrt{8\pi}\, M_{\mathrm{P}} = 1/\sqrt{G_{\mathrm{N}}}$ appears only where the Hubble parameter or the expansion history needs $1/\sqrt{G_{\mathrm{N}}}$ in the standard normalisation.

The role of G_{N} is therefore narrowly circumscribed. It converts the dimensionless output of the spectral sums into physical units, and it provides the comparison anchor for the gravitational sector. It does not enter the computation of any dimensionless coupling, any mass ratio, or any spectral eigenvalue. Those quantities are computed from the spectral datum within the stated closure conventions, and G_{N} is used only at the end to give them units.

The inputs of the whole computation fall into three disjoint classes. The first class is the single observed anchor, the Newton constant G_{N}. The second class is the structural numbers, the pure mathematics 2, π, $3/2$, and $1/8$, together with the geometric factors $\sqrt{2\pi}$ and $\sqrt{3/\tau}$ recorded where they first appear. The third class is the derived intermediates, the window capacity kL, the torsion modulus τ, the emergence scale M_{G}, and every later quantity. Each member of the third class is computed from the first two classes by a derivation recorded in this paper, with the full algebra in Appendix C. The chain is therefore fixed from the single anchor and the structural numbers through the stated algebra.

2.2 The spectral-sum representation

The companion paper defines physical quantities as convergent spectral sums over the spectrum of the internal Laplacian, together with their closed forms where a regularisation is available [1]. This paper uses that representation throughout. The internal spectrum is

$$\lambda_l = \frac{l(l+2)}{L^2}, \qquad d_l = (l+1)^2, \qquad l = 0, 2, 4, \ldots$$

for the scalar Laplacian on RP^3 of radius L, and

$$\pm\frac{n+3/2}{L}, \qquad n = 0, 2, 4, \ldots$$

for the Dirac operator in the spin structure selected by the chiral content. The mass gap parameter m_δ of the companion paper appears as the positive spectral lower bound of the colour-singlet scalar sector. Quantities are dimensionless ratios or products of spectral sums, so the dependence on L cancels in the dimensionless output and reappears only through the anchor M_{P}.

2.3 Internal closure and the structural numbers

The computation is closed in the following sense. A quantity is declared derived when it is returned by a computation from the anchor and the spectrum, with no continuous free parameter and no adjusted phenomenological coefficient once the structural conventions are fixed. The explicit numerical inputs are limited to the structural numbers 2, π, $3/2$, and $1/8$, together with the geometric factors $\sqrt{2\pi}$ and $\sqrt{3/\tau}$, whose origin is recorded where

Class	Meaning and formulation
Structural input	Inherited from the companion paper, namely the internal geometry RP^3, the gauge algebra, and the fermion representation content. Formulated as *assumed input / inherited structure*.
Scheme-dependent closure	The Gaussian window, the transverse-traceless kernel, the pole threshold, and the closed value of the window capacity kL. Formulated as *prescription /closure choice*.
Derived output	Quantities computed within the scheme, namely the gauge couplings, the electroweak scale v, the electron mass m_e, the Hubble rate H_0, and the mass ratios. Formulated as *derived quantity within the stated scheme*.
Phenomenological comparison	The more remote, model-dependent applications, namely the CKM and CP relations, the primordial abundances, the dark-energy density, and the acceleration scale. Formulated as *comparison /downstream implication*.

Table 1. The four epistemic classes of the computed quantities. The later sections identify their outputs by class, and no quantity is reported in a class higher than the one it actually occupies.

they first appear. Every other number that enters the computation is itself derived earlier in the chain. This discipline rules out the two operations that would otherwise inflate the apparent success of the comparison, namely fitting a parameter to the data and importing an observed value into the middle of the calculation. Neither is used here.

2.4 Status of the computed quantities

The quantities computed in this paper do not all carry the same epistemic status, and the comparison with observation is read differently at each level. Four classes are distinguished, and each subsequent section states the class of its outputs rather than leaving the level implicit.

The numerical results of Section 10 are read through this table at three tiers of decreasing directness, namely *primary outputs* (the window capacity, the generation count, the torsion modulus, the gauge couplings, the electroweak scale, and the primary mass ratios), *downstream consequences* (the QCD scale, the proton and glueball mass estimates, and selected CKM quantities, each obtained through an additional effective relation), and *exploratory implications* (the baryon asymmetry, the dark-energy density, the endpoint residual, the acceleration scale, and the primordial helium fraction). The Status column of Table 3 and the Tier column of Appendix A record the class of each row.

After the structural inputs and closure prescriptions are specified, the numerical chain contains no continuously adjusted phenomenological parameter. The result nevertheless depends on the discrete and functional choices, namely the window profile, the pole criterion, and the normalisation conventions, whose sensitivity is quantified in Appendix D.

2.5 Reproducibility

The derivations form an acyclic chain. Each quantity is defined in terms of quantities that precede it, and no quantity is defined in terms of a later one, so the computation contains no circularity. Every quantity is either derived, computed from earlier quantities by a derivation recorded in Appendix C, or observed, the single Newton constant of Section 2.1. The parameter table of Section 10 is produced from these closed forms rather than stated independently. Appendix B records the closure convention in detail.

2.6 The use of renormalisation group equations

In this paper, the standard two-loop RG equations are used to evolve couplings from the emergence scale M_{G} down to the Z-boson mass M_Z. This evolution occurs in the regime where the spectral sums are dominated by the zero modes and the lowest Kaluza–Klein excitations, the same regime in which the spectral-sum definition and the standard measure-theoretic formulation coincide, as shown in Appendix C of the companion paper [1]. The RG equations are therefore used as the low-energy expression of the spectral-scale flow in the shared effective regime. The interacting corrections above the emergence scale are handled by the spectral sums themselves. Below that scale, the standard perturbative running is a controlled approximation whose accuracy is recorded in Section 10.4. The precise sense in which the spectral-scale flow matches the RG equations is recorded in Appendix D.2.

The dependency structure of the whole computation is summarised in Figure 1.

3 The internal scale and the window capacity

This section fixes the internal scale of the construction and derives the generation count from it. It is the anchor of the computation, in the sense that every later section reads its numbers from here. The section records the spectral datum, the Kaluza–Klein descent from the Planck anchor, the closure of the window capacity, and the resulting generation count.

3.1 Scope and sensitivity of the closure

Before the window capacity is closed, four points fix the status of this section and of the later results. First, kL is the pivotal intermediate of the whole computation, from which the emergence scale, the gauge couplings, the fermion masses, and the electroweak scale all descend, so its status carries the status of the hierarchy. Second, kL is fixed by a closure condition whose ingredients are scheme choices, namely the Gaussian window profile, the transverse-traceless kernel, and the pole threshold $4/27$, recorded in Section 2. The fixed point is a consequence of those choices, not an independently measured quantity. Third, the downstream dimensionful hierarchy is exponentially sensitive to kL. Because $v \propto e^{-4\pi kL}$ and $m_e \propto e^{-20kL}$, a one-percent shift of kL moves v by -32%, m_e by -50%, and ρ_Λ by

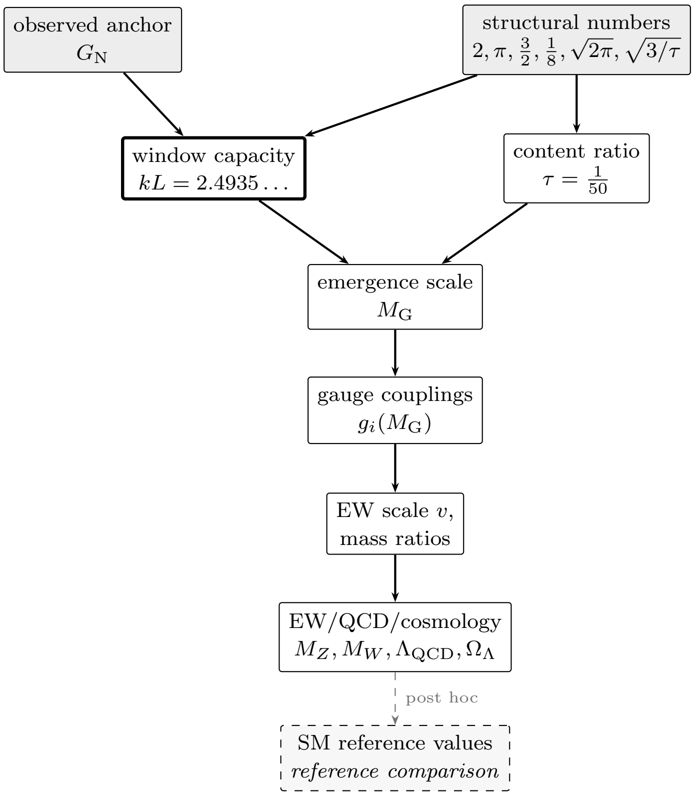

Figure 1. Single-anchor dependency graph. The computation is a directed dependency graph rooted in one observed anchor $G_{\rm N}$ and the structural numbers; every downstream quantity is a closed-form function of these and of the self-consistent window capacity kL (the central node, highlighted). The Standard-Model reference values (dashed) are used only after the internal quantities have been computed.

-262%, while the pure content ratios such as Ω_Λ have zero elasticity. Fourth, and in consequence, the dimensionful outputs reported in this paper are *scheme-conditioned outputs*, not scheme-independent predictions. They are fixed within the adopted closure prescription, and their physical robustness is assessed by the sensitivity analysis of Appendix D rather than asserted. That appendix records the elasticity matrix and the error band of Table 12.

3.2 The spectral datum

The internal space is $RP^3 = S^3/\mathbb{Z}_2$, the real projective space of radius L. Its relevant spectral data are fixed in the companion paper [1] and collected here. The scalar Laplacian has the spectrum

$$\lambda_l = \frac{l(l+2)}{L^2}, \qquad d_l = (l+1)^2, \qquad l = 0, 2, 4, \ldots,$$

where the restriction to even l is the antipodal projection, and d_l is the multiplicity. The Dirac operator in the spin structure selected by the chiral content has the spectrum

$$\pm\frac{n+3/2}{L}, \qquad n = 0, 2, 4, \ldots,$$

again restricted to the even sector of the tower. The volume is $\mathrm{Vol}(RP^3) = \pi^2 L^3$, half the round-sphere volume. The number of Killing vectors is six, the dimension of the isometry algebra $\mathfrak{so}(4)$. The first cohomology vanishes, $H^1(RP^3, \mathbb{R}) = 0$, so there is no harmonic one-form, and the scalar sector has no zero mode beyond the constants.

3.3 Kaluza-Klein descent from the Planck anchor

The spectral datum has one free scale, the radius L. This scale is fixed by the descent from the Planck anchor. The companion paper records the product-space construction in which the internal spectrum is lifted to four dimensions and the heavy modes decouple, leaving a four-dimensional effective theory whose mass scales are the internal eigenvalues measured in Planck units [14, 15]. The descent chain is

$$G_{\mathrm{N}} \;\to\; M_{\mathrm{P}} = \frac{1}{\sqrt{8\pi G_{\mathrm{N}}}} \;\to\; M_{\mathrm{G}} = \frac{M_{\mathrm{P}}\sqrt{\pi}}{kL},$$

where M_{G} is the emergence scale and kL is the dimensionless window capacity introduced below. The Kaluza-Klein masses are

$$m_n = \frac{n+3/2}{kL}\, M_{\mathrm{G}},$$

and the window edge is

$$kL\, M_{\mathrm{G}} = M_{\mathrm{P}}\sqrt{\pi}.$$

The dimensionless quantity kL is the only number that remains to be fixed, and it is fixed by the closure condition of the next subsection.

3.4 Closure of the window capacity

The window capacity kL is not chosen by hand. It is the self-consistent fixed point of the spectral-pole condition of the spin-2 channel of the improved energy-momentum tensor. The object is the transverse-traceless two-point amplitude at zero external momentum, evaluated as a spectral sum over the discrete RP^3 spectrum,

$$\Pi^2_{\mu\nu} = \frac{1}{V_3} \sum_{\text{fields}} \sum_{\text{modes}} d_n\, w \int_0^\infty \frac{d\omega}{\pi}\, \frac{(\omega^2+\lambda_n)^2}{(\omega^2+\lambda_n+m_{\text{eff}}^2)^2}\, e^{-(\omega^2+\lambda_n)/\Lambda^2},$$

with $V_3 = \pi^2 L^3$ the internal volume, d_n the mode multiplicity, and the weights $w = 4$ for each gauge polarisation, $w = 4$ for each Weyl fermion, and $w = 1$ for each scalar. The kernel vanishes at zero mass by the Ward identity of the improved tensor, so only the curvature and torsion masses of the RP^3 modes activate the channel. The full definition is recorded in Appendix C.1.

The massless-pole threshold is the normalisation extremum

$$\frac{4}{27} = \max_{0\le y\le 1} y(1-y)^2 \qquad \text{at} \qquad y = \frac{1}{3},$$

the normalisation of the spin-2 spectral-pole condition, where $y = m^2/(k^2 + m^2)$ is the mass fraction of the transverse-traceless kernel $k^4/(k^2 + m^2)^2 = (1 - y)^2$. The emergence scale is reached when the spin-2 mode becomes massless, which is the fixed point

$$\frac{V_3\,\Pi^2_{\mu\nu}(kL, k^2/M_{\rm P}^2)}{32\pi^2} = \frac{4}{27} \qquad \text{at} \qquad k = M_{\rm G}.$$

On the self-similar flow $L(k) = C/k$ with $C = M_{\rm P}\sqrt{\pi}$ this is one equation in one unknown. It is solved by bisection on the interval $[0.3\,M_{\rm G}, 3\,M_{\rm G}]$ with the stopping width 10^{-15}, and the self-consistency between $M_{\rm G} = C/kL$ and the pole scale is iterated to the displayed precision. The fixed point is

$$kL = 2.49353.$$

Proposition 3.1 (Unique window-capacity fixed point)**.** *On the self-similar flow $L(k) = C/k$ with $C = M_{\rm P}\sqrt{\pi}$, the fixed-point function*

$$F(kL) = \frac{V_3\,\Pi^2_{\mu\nu}(kL, k^2/M_{\rm P}^2)}{32\pi^2} - \frac{4}{27}, \qquad k = M_{\rm P}\sqrt{\pi}/kL,$$

is monotonic on the interval $kL \in [0.3, 3]$ in the numerical evaluation of the spectral sum, and changes sign once. The spectral-pole closure $F(kL) = 0$ therefore has a unique solution, to which the bisection with the self-consistency iteration converges.

Proposition 3.2 (Weak-coupling comparison of the window capacity)**.** *The window capacity is determined by the spectral-pole closure $F(kL) = 0$ (Proposition 3.1). After that closure has been made, the window-capacity form $\alpha_W = 2/kL^5$ can be inverted using the reference weak coupling; the resulting post hoc comparison value of kL agrees with the spectral-pole value to within 0.0002%.*

The cross-check does not enter the closure. The spectral-pole condition fixes kL from the spin-2 channel alone; the weak-coupling inversion uses a Standard Model reference value only after the spectral value has been computed. Appendix B records the reproducibility convention. The endpoint geometry is first-principles in the sense that the Gaussian width of the scale field is $L_{\rm Cg} = \sqrt{\pi}$, which fixes the critical curvature

$$R_{\rm c} = \frac{6}{\pi}.$$

The window capacity is then

$$(kL)^2 = 6.2177,$$

which sits below the Euclidean-period value $2\pi = 6.2832$ by 1.04%.

3.5 The generation count

The number of fermion generations is the number of spinor modes of the internal tower that fit inside the coarse-graining window. The criterion is

$$n + \frac{3}{2} < (kL)^2.$$

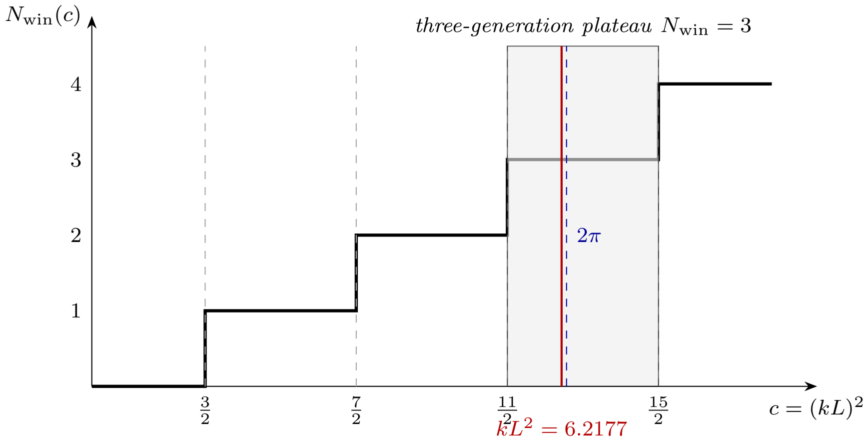

Figure 2. Window-capacity plateau. The generation count $N_{\rm win}(c)$ is a step function of the window capacity $c = (kL)^2$, with thresholds at $n + \frac{3}{2}$ for $n = 0, 2, 4, 6$. The value $N_{\rm win} = 3$ is a plateau over the whole interval $\frac{11}{2} < c < \frac{15}{2}$, so the count is insensitive to the precise value of c. Both the self-consistent closure value $kL^2 = 6.2177$ and the Gaussian benchmark $2\pi = 6.2832$ lie inside this plateau.

With $(kL)^2 = 6.2177$ the retained modes are

$$n = 0\ (1.5), \qquad n = 2\ (3.5), \qquad n = 4\ (5.5),$$

while $n = 6$ gives 7.5, which lies above the edge and is excluded. The window therefore contains exactly three modes, and the generation count is

$$n_g = 3.$$

The mode mass ladder is $m_n \propto e^{-\alpha n}$ with the extrusion order $n = \{0, 2, 4\}$, so the three generations are ordered by their extrusion order along the scale flow, and the hierarchy among them is the geometric hierarchy recorded in Section 6. This closes the two quantities that the companion paper left open, the numerical value of the window capacity and the generation count that follows from it. The step structure of the criterion is shown in Figure 2.

4 Content symmetries and the torsion parameter

This section records the counting symmetries of the fermion content and derives the torsion parameter from them. The content is fixed by the companion paper, which obtains the five representations and the hypercharge assignment from anomaly cancellation and the Yukawa structure. The present section turns that content into the numerical identities that the later sections read.

4.1 Content counting

The five representations per generation are $Q_L = (3, 2)$, $u_R = (3, 1)$, $d_R = (3, 1)$, $L_L = (1, 2)$, and $e_R = (1, 1)$. The left-handed components are the two doublets, Q_L with $3 \times 2 = 6$

components and L_L with $1 \times 2 = 2$ components, so

$$N_L = 6 + 2 = 8.$$

The right-handed singlets give

$$N_R = 3 + 3 + 1 = 7.$$

The number of Weyl fermions per generation is

$$N_f = N_L + N_R = 15,$$

and the total over three generations is 45. The left-handed count equals the number of colour generators

$$N_L = N_g = N_c^2 - 1 = 8,$$

and the right-handed count is one less. This is the content-gauge symmetry recorded and proved in the companion paper. The chiral asymmetry is

$$N_L - N_R = 1,$$

an odd integer, and its parity is the non-trivial element of $\pi_1(RP^3) = \mathbb{Z}_2$, the non-trivial spin structure of the internal space.

The hypercharge assignment satisfies

$$\sum Y = 0,$$

the vanishing of the mixed gravitational anomaly, and its first non-zero moment is

$$\sum Y^2 = \frac{10}{3},$$

the sum $6 \cdot (1/6)^2 + 3 \cdot (2/3)^2 + 3 \cdot (1/3)^2 + 2 \cdot (1/2)^2 + 1 \cdot (1)^2$. The fermion content also obeys the identity

$$N_f = 2 \sum Y^2 \Delta_f^2 = 15,$$

with the fermion conformal weight $\Delta_f = 3/2$, a relation between the Weyl count, the hypercharge moment, and the conformal weight that the torsion parameter of Section 4.3 uses.

4.2 Closure of the conformal-gauge duality

The companion paper fixes the colour algebra by the minimality principle, its third application after $SU(2)$ over $U(2)$ and RP^3 over the lens spaces, and derives the conformal-gauge duality as an identity of those selections. The present section verifies its numerical content. The conformal coupling of a scalar in dimension d is

$$\xi = \frac{d-2}{4(d-1)},$$

and with $d = 3$ it is $\xi = 1/8$. The duality states

$$N_g \xi = 1,$$

equivalently $N_g \Delta = 2(d-1)$ with the scalar conformal weight $\Delta = (d-2)/2 = 1/2$. Substituting $\xi = 1/8$ gives $N_g = 8$, and $N_c^2 - 1 = 8$ has the single positive solution $N_c = 3$. The duality is therefore the statement that the conformal weight per generator is the conserved unit, and it ties the spacetime dimension to the colour number through $d = N_c = 3$.

4.3 The torsion parameter

The internal space carries an Einstein-Cartan geometry with a totally antisymmetric torsion

$$T^a{}_{bc} = \frac{\tau}{L} \varepsilon^a{}_{bc},$$

where τ is the dimensionless torsion modulus. It is not a free input. The torsion field equation sources the torsion by the chiral current and screens it by the hypercharge polarisation. The chiral drive is the intensive quantity

$$\langle \chi \rangle = \frac{N_L - N_R}{N_f} = \frac{1}{15},$$

the chiral asymmetry per unit fermion content, and the renormalised hypercharge polarisation is

$$\Pi_{\rm ren} = \sum Y^2 = \frac{10}{3}.$$

The torsion modulus is then

$$\tau = \frac{\langle \chi \rangle}{\Pi_{\rm ren}} = \frac{N_L - N_R}{N_f \sum Y^2} = \frac{1}{15 \cdot 10/3} = \frac{1}{50}.$$

The two field-theoretic layers of this closure are the following, with the complete seven-layer statement in Appendix C.2. The bare torsion field equation carries the window geometry $1/(2\pi k L^4)$, and the hypercharge polarisation is renormalised at the emergence scale by the condition

$$\Pi_{\rm ren}(M_{\rm G}) = \sum Y^2 = \frac{10}{3},$$

the coupling convention of the construction, the analogue of a fixed renormalisation point. The counter-term absorbs the bare one-loop polarisation, whose magnitude is $0.0014 \times \sum Y^2$ (Appendix C.2). The window capacity $2\pi k L^4$ is a notation common to the bare field equation and the screening factor. It carries no independent spectral sum and cancels identically, so the surviving value is the content ratio fixed by the three-layer skeleton of Appendix C.2, the $\mathbb{Z}_2$ topology in the numerator, the anomaly normalisation in the denominator, and the Einstein-Cartan field equation as the bridge.

4.4 Geometric dynamics

The torsion enters the spectrum and the effective potential through the geometric quantities that the later sections use. The $J = 2$ Lichnerowicz eigenvalue of the Einstein-Cartan sector is

$$\lambda_{\rm EC} = N_g\Big(1 + \frac{\tau}{2}\Big)^2 + 6 = 14 + 8\tau + 2\tau^2,$$

and the transverse-traceless eigenvalue is

$$\lambda_{\rm TT} = 14 = 2N_R.$$

The squash amplitude is

$$s_0 = n_{\rm broken}\,\tau = 2\tau,$$

with $n_{\rm broken} = 2$ the number of broken generators of $\mathfrak{su}(2)_R$, and its content ratio is

$$\frac{s_0}{N_R} = \frac{1}{175}.$$

The curvature relation of the Einstein-Cartan geometry is

$$R(\omega) = R_{\rm LC} - \frac{3}{2}\Big(\frac{\tau}{L}\Big)^2,$$

with $R_{\rm LC} = 6/L^2$, so the torsion corrects the curvature at order $\tau^2 \sim 10^{-4}$. These quantities are the geometric inputs to the gauge couplings of Section 5 and the electroweak order parameters of Section 7.

5 Gauge couplings

This section computes the three gauge couplings from the endpoint geometry of Section 3 and the content symmetries of Section 4. The couplings are not fitted to the observed values. They are the closed forms of the geometric reduction of the isometry zero modes, corrected by the content ratios that the companion paper fixes.

5.1 The weak coupling and its conservation law

The $\mathfrak{su}(2)_L$ coupling descends from the Kaluza-Klein reduction of the isometry zero modes on RP^3. The Killing normalisation gives the bare coupling

$$g_2^2 = 8\Big(\frac{M_{\rm G}}{M_{\rm P}}\Big)^2 kL^{-3}.$$

The window-edge identity $kLM_{\rm G} = M_{\rm P}\sqrt{\pi}$ of Section 3.3 collapses this to

$$g_2^2 = \frac{8\pi}{kL^5}, \qquad \alpha_W(M_{\rm G}) = \frac{g_2^2}{4\pi} = \frac{2}{kL^5},$$

so the weak coupling at the emergence scale is a pure function of the window capacity, with no mass scale entering. This is the bare geometric value. It is corrected by the geometric-dynamics conservation law

$$\frac{1}{\alpha_{\rm SM}} = \frac{1}{\alpha_W} + \frac{1}{N_c} - \frac{\tau^2\pi}{2}, \qquad N_c\Big(\frac{1}{\alpha_{\rm SM}} - \frac{1}{\alpha_W} + \frac{\tau^2\pi}{2}\Big) = 1,$$

where $\alpha_W = 2/kL^5$ is the window-capacity weak coupling and the terms $1/N_c$ and $\tau^2\pi/2$ are the colour-number and torsion content ratios. The conservation law is equivalent to the conformal-gauge duality $N_g\xi = 1$ of Section 4.2, and its complete derivation is given in Appendix C.4. The resulting central value is

$$g_2(M_{\mathrm{G}}) = \sqrt{\frac{4\pi}{1/\alpha_W + 1/N_c - \tau^2\pi/2}} = 0.508848,$$

which matches the Standard Model comparison value 0.508845 to +0.0005%. The bare Killing value overshoots by +0.34%, and the two content-ratio terms shift the result to the comparison value without introducing a fitted coefficient.

5.2 The hypercharge coupling

The $\mathrm{U}(1)_Y$ coupling descends from the $\mathfrak{su}(2)_R \to \mathrm{U}(1)_R$ breaking by the $J = 2$ squash. The squash rescales the U(1) normalisation by the metric factor

$$\kappa^2(s) = \frac{1+s}{(1-2s)^{5/2}},$$

evaluated at the squash amplitude $s_0 = 2\tau$, which gives $\kappa(2\tau) = 1.13183$. The mixing acts at the breaking scale k_{GUT}, and the coupling then runs down to M_{G}. A further correction of structural origin enters through the content ratio

$$\delta_{g_1} = -\tau\, r_{23} \sum Y^2 \Delta_f \xi,$$

where $r_{23} = 3/(10\sqrt{3})$ is the hypercharge-trace hierarchy ratio, and the product $\sum Y^2\Delta_f\xi = (10/3)\cdot(3/2)\cdot(1/8) = 5/8$ is the hypercharge capacity times the fermion conformal weight times the conformal coupling. This is the same conformal-gauge duality that closes the $1/N_c$ term of Section 5.1. The resulting coupling is

$$g_1(M_{\mathrm{G}}) = 0.60499.$$

Together with g_2, it determines the fine-structure constant by the two-loop geometric renormalisation group (the same running as $\alpha_s(M_Z)$ of Section 5.3), giving $\alpha^{-1}(M_Z) = 128.208$ against the observed 127.952, a deviation of +0.20%.

5.3 The colour coupling

The colour coupling shares the $\mathfrak{su}(2)$ Killing normalisation at order α^0, so at the breaking scale the two couplings coincide up to the long-root bifurcation

$$g_3(k_{\mathrm{GUT}}) = g_2(k_{\mathrm{GUT}}) \left(1 + \frac{\alpha_{\mathrm{GUT}}^2}{K}\right),$$

where $\alpha_{\mathrm{GUT}} = g_2(k_{\mathrm{GUT}})^2/(4\pi)$ and

$$K = \frac{J(J+2)}{d} = \frac{8}{3}$$

is the long-root condensation coefficient, the kinetic eigenvalue $J(J+2) = 8$ of the $J = 2$ squash over the internal dimension $d = 3$. The bifurcation is the two-loop self-coupling of SU(3) diluted by the long-root condensation, so g_3 sits above g_2 at the breaking scale. Running both couplings with the two-loop renormalisation group, with the geometric Yukawa $y_0 = 1$ of Section 6.2, gives

$$g_3(M_\mathrm{G}) = 0.49776,$$

and the strong coupling at the electroweak scale is

$$\alpha_s(M_Z) = 0.11799,$$

matching the observed 0.1179 to +0.073%.

5.4 The two-loop coefficients

The renormalisation group running of Section 5.3 uses the two-loop gauge beta functions. Their coefficients are not imported from a table. They are computed from the group-theoretic content, the quadratic Casimir and Dynkin index of each representation. The one-loop coefficients are

$$b_1 = \frac{41}{10}, \qquad b_2 = -\frac{19}{6}, \qquad b_3 = -7,$$

and the two-loop matrix is

$$B = \begin{pmatrix} 199/50 & 27/10 & 44/5 \\ 9/10 & 35/6 & 12 \\ 11/10 & 9/2 & -26 \end{pmatrix}.$$

These reproduce the standard values, but they are derived here from the representation content recorded in Section 4.1, so no external coupling table enters the computation. The top-Yukawa and Higgs-quartic beta functions entering the running are evaluated on the same content (the one-loop $9/2 = 3/2 + N_c$, the gauge terms $-3[C_2(Q_L) + C_2(u_R)]$ per group, and the full two-loop coefficients). Their derivation is given in Appendix C.6. The running is carried out with the geometric Yukawa $y_0 = 1$ as the scalar input, and the scale $k_\mathrm{GUT} = M_\mathrm{P}\sqrt{\pi}/L_\mathrm{GUT}$ with $L_\mathrm{GUT} = \sqrt{3}/\tau$ is the isometry-breaking scale of the $J = 2$ squash.

6 Flavour structure

This section computes the fermion masses from the window capacity of Section 3 and the torsion parameter of Section 4. The masses are not a set of free Yukawa couplings. They are the geometric ladder of the extrusion order along the scale flow, fixed by the window and its non-adiabatic corrections. The neutrino sector is a phenomenological extension of this ladder and is recorded in Appendix E.

6.1 The ladder

The generation hierarchy is the extrusion order $n = \{0, 2, 4\}$ of Section 3.5, and the mass ladder is the geometric exponent attached to each rung. The first exponent is the window width reduced by the non-adiabatic correction

$$\alpha_{\rm up} = kL - 2\tau,$$

the rung spacing is

$$\Delta = 6(1 - n_s)kL_{\rm CMB},$$

where the tilt decrement is $1 - n_s = \tau w_{\rm W}$ with $w_{\rm W} = (1 + 2 + 1 + 3)/(d + 1) = 7/4$ for $d = 3$ (the RP^3 Weyl-window ratio; see App. C.14), and $kL_{\rm CMB} = kL(1 - \tau/4)$ is the local window at the cosmic-microwave pivot. Below we write the positive decrement as $n_s^{\rm tilt} \equiv 1 - n_s$. The down and lepton ladders split by the hypercharge ratio

$$\frac{9}{8} = \frac{1}{1 - (Y_d/Y_l)^2},$$

and the rung exponents are

$$\alpha_{\rm dn} = \alpha_{\rm up} - \frac{18}{17}\Delta, \qquad \alpha_{\rm lp} = \alpha_{\rm up} - 2\Delta,$$

with the down first-generation step carrying the colour-dilution correction

$$\alpha_{\rm sd} = \alpha_{\rm dn} - \frac{kL_{\rm CMB}}{6}.$$

These ladder indices are intermediate definitions: each exists only to carry the mass ratios that follow, and their numerical values are fixed in Appendix C.7. The ladder and its colour-dilution step are derived in Appendix C.7.

6.2 The top-quark anchor

The top quark is the anchor of the ladder. Its Yukawa coupling is the $(0, 0)$ diagonal overlap of the $n = 0$ mode, which is exactly normalised

$$y_0 = 1,$$

and it is scale-invariant, in the sense that it does not run along the geometric renormalisation group. The top mass is then fixed by the vacuum expectation value of Section 7,

$$m_t = \frac{y_0 v}{\sqrt{2}} = 174.1 \text{ GeV}.$$

The normalisation $y_0 = 1$ is derived in Appendix C.8.

6.3 The lower masses

The bottom Yukawa is the geometric average

$$\frac{y_b}{y_t} = e^{-(2\alpha_{\rm dn} - n_s^{\rm tilt}(kL_{\rm CMB}+2\tau))} = 0.024344,$$

where $n_s^{\rm tilt}$ is the positive spectral-tilt decrement. This gives the bottom mass

$$m_b = \frac{y_b}{y_t}\, m_t = 4.238 \text{ GeV}.$$

The mass ratios within each sector are the geometric exponents

$$\frac{m_s}{m_d} = e^{2\alpha_{\rm sd}} = 19.62, \qquad \frac{m_b}{m_s} = e^{2\alpha_{\rm dn}} = 44.87, \qquad \frac{m_\mu}{m_e} = e^{2\alpha_{\rm lp}+\sqrt{2\pi}} = 206.36,$$

and the charm and up ratios are

$$\frac{m_t}{m_c} = e^{2\alpha_{\rm up}} = 135.2, \qquad \frac{m_t}{m_u} = e^{2\alpha_{\rm up}} e^{2kL_{\rm CMB}+\ln 4} = 77304.$$

The electron mass is the Planck-anchored closed form

$$m_e = M_{\rm P}\, e^{-20kL}\,(1 - s_0\kappa) = 0.510 \text{ MeV},$$

where $s_0\kappa$ is the squash correction of Section 4.4. The down quark, the strange quark, the muon, and the tau follow from these ratios and the sector anchors. The full set of absolute masses descends from the single Planck anchor through the window capacity and the ladder exponents, with no mass inserted by hand. The closed forms of this subsection are derived in Appendix C.9.

7 Electroweak breaking

This section computes the quartic order parameter, the mass gap, and the vacuum expectation value. The CKM and CP content is recorded in Appendix E. These are the quantities that turn the condensation structure of the companion paper into numbers.

7.1 The quartic order parameter

The scalar order parameter is the quartic coupling of the colour-singlet sector. It is fixed by the stationarity of the effective potential at the onset of the symmetry breaking, together with the Einstein-Cartan torsion algebra. The closed form is

$$\lambda = \frac{\xi(R_{\rm c} - R_{\rm GUT})}{(2\tau)^2} = 149.145,$$

where $\xi = 1/8$ is the conformal coupling, $R_{\rm c} = 6/\pi$ is the critical curvature, and $R_{\rm GUT} = 6/L_{\rm GUT}^2$ with $L_{\rm GUT} = \sqrt{3}/\tau$ is the curvature at the breaking onset. The torsion algebra fixes the three coefficients of the parity-preserving torsion Lagrangian as $a = M_{\rm G}^3/4$, $b = 4a$ by the algebraic-torsion condition, and $c = -(7/3)a$ by the trace term, and the stationarity

$\partial V/\partial\varphi = 0$ then selects the quartic coupling. The value 149 is the dimensionless self-coupling that the stationarity and the torsion algebra produce, with no scalar mass or coupling inserted by hand. This λ is the colour-singlet order parameter. It is distinct from the Higgs quartic λ_H that enters the tree-level Higgs mass $m_H = \sqrt{2\lambda_H}\, v$ of Section 10.6; λ_H itself is closed by the pseudo-dilaton identity of Appendix C.12. The complete derivation is in Appendix C.11.

7.2 The mass gap

The companion paper derives the mass-gap structure from the condensation of the colour-singlet scalar. The present section records its numerical content. The condensate trigger is the long-root tachyon

$$m^2_{\text{long}} = K(R - R_{\text{c}}) < 0 \qquad \text{for} \qquad R < R_{\text{c}},$$

with $R_{\text{c}} = 6/\pi$. Below the critical curvature the mode condenses, and the lowest colour-singlet excitation acquires the positive squared mass

$$m^2_\delta = \frac{16}{3}(R_{\text{c}} - R) > 0.$$

At the flow endpoint the condensation condition is realised numerically, not left conditional. The endpoint is $\sigma_C = 6.95 \times 10^{41}$ GeV^{-1}, the Hubble scale of Section 8.2, and the long-root mode is already tachyonic at the emergence scale, $m^2_{\text{long}}(M_{\text{G}}) = K(R(M_{\text{G}}) - R_{\text{c}}) = -2.52 < 0$. Since $R = 6/L^2$ decreases monotonically along the self-similar flow $L(k) = C/k$, the endpoint curvature satisfies $R(\sigma_C) < R(M_{\text{G}}) < R_{\text{c}} = 6/\pi$, so the endpoint lies in the condensed phase and the positive scalar mass $m^2_\delta > 0$ is realised there.

The condensate expectation value is fixed by

$$\langle E\rangle^2 = \frac{8}{3}\frac{R_{\text{c}} - R}{\lambda},$$

and the gap is $m^2_\delta = 2\lambda\langle E\rangle^2$. The physical value of this gap is the glueball tower of Section 9.2, whose lightest member sits at $8\Lambda_{\text{QCD}}$. The colour-singlet sector is therefore gapped, with no zero mode beyond the constants, and the gap is a positive spectral lower bound rather than a fitted scale.

7.3 The vacuum expectation value

The electroweak vacuum expectation value is the emergence scale times the window-squared line. Its closed form is

$$v = M_{\text{G}}\,\frac{3\alpha}{\pi}\,e^{-4\pi kL}\,(1 - s_0\kappa) = 246.19\ \text{GeV},$$

where the squash correction $s_0\kappa$ with $s_0 = 2\tau$ and $\kappa = 1.13183$ is the $J = 2$ isometry-breaking normalisation of Section 5.2. The value matches the observed 246.22 GeV to -0.012%. The hierarchy $v/M_{\text{G}} \sim 10^{-16}$ is the geometric window-squared line, the analogue for the electroweak scale of the gauge-coupling conservation law of Section 5.1, and it is produced by the exponential $e^{-4\pi kL}$ rather than by a fine-tuned scalar mass.

8 Cosmology

This section computes the homogeneous cosmological quantities from the same spectral datum. Two Gaussian entropies, one at the Planck anchor and one at the vacuum floor, fix the entropy integral, and from it the Hubble rate, the dark-energy fraction, and the microwave temperature follow. The endpoint residual is then propagated as a cold source in the linear cosmological comparison. No background cosmological parameter is taken from observation in these closed forms.

8.1 Two Gaussian entropies

The entropy integral of the scale flow is the logarithm of the ratio of two Gaussian widths. The Planck Gaussian has width M_{P}^2, and the vacuum-floor Gaussian has width $\sqrt{\rho_\Lambda}$, where the vacuum energy density is the dark-energy content, a closure relation of the spectral datum

$$\rho_\Lambda = Y_u\, m_{\nu_1}^4\,(1 - 4s_0\kappa) = \frac{2}{3} m_{\nu_1}^4 (1 - 4s_0\kappa).$$

The factor $(1 - 4s_0\kappa)$ is the dark-energy-weight symmetry correction of the $J = 2$ squash. The Weinberg mass m_{ν_1} carries $+s_0\kappa$, so its fourth power carries $+4s_0\kappa$, and the weight cancels it to order $s_0^2\kappa^2$, keeping ρ_Λ conserved under the squash level transfer (Appendix C.14). The entropy integral is

$$\int \gamma_M = \ln \frac{M_{\mathrm{P}}^2 \sqrt{2\pi + r_{23}}}{\sqrt{\rho_\Lambda}} = 139.2537,$$

with $r_{23} = 3/(10\sqrt{3})$ the hypercharge-trace hierarchy ratio. This single number, the logarithm of the window span, is the bridge between the ultraviolet Planck anchor and the infrared vacuum floor, and it encodes the Boltzmann entropy of the flow as $S = \ln W$ with W the window span.

8.2 The Hubble rate and the dark-energy fraction

The Hubble rate is the infrared endpoint of the emergence window,

$$H_0 = M_{\mathrm{P}} \sqrt{\pi}\, e^{-\int \gamma_M} = 1.4388 \times 10^{-42}\ \mathrm{GeV},$$

which corresponds to $67.45\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. The dark-energy fraction is

$$\Omega_\Lambda = \frac{\rho_\Lambda}{3H_0^2 M_{\mathrm{P}}^2} = \frac{2}{3} + \frac{r_{23}}{3\pi} = 0.68504.$$

The second form shows that Ω_Λ is a content ratio, the $2/3$ of the maximum-entropy closure plus the hypercharge-trace correction $r_{23}/(3\pi)$. The value matches the Planck determination to $+0.05\%$. The full derivation of this section is in Appendix C.14.

The zero-mode effective action, projected onto the transverse-traceless zero mode, has the low-energy tensor structure of the Einstein–Hilbert action with a vacuum term,

$$S_{\mathrm{eff}} = \frac{M_{\mathrm{P}}^2}{2} \int d^4x \sqrt{-g}\, R - \int d^4x \sqrt{-g}\, \rho_\Lambda,$$

with $M_{\rm P}^2 = 1/(8\pi G)$. This statement is not a derivation of H_0 from a Friedmann equation. In the computation, H_0 is fixed by the entropy integral and ρ_Λ by the neutrino floor. Their ratio is the dark-energy content fraction

$$\Omega_\Lambda = \frac{\rho_\Lambda}{3H_0^2 M_{\rm P}^2} = \frac{2}{3} + \frac{r_{23}}{3\pi},$$

so ρ_Λ alone corresponds to $\Omega_\Lambda = 0.68504$, not to a universe with $\Omega_\Lambda = 1$. A homogeneous Friedmann reading is recovered only after the full endpoint budget is included,

$$\rho_{\rm tot} = \rho_\Lambda + \rho_b + \rho_\Sigma = 3H_0^2 M_{\rm P}^2, \qquad \Omega_b + \Omega_\Sigma + \Omega_\Lambda = 1.$$

Thus the zero-mode action is compatible with the standard homogeneous cosmological description once the baryon component and the endpoint residual are included. The Friedmann form is a consistency reading of the closed outputs, not an additional input and not the step from which H_0 is obtained.

8.3 The microwave background temperature

The cosmic-microwave temperature is the photon floor set by the lightest neutrino. The raw floor is

$$T_{\rm CMB}^{\rm raw} = \frac{m_{\nu_1} r_{12}}{\pi}(1 - s_0\kappa)(1 - \tau\Delta_s) = 2.7310 \text{ K},$$

where $r_{12} = 3/10$ is the hypercharge ratio and $\Delta_s = 1/2$ is the scalar conformal weight, and the factor $(1 - s_0\kappa)$ is the same squash correction as the electroweak scale. The photon floor carries $-s_0\kappa$ while the seesaw mass carries $+s_0\kappa$ (the level transfer). The finite photon zero-mode factor at the endpoint is

$$C_\gamma = 1 - \frac{\tau}{\pi^2}, \qquad T_{\rm CMB} = C_\gamma T_{\rm CMB}^{\rm raw} = 2.72547 \text{ K}.$$

The corrected value is used for the CMB-monopole comparison and for the linear comparison propagation. This endpoint photon correction is not fed back into the internal baryon-density closure of Section 8.4; that closure uses the raw photon floor together with the raw Sakharov value of η_B.

8.4 Flatness from the closure relation

The three energy fractions close to unity by the closure relation. The baryon fraction is

$$\Omega_b = \frac{\eta_B^{\rm raw} n_\gamma^{\rm raw} m_p}{\rho_{\rm crit}} = 0.04925,$$

built from the raw Sakharov baryon asymmetry η_B of Appendix E, the raw photon density

$$n_\gamma^{\rm raw} = \frac{2\zeta(3)}{\pi^2}\left(T_{\rm CMB}^{\rm raw}\right)^3,$$

the proton mass of Section 9.1, and the critical density $\rho_{\rm crit} = 3H_0^2 M_{\rm P}^2$ with the Hubble rate of Section 8.2. The C_γ factor of Section 8.3 is a finite endpoint correction used for the

Quantity	Comparison value
$\Omega_m = \Omega_b + \Omega_\Sigma$	0.314956
r_{drag}	146.980 Mpc
$100\,\theta_*$	1.041595
z_{eq}	3414.88
σ_8	0.81423
S_8	0.83428

Table 2. Compressed linear-cosmology quantities obtained by propagating the closed endpoint set with $\Omega_{\text{cdm}} := \Omega_\Sigma$ and the corrected photon monopole through CAMB. The propagation is a comparison step only: no CAMB output is used to set a closure parameter. The reionization optical depth used for the CMB spectra is a standard visibility parameter, not a fundamental closure.

microwave-monopole comparison, not a multiplier in this internal Ω_b computation. The dark-energy fraction is 0.68504 of Section 8.2, and the remaining fraction is fixed by flatness

$$\Omega_\Sigma = 1 - \Omega_\Lambda - \Omega_b = 0.26570.$$

The sum

$$\Omega_b + \Omega_\Sigma + \Omega_\Lambda = 1.00000$$

is imposed by the closure relation, not fitted, because the three fractions are built from the same content and the flatness closure fixes the third. The remaining fraction Ω_Σ is the maximum-entropy endpoint residual. It occupies the cold-matter slot in the linear cosmological propagation. The observational cold-dark-matter fraction is used only as the reference column in comparisons. Its reading, together with the low-acceleration scale a_0, is recorded in Appendix G.

8.5 Endpoint residual in linear propagation

For homogeneous and linear cosmology the endpoint Hamiltonian residual is propagated as a conserved cold source,

$$T^{\Sigma}_{\mu\nu} = \rho_\Sigma u_\mu u_\nu, \qquad \nabla_\mu(\rho_\Sigma u^\mu) = 0, \qquad p_\Sigma = 0, \quad c^2_{s,\Sigma} = 0, \quad \pi_\Sigma = 0.$$

Equivalently, in a standard Boltzmann propagation it fills the cold-dark-matter slot as $\Omega_{\text{cdm}} := \Omega_\Sigma$, without adding a particle species or a fitted dark-sector parameter. The local acceleration projection of the same endpoint is not double-counted in this linear propagation; it is the separate low-acceleration branch discussed in Appendix G. With the closed outputs and the corrected photon monopole fixed, a CAMB propagation used only for comparison gives the compressed quantities in Table 2. CAMB supplies no closure parameter; it is used only to propagate the internally closed set to standard compressed observables. These numbers are comparison quantities, not fitted constraints, and are not read back into the closure chain.

8.6 The gravitational sector

The gravitational sector is the transverse-traceless zero mode of the spectral construction. The gravitational coupling takes the form

$$G_{\mathrm{N}} = \frac{1}{8\pi Z_{\mathrm{phys}} M_{\mathrm{P}}^2},$$

where the matter back-reaction factor is

$$Z_{\mathrm{phys}} = \frac{\lambda}{\lambda + \sigma} = 1.000000,$$

with $\lambda = 8M_{\mathrm{G}}^2/kL^2$ the transverse-traceless eigenvalue at the emergence scale and σ the five-channel matter self-energy,

$$\sigma = \Big|\sum_{\text{channels}} V_3\,\Pi^2_{\text{channel}}\Big| \frac{k^2}{M_{\mathrm{P}}^2},$$

the sum of the five channel amplitudes used in the spectral-pole closure of Appendix C.1: the $T_{\mu\nu}$ spin-2, $T_{\mu\nu}$ spin-0, F^2, G^2, and J^μ channels, each the discrete spectral sum over RP^3. At the emergence scale the ratio is $\sigma/\lambda = 3.15 \times 10^{-37}$, so $Z_{\mathrm{phys}} = 1$ to the displayed precision. The self-energy is tiny compared with the eigenvalue, so the back-reaction does not shift the gravitational normalisation. The Newton constant is the single observed anchor of the construction, and $G_{\mathrm{N}} = 1/(8\pi M_{\mathrm{P}}^2)$ is the identity that defines the anchor. With the anchor fixed by the observed value, the residue $Z_{\mathrm{phys}} = 1$ shows that matter back-reaction is negligible at the stated level.

The low-acceleration scale a_0 that follows from this zero-mode structure, and the endpoint-residual reading, are recorded in Appendix G.

9 QCD scale and confinement

This section computes the QCD scale, the proton mass, the glueball spectrum, and the confinement quantities. The primordial helium abundance is recorded in Appendix F. The QCD scale descends from the colour coupling of Section 5.3, and the spectral quantities descend from it.

9.1 The QCD scale and the proton mass

The QCD scale is fixed by the two-loop running of the colour coupling $g_3(M_{\mathrm{G}})$ of Section 5.3 down to the point where the coupling becomes strong, with the top-quark threshold matched at the geometric mass m_t. The extracted scale carries the correction $(1 - s_0\kappa/N_g)$ for the Yukawa difference between the scale-invariant geometric Yukawa $y_0 = 1$ and the running Standard Model top Yukawa in the two-loop gauge beta function, with $s_0 = 2\tau$ the squash amplitude and $N_g = 8$ the generator count. The correction is included explicitly in the scale definition (Appendix C.15). The result is

$$\Lambda_{\mathrm{QCD}} = 0.2074 \text{ GeV},$$

a deviation of -1.25% from the observed 0.210 GeV. The proton mass is the three-quark bound state expressed through the symmetry content, a spectral estimate of the confinement scale

$$m_p = N_c\Delta_f\Big(1-\frac{1}{N_g^2\Delta_s}\Big)\Lambda_{\rm QCD}\Big(1+\tau\kappa\sum Y^2\Delta_s\Big) = \frac{279}{64}\Lambda_{\rm QCD}\Big(1+\frac{5\tau\kappa}{3}\Big) = 0.9382 \text{ GeV},$$

where $N_c = 3$, $\Delta_f = 3/2$, and the factor $1 - 1/(N_g^2\Delta_s) = 31/32$ is the conformal-duality correction[1], with the final bracket the scheme correction between the constituent-quark scale and the $\overline{\rm MS}$ scale. The value matches the observed proton mass to -0.012%.

9.2 The glueball spectrum

The colour-singlet scalar sector of the companion paper has a discrete spectrum whose lightest level is the mass gap. The glueball tower is the geometric ladder

$$\lambda = 2\lambda_{\rm gluon} + C_2(J) + n\,N_g\xi,$$

with $N_g\xi = 1$, and the lightest scalar, a spectral estimate of the mass gap, is

$$m_G = \lambda(0^{++})\Lambda_{\rm QCD} = 8\Lambda_{\rm QCD} = 1.659 \text{ GeV}.$$

The tower ratios are geometric,

$$2^{++}/0^{++} = \sqrt{2}, \qquad 0^{++*}/0^{++} = 1.50, \qquad 0^{-+}/0^{++} = 1.46,$$

so the tensor state sits at 2.346 GeV, the first conformal excitation at 2.489 GeV, and the pseudoscalar at 2.418 GeV. The pseudoscalar level 0^{-+} at 2.418 GeV is compared with the quenched lattice-QCD prediction $m_{0^{-+}} = 2560$ MeV [16] and with the glueball-like state $X(2370)$, whose mass 2359^{+13}_{-14} MeV and spin-parity $J^{PC} = 0^{-+}$ were reported by the BESIII Collaboration in J/ψ radiative decays [17–19]. The level sits 5.5% below the pure-gauge lattice value and 2.5% above the $X(2370)$ mass. The pure-gauge comparison is not a full hadronic calculation: in the physical spectrum the 0^{-+} glueball component can mix with flavour-singlet pseudoscalar channels, including η' and η_c, and such mixing can shift the observed resonance relative to the quenched value. The present value is therefore listed as the pseudoscalar comparison target of the tower, while the scalar 0^{++} level remains a lattice-scale comparison [16]. The $\sqrt{2}$ ratio is the two-gluon tensor over the two-gluon scalar, a pure spectral ratio. This closes the mass-gap structure of Section 7.2 at the numerical level, with the gap given by the lightest glueball rather than by a fitted scale.

[1] The factor 31/32 follows from the two identities $N_g\xi = 1$ and $N_g\Delta_s = 2(d-1)$ of Section 4.2, since $1/(N_g^2\Delta_s) = 1/(8^2\cdot 1/2) = 1/32 = \xi/4$, with $\Delta_s = (d-2)/2 = 1/2$ the scalar conformal weight. The derivation is given in Appendix C.15.

9.3 String tension and deconfinement

The confinement quantities are the spectral eigenvalues of the transverse-traceless and vector sectors. The string tension is

$$\sigma = \frac{\lambda_{\mathrm{TT}}}{\pi}\Lambda_{\mathrm{QCD}}^2 = \frac{14}{\pi}\Lambda_{\mathrm{QCD}}^2 = 0.1917\ \mathrm{GeV}^2,$$

where $\lambda_{\mathrm{TT}} = 14$ is the lowest transverse-traceless eigenvalue. The deconfinement temperature is

$$T_d = \frac{\lambda_{\mathrm{vector}}}{N_c}\Lambda_{\mathrm{QCD}}(1-\tau\kappa) = \frac{4}{3}\Lambda_{\mathrm{QCD}}(1-\tau\kappa) = 270\ \mathrm{MeV},$$

where $\lambda_{\mathrm{vector}} = 4$ is the lowest vector eigenvalue and $(1-\tau\kappa)$ the chiral-squash correction, since deconfinement is accompanied by chiral restoration, so the deconfinement scale inherits the chiral-squash content of Section 4.4. The ratio

$$\frac{\sigma}{T_d^2} = \frac{14}{\pi}\cdot\frac{9}{16}(1-\tau\kappa)^{-2} = \frac{126}{16\pi(1-\tau\kappa)^2} = 2.6242$$

is the self-consistency of the confinement scale. The content ratio $126/(16\pi) = 2.5068$ between the area law and the deconfinement temperature, softened by the chiral-squash factor $(1-\tau\kappa)^2$. The full spectral derivation of this subsection is in Appendix C.15.

10 Numerical results and comparison

This section collects the numerical results and compares them with observation. The full parameter table is given in Appendix A, produced from the closed forms of Appendix C rather than stated independently. The present section summarises the table by sector and records the deviation distribution.

The numerical comparisons reported below are evaluations of a fixed closure chain. The reference values are not used in generating the computed outputs, and scheme-sensitive quantities are labelled by their perturbative and renormalisation status. Agreement is therefore interpreted as a correlated consistency test of the closed numerical construction, rather than as a collection of independent precision predictions.

10.1 The interface chain

The interface with the Standard Model is a boundary-condition problem rather than an embedding. At the emergence scale M_{G} the spectral sums fix the ultraviolet boundary conditions, namely the gauge couplings $g_i(M_{\mathrm{G}})$ of Section 5, the geometric Yukawa $y_0 = 1$ of Section 6.2, the Higgs quartic λ_H of Appendix C.12, the electroweak scale v of Section 7.3, and the cosmological densities, all closed forms of the spectral datum. Below M_{G} the geometric renormalisation group supplies the scale flow. The gauge couplings run with the two-loop coefficients of Section 5.4, which are derived from the group-theoretic content rather than imported, while the geometric Yukawa $y_0 = 1$ and the Higgs quartic λ_H are scale-invariant by the exact overlap of Section 6.2 and the pseudo-dilaton closure of Appendix C.12. Only the gauge couplings run, in contrast to the full parameter flow of

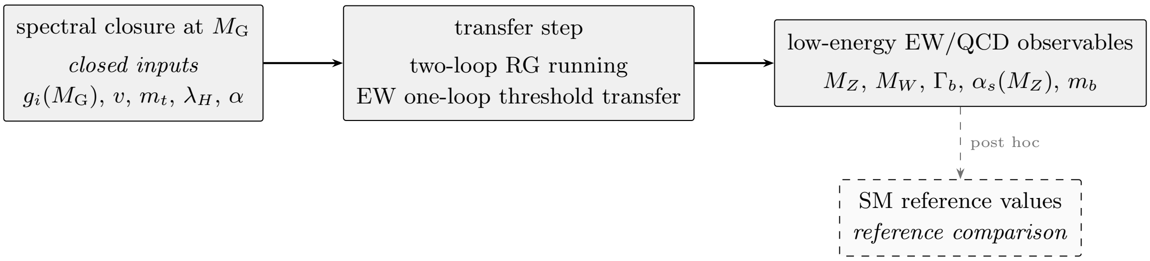

Figure 3. The three-stage interface chain of Section 10.1. The spectral construction supplies the ultraviolet boundary conditions at M_{G} (left); the transfer step applies the two-loop renormalisation-group running and the electroweak one-loop threshold transfer (centre); the low-energy electroweak and QCD observables emerge for comparison (right). The Standard-Model formulas (including the Denner electroweak reduction) enter only as the transfer map in the centre, never as a source of parameters; the reference values (dashed) enter only at the final comparison.

the Standard Model. The flow terminates at the internal Z mass M_Z, the self-consistent fixed point of the mass formula of Section 10.6, and the low-energy observables, namely m_t, m_b, $\alpha_s(M_Z)$, $\alpha(M_Z)$, and the electroweak precision quantities of Section 10.6, are compared with observation. The identification of the spectral-scale flow with the standard renormalisation-group equations in the shared low-energy regime is recorded in Appendix D.2. The spectral construction supplies boundary conditions; the Standard Model equations supply the low-energy transfer map. The two are not merged, and neither absorbs the other. The chain is summarised in Figure 3.

10.2 The parameter table

The computation produces the full set of quantities listed in Appendix A. One of them, the Newton constant G_{N}, is the observed anchor of Section 2.1. The remainder are computed from the spectral datum, each returned by a closed-form derivation recorded in Appendix C. Only a smaller subset of these quantities are independent comparison channels, however. Many are correlated downstream quantities built from the same few intermediates, and their epistemic status is recorded in Table 1 of Section 2, while the Tier column of Appendix A assigns each comparable quantity to one of the four tiers of Table 4. The words "closure", "derived", "scheme-sensitive", and "exploratory" in the sector summary refer to this status-and-tier classification. The table of Appendix A lists, for each quantity, its closed form, its computed value, the reference value where one exists, and the signed deviation. The sectors and their representative values are collected in Table 3.

10.3 The deviation distribution

The signed deviations are recorded in the table. They are small for most entries. The three gauge couplings agree to better than 0.01%, the fine-structure constant to 0.20%, the electroweak scale to 0.01%, the charged-fermion mass ratios and neutrino mass-squared splittings to within a percent, the cosmological fractions to within a percent, and the QCD scale to about one percent. The quantities that exceed one percent are discussed below. No computed value is adjusted after the comparison. The deviations are reported with

Sector	Representative quantity	Computed value	Status
Internal scale	generation count n_g	3	closure
Gauge couplings	weak coupling $g_2(M_{\mathrm{G}})$	0.50885	derived
Flavour	top mass m_t	174.1 GeV	derived
Electroweak	vacuum expectation v	246.19 GeV	derived
Electroweak precision	Z mass M_Z	91.124 GeV	derived
Cosmology	Hubble rate H_0	$67.45\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	exploratory
QCD	proton mass m_p	0.9382 GeV	spectral est.

Table 3. The sectors of the computation and one representative value per sector, with the epistemic status of Table 1. The full table is Appendix A.

Tier	Content	Reading
A	anchor and closure	the fixed data of the computation (the Newton constant, the structural constants, the generation count), not predictions
B	core outputs	gauge couplings, mass ratios, v, the electroweak masses, the top and charged-lepton masses, compared post hoc
C	scheme-sensitive	m_b, Λ_{QCD}, Γ_b, $\sin^2\theta_{\mathrm{eff}}$, consistency checks labelled by their perturbative and renormalisation status
D	exploratory	the baryon asymmetry, the cosmological fractions and T_{CMB}, the hadronic scales, phenomenological extensions

Table 4. The four tiers of the comparison. Tier A collects the fixed data of the computation, Tier B the core outputs, Tier C the scheme-sensitive quantities, and Tier D the exploratory downstream extensions. The tiers separate the fixed data from the outputs and the scheme-sensitive quantities from the exploratory extensions, so that distinct comparison channels are not combined into a single undifferentiated list.

their signs. The numerical stability of the chain and the isolation of the reference values from the computation are recorded in Appendix B. The full signed-deviation distribution is collected in Figure 4.

10.4 Deviations above one percent

Three derived quantities outside the electroweak precision block of Section 10.6 deviate from the observed value by more than one percent. Each is traced to an identified source. One further quantity is excluded because its reference is an upper bound rather than a

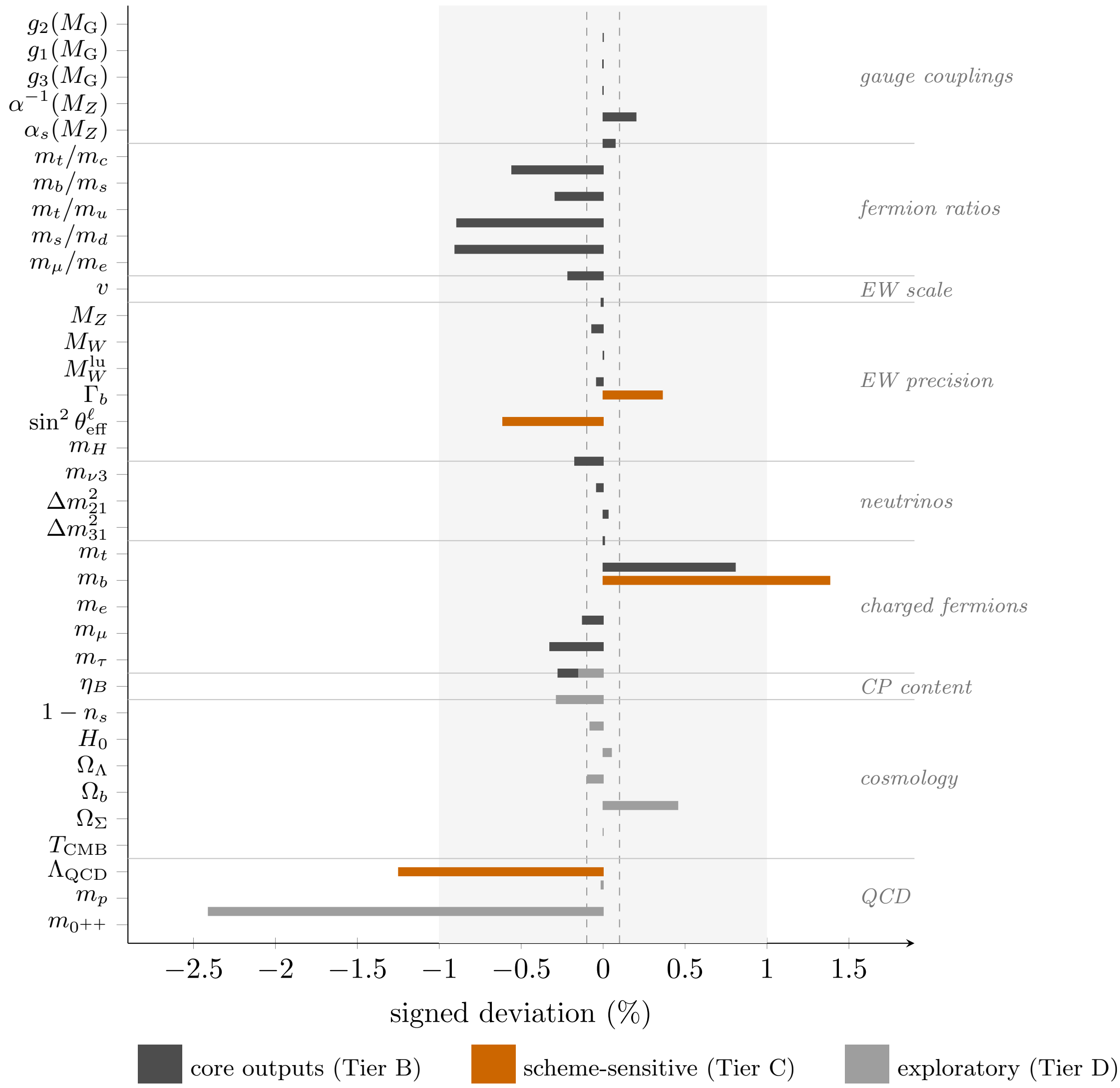

Figure 4. Signed deviations of the comparable quantities of Appendix A, grouped by sector and coloured by the tier of Table 4. The shaded band marks ±1%, the inner dashed lines ±0.1%. The majority of the entries cluster near zero across the gauge, flavour, electroweak, neutrino, cosmological, and QCD sectors; the three entries outside ±1% (m_b, $\Lambda_{\rm QCD}$, $m_{0^{++}}$) are discussed in Section 10.4.

central value. The tensor-to-scalar ratio $r = 0.0253$ of Appendix C.14 lies below the current combined bound $r < 0.036$ [20], so it is consistent with observation and carries no signed deviation. It is recorded in Appendix C.14. The neutrino entries are compared through the oscillation observables Δm^2_{21} and Δm^2_{31} rather than by treating m_{ν_2} as a proxy for $\sqrt{\Delta m^2_{21}}$. The raw light-pair rest split is 6.874×10^{-5} eV^2; the signed table uses the tilt-dressed propagation value of Appendix C.10.

- The bottom mass m_b deviates by +1.38%. It is the geometric average of Section 6.3, and the deviation reflects the scale-invariant anchor $y_0 = 1$ rather than a fitted bottom Yukawa.

- The QCD scale Λ_{QCD} deviates by -1.25%. It is fixed by the two-loop running, and the deviation is the loop-order accuracy of the two-loop against the four-loop extraction.
- The lightest glueball mass deviates by -2.41%. It is the spectral eigenvalue $8\Lambda_{\mathrm{QCD}}$, and the deviation combines the loop-order of Λ_{QCD} with the spectral-level approximation of the tower.

None of these is a fitted quantity, and none is adjusted to close the comparison. The three quantities, namely the bottom mass, the QCD scale, and the lightest glueball, are core quantities of Sections 6 and 9. The other flavour- and cosmological-extension relations, namely the PMNS mixing angles, the CKM phase, and the primordial abundances, are stated as formal closure in Appendices E and F rather than as precision comparisons, and carry no signed deviation here. The remaining core quantities agree with observation to within one percent.

The comparatively scheme-sensitive entries, such as m_b, Λ_{QCD}, m_t, M_W, and Γ_b, test the same closure data after QCD running, heavy-threshold matching, and electroweak radiative transmission.

10.5 Theoretical sensitivity

Fixing the central values within the adopted closure prescription does not by itself quantify how the outputs respond to the structural choices that carry the chain. The full response is recorded in Appendix D, and its conclusion is stated here. The dimensionless hierarchy is carried by exponentials of the window capacity, $v \propto e^{-4\pi kL}$ and $m_e \propto e^{-20kL}$, so the dimensionful outputs are exponentially sensitive to kL. A one-percent shift of kL moves v by -32%, m_e by -50%, and ρ_Λ by -262%, while the dark-energy fraction Ω_Λ, a pure content ratio, has elasticity exactly zero. This sensitivity does not introduce an additional continuous fit. The three conventions upstream of kL are each fixed within the stated prescription. The transverse-traceless kernel is fixed by the Ward identity and tracelessness, the threshold $4/27$ is the unique interior extremum of $y(1-y)^2$ selected by the maximum-entropy criterion and by the spin-2 content symmetry, and the window capacity $2\pi kL^4$ cancels out of the torsion modulus, leaving $\tau = 1/50$ fixed by the content ratio alone. The regularisation scheme is likewise selected by the unbiasedness clause of the coarse-graining principle, as recorded in Appendix D. The central values are therefore fixed within the adopted closure prescription, and the hierarchy is convention-sensitive. The quantitative statements behind each of these claims are in Appendix D. The exponential amplification of the hierarchy is summarised in Figure 5.

10.6 Electroweak precision observables

The interface block of Section 10.1 ends in the electroweak observables measured at high precision. Their computation is a standard electroweak reduction on the closed inputs, namely v, $G_F = 1/(\sqrt{2}\,v^2)$, $\alpha(M_Z)$ of Section 5, $\alpha(0)$ obtained from the one-loop QED

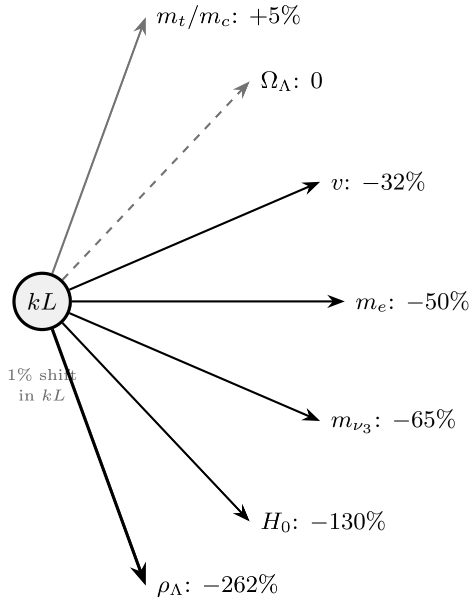

Figure 5. The sensitivity fan of the window capacity kL. A one-percent shift of kL propagates through the exponential hierarchy $e^{-4\pi kL}$, e^{-20kL}, and their powers into the dimensionful outputs, moving v by -32%, m_e by -50%, and ρ_Λ by -262%; the mass ratios carry only order-unity elasticities, and the dark-energy fraction Ω_Λ, a pure content ratio (dashed, grey), has elasticity exactly zero. The full elasticity matrix is Table 10 of Appendix D.

vacuum-polarisation descent of Appendix C.16, the geometric couplings at $M_{\rm G}$, the fermion masses of Section 6, and the scale-invariant Higgs quartic λ_H of Appendix C.12. The perturbative level of each quantity is specified below. The numerical values are collected below.

The internal Z mass is the self-consistent fixed point of the tree-level mass formula evaluated on the two-loop geometric running,

$$M_Z = \frac{v}{2}\sqrt{g_2^2 + \tfrac{3}{5}g_1^2}\,\Big|_{\mu=M_Z},$$

which gives $M_Z = 91.124$ GeV against the observed 91.1876 GeV (-0.069%), with $\alpha^{-1}(M_Z) = 128.208$ ($+0.20\%$). The W mass is the on-shell relation

$$s^2c^2 = \frac{\pi\alpha(0)}{\sqrt{2}\,G_F M_Z^2}\frac{1}{1-\Delta r}, \qquad \Delta r = \Delta\alpha - \frac{c^2}{s^2}\Delta\rho,$$

with $\Delta\alpha = 1-\alpha(0)/\alpha(M_Z)$ and $\Delta\rho$ the exact one-loop t-b doublet contribution (Veltman). The result is $M_W = 80.371$ GeV against 80.369 GeV ($+0.003\%$), and the on-shell mixing angle $\sin^2\theta_W = 1 - M_W^2/M_Z^2 = 0.22208$ (-0.50% against the value from the observed masses). The leading universal higher-order corrections of the on-shell scheme are then applied. These are the two-loop ρ parameter $\bar\rho = \Delta\rho\big[1+\frac{G_F m_t^2}{8\sqrt{2}\pi^2}(19-2\pi^2)\big]$ (Denner eq. 8.22) and the resummed W-mass relation $\pi\alpha(M_W)/\bar s_W^2 = \sqrt{2}G_F M_W^2$ with $\bar s_W^2 = s_W^2 + c_W^2\Delta\bar\rho$ (Denner eq. 8.24, 8.21), giving $M_W = 80.337$ GeV (-0.040%). Evaluated on conventional

Standard-Model benchmark values, the same formulas return $M_W = 80.369$ GeV against the known full one-loop result 80.36 GeV. The residual non-universal remainder $\Delta r_{\rm rem}$ (bosonic and box) is not included and is stated in the level table. The MS-bar-like mixing from the geometric couplings at the internal M_Z is $\sin^2\theta(M_Z) = 0.23328$ (+0.89% against the MS-bar value 0.23122). The ρ parameter in the improved-Born form is $\rho = 1/(1-\Delta\rho) = 1.00951$ against the Standard-Model value 1.00936 (+0.015%), and the two-loop value is $\bar{\rho} = 0.00948$.

The Z widths are computed at Born level with the QCD radiator $\alpha_s/\pi + 1.409(\alpha_s/\pi)^2$ for quarks and the QED radiator $3Q_f^2\alpha(M_Z)/(4\pi)$ for charged fermions,

$$\Gamma_f = N_c \frac{G_F M_Z^3}{6\pi\sqrt{2}} (g_{Vf}^2 + g_{Af}^2)(1+\delta_{\rm QCD})(1 + \tfrac{3Q_f^2}{4\pi}\alpha), \qquad g_{Vf} = T_3 - 2Q_f\sin^2\theta, \quad g_{Af} = T_3.$$

The leading electroweak vertex correction, the top-quark (t, W) loop in $Z \to b\bar{b}$, is included at its improved-Born value $\delta_b = -G_F m_t^2/(4\sqrt{2}\pi^2) = -(2/3)\Delta\rho = -0.63\%$ (relative width, the standard result of the m_t^2-enhanced non-universal vertex, cf. Denner section 8.2.3). The remaining light-fermion phase-space factors are $O(m_f^2/M_Z^2) \leq 10^{-3}$ and are not included. The results, with the level stated, are collected in Table 5. The total width $\Gamma_Z = 2.480$ GeV (−0.62%) inherits the −0.07% of the internal M_Z through its cube and the absence of the vertex corrections in the other channels. The top-loop vertex reduces Γ_b from 0.3808 to 0.3784 GeV, i.e. from +1.00% to +0.36% of the observed value, with the remaining offset tied mainly to the closed top mass. The hadronic peak is $\sigma^0_{\rm had} = 41.51$ nb (+0.07%). The tree-level Higgs mass $m_H = \sqrt{2\lambda_H}\, v = 124.98$ GeV (−0.17%), where $\lambda_H = 0.129$ is the Higgs quartic, distinct from the colour-singlet order parameter $\lambda = 149$ of Section 7.1. The absolute muon and tau masses of the internal ladder (105.31 MeV and 1.772 GeV, −0.33% and −0.28%) complete the block.

11 Discussion and conclusion

This paper has carried out the numerical evaluation of the spectral construction whose structural content is recorded in the companion paper [1]. The outcome is the full set of computed quantities recorded in Appendix B, one observed anchor and the remainder computed from it, attached to a single spectral datum, the Laplace and Dirac spectrum of RP^3. After the structural inputs and closure prescriptions are specified, the numerical chain contains no continuously adjusted phenomenological parameter, and the explicit numerical inputs are limited to the stated structural constants and closure conventions. The result nevertheless depends on the discrete and functional choices recorded in Section 2, whose sensitivity is quantified in Appendix D.

The most direct outputs of the construction are the window-capacity fixed point $kL = 2.49353$, the associated generation count, the torsion modulus $\tau = 1/50$, and the effective gauge-coupling normalisations. From these descend the fermion mass ratios, the electroweak scale, and a number of low-energy scales, each read from the quantities that precede it in the chain, with the single anchor $G_{\rm N}$ entering only to convert the dimensionless output into physical units. Their numerical proximity to reference values is a fact

Quantity	Computed	Observed	Deviation
M_Z (GeV)	91.124	91.1876	-0.069%
$\alpha^{-1}(M_Z)$	128.208	127.952	$+0.20\%$
$\sin^2\theta(M_Z)$	0.23328	0.23122	$+0.89\%$
M_W (GeV)	80.371	80.369	$+0.003\%$
M_W leading-univ. (GeV)	80.337	80.369	-0.040%
$\sin^2\theta_W$ (on-shell)	0.22208	0.22321	-0.50%
ρ (improved Born)	1.00951	1.00936	$+0.015\%$
$\sin^2\theta^l_{\text{eff}}$	0.23011	0.23153	-0.61%
Γ_Z (GeV)	2.4798	2.4952	-0.62%
Γ_{had} (GeV)	1.7331	1.7444	-0.65%
Γ_b (GeV)	0.3784	0.3771	$+0.36\%$
Γ_l (GeV)	0.2499	0.2519	-0.78%
Γ_{inv} (GeV)	0.4967	0.4990	-0.45%
σ^0_{had} (nb)	41.51	41.481	$+0.07\%$
R_l	20.80	20.767	$+0.17\%$
R_b	0.2197	0.21629	$+1.59\%$
m_H (GeV)	124.98	125.20	-0.17%

Table 5. The electroweak precision block of Section 10.1. Every input is a closed value, and the observed values appear only for comparison. The level of each observable is stated in the text. The Z mass is tree-level on the two-loop running, the W mass is at one-loop ρ and at the leading-universal (two-loop ρ) level, Γ_f is at Born plus QCD/QED radiators with the top-loop vertex on Γ_b, $\sin^2\theta^l_{\text{eff}}$ is at improved Born, and m_H is at tree level.

about the chosen closure. It does not by itself establish the physical correctness of the prescription. The proximity is conditional on the specified window, pole, and matching prescriptions.

The standard-model field content is recovered from the spectral datum through zero-mode projection. The gauge fields are the Kaluza–Klein zero modes of the internal connection, their kinetic terms the zero-mode projection of the field strength squared, and their self-interaction vertices the zero-mode overlap of the cubic and quartic terms. The fermions are the spinor modes of the chiral clause, their kinetic terms the Dirac operator restricted to the zero modes, and their gauge interactions the zero-mode coupling of the connection to the spinor currents. The scalar is the trace-zero mode, its kinetic term the zero-mode projection of the Laplacian, and the Yukawa couplings the overlap integrals of two spinor zero modes with the scalar zero mode. This is the same overlap structure O_e that fixes the charged-fermion masses of Section C.9. Each term is therefore a specific zero-mode projection or overlap integral of the internal geometry, as recorded in the product-space correspondence of the companion paper [1]. The construction is thus not a competing formulation of the Standard Model but a Lagrangian of the same form obtained from the spectrum. The zero modes project the internal geometry onto the four-dimensional fields,

Observable	Level	Not included
M_Z	tree level on the two-loop running	one-loop electroweak self-energy
M_W	one-loop ρ + leading universal (two-loop ρ, resummed α)	non-universal $\Delta r_{\rm rem}$ (bosonic, box), ≈ 0.03 GeV
$\sin^2\theta(M_Z)$	geometric couplings at M_Z	full one-loop $\sin^2\theta_{\rm eff}$
$\sin^2\theta^l_{\rm eff}$	improved Born (two-loop ρ)	process-dependent κ_l, higher order
Γ_f	Born + QCD/QED radiator, top vertex on Γ_b	light-fermion phase space, other-channel EW vertices
m_H	tree level	one-loop

Table 6. The level of each electroweak precision observable of Table 5. The column "Not included" records the dominant corrections omitted at the stated level, so the block is a consistency comparison rather than a high-precision prediction. No scheme or threshold uncertainty budget is claimed.

and the overlap integrals supply the couplings.

The comparison with observation is reported without adjustment. The three gauge couplings agree to better than 0.01%, the fine-structure constant to 0.20%, the electroweak scale to 0.01%, the electroweak precision observables of Section 10.6 to within a percent (M_W to 0.003% at Born-plus-ρ and to 0.04% at the leading-universal level, the Z mass to 0.07%, the total Z width to 0.6%, and Γ_b to 0.4% after the top-loop vertex), the fermion mass ratios and neutrino mass-squared splittings to within a percent, the cosmological fractions to within a percent, and the QCD scale to about one percent. Three quantities deviate by more than one percent, and each is traced to an identified source. These are the scale-invariant anchor for the bottom mass and the loop order for the QCD scale and the lightest glueball. No value is fitted, and no deviation is adjusted after the fact. The neutrino sector is reported through its absolute mass spectrum and, separately, through the oscillation splittings; the solar splitting uses the finite spectral-tilt propagation factor stated in Appendix C.10.

The core computation closes at the quantities of Sections 5, 6, 7, 8, and 9. The same closure, extended by the effective relations recorded in Appendices E and F, reaches the neutrino sector, the CKM parameters, and the primordial abundances. These extensions are reported in the appendices for completeness, as formal closure rather than as precision comparisons.

The robustness of these central values sets the boundary of the claim. The dimensionless hierarchy is carried by exponentials of the window capacity, so the dimensionful outputs are exponentially sensitive to kL. A one-percent shift of kL moves the electroweak scale by -32%, the electron mass by -50%, and the dark-energy density by -262%, while the dark-energy fraction, a pure content ratio, has elasticity exactly zero (Appendix D).

The quoted central values are therefore fixed within the adopted closure prescription, and their physical robustness is assessed by the sensitivity analysis rather than asserted. The conventions upstream of kL are recorded in Appendix D; they are fixed by the adopted prescription, but the hierarchy they carry is convention-sensitive by construction.

Beyond the individual numbers, the computation produces structural relations that the Standard Model itself does not fix. The baryon fraction, dark-energy fraction, and endpoint residual close to unity by the closure relation, with the residual occupying the cold-source slot in the linear cosmological comparison. The gravitational coupling is the transverse-traceless zero mode with a back-reaction factor of unity, so gravity is a spectral zero mode with no additional degree of freedom. The glueball tower of Section 9.2 provides an additional comparison: its pseudoscalar level $m_{0^{-+}} = 2.418$ GeV is fixed by the spectral tower without glueball data as input, and can be compared with the $X(2370)$ measured by the BESIII Collaboration at 2359 MeV [17, 18]. Relations of this kind distinguish the construction from a simple reparametrisation of the observed values. Their dynamical status is stated individually in the sections where they are introduced.

Several of the closed quantities are accessible to current or planned experiments. The most direct tests are listed in Table 7. The table is not a likelihood forecast; it records only the numerical value fixed by the present prescription and the qualitative observation that would be incompatible with it. The tensor-to-scalar ratio, the neutrino ordering, the atmospheric octant, and the summed neutrino mass are the cleanest entries because their interpretation is least dependent on hadronic modelling. The PMNS phase belongs to the formal flavour closure, while the glueball and acceleration-scale entries are phenomenological comparisons whose status is stated in Sections 9.2 and G.2.

The acceleration-scale relation has a more limited status. The construction gives the scale $a_0 = cH_0/(2\pi)\sqrt{4/3}$ of Appendix G, which lies close to the Milgrom scale with the coefficient $\sqrt{4/3}/(2\pi)$ fixed rather than fitted, and the endpoint branch gives a fixed local response $\mu(y) = y/\sqrt{1+y^2}$. In this paper that response, the SPARC comparison, and the cluster residual reading are comparisons at fixed closed parameters. Full nonlinear structure formation, cluster weak-lensing map likelihoods, and joint CMB/BAO/SN/RSD inference are left to separate work.

The relation to the companion paper is as follows. The companion paper supplies the structure, the gauge algebra, the fermion content, and the mass-gap form, each derived from the coarse-graining principle by the chain recorded there. The colour factor distinct from the electroweak one follows from the minimality principle and the chiral clause together with colour vectoriality, the same economy criterion that selects SU(2) over U(2), $\mathfrak{su}(3)$ over $\mathfrak{so}(5)$ and G_2, and RP^3 over the lens spaces. The present paper supplies the numerical evaluation and the comparison with observation. The flavour, QCD, and cosmological applications are extensions of the core construction rather than consequences of the spectrum alone. They are reported as consistency checks of the extended relations, not as independent predictions. The core outputs are the window-capacity fixed point, the generation count, and the gauge-coupling normalisations. These are the quantities whose derivation from the spectrum is most direct. This paper is a numerical evaluation of a fixed spectral construction, not an independent claim that the closure principle uniquely determines nature.

Quantity	Value fixed here	Incompatible outcome
Tensor-to-scalar ratio	$r = 0.0253$	a robust upper bound well below this value
Neutrino mass ordering	normal	inverted ordering
Atmospheric octant	$\sin^2\theta_{23} = 0.551$	lower octant
Summed neutrino mass	$\sum m_\nu = 0.0615$ eV	a robust upper bound below this value
PMNS phase (formal closure)	$\delta_{\rm PMNS} \simeq (8/7)\pi$	value far from this branch
Pseudoscalar glueball target	$m_{0^{-+}} = 2.418$ GeV	no compatible flavour-singlet 0^{-+} state
Low-acceleration branch	$a_0 = 1.204 \times 10^{-10}\ \mathrm{m\,s^{-2}}$, $\mu(y) = y/\sqrt{1+y^2}$	rotation-curve or lensing tests incompatible with the fixed response

Table 7. Near-term comparison targets. The first four rows are direct tests of closed quantities. The PMNS phase, glueball, and acceleration-scale rows test downstream or phenomenological closures, and therefore carry the lower status stated in the text.

A The parameter table

The computation is carried by a small set of structural constants and a set of quantities that admit comparison with observation. The structural constants are the closure fixed point and the integer content identities of the construction. They carry no reference value of their own, but every comparable quantity descends from them, and they are recorded in Table 8. The comparable quantities are listed in Table 9, produced from the closed forms of Appendix C rather than stated independently. Each row gives the quantity, its computed value, the reference value where one exists, the signed deviation, a short record of the derivation, and the tier (A–D) defined in Table 4 of Section 10. One quantity, the Newton constant, is the observed anchor of Section 2.1. The remaining quantities are computed, and none is fitted. The computed values are conditional on the structural choices recorded in the companion paper and on the closure prescriptions of Section 2. The comparison reports the signed deviations of these values against reference values that are kept outside the closure chain and are never inputs to the computation. Where a direct measurement exists the reference value is the Particle Data Group central value. Where the comparison is made at the emergence scale, as for the gauge couplings $g_i(M_{\rm G})$, the reference value is the two-loop Standard-Model extrapolation used in this paper, so the two sides are compared at the same scale.

The derived quantities that are not tabulated here fall into three groups. The intermediate quantities of the chain, namely the ladder indices, the sector steps, the dilaton line, and the correction factors, are auxiliary definitions. Each exists only to carry a later quan-

Structural constant	Value	Origin
window capacity kL	2.49353	closure fixed point, Sec. 3.4
emergence scale $M_{\rm G}$	1.731×10^{18} GeV	$M_{\rm G} = M_{\rm P}\sqrt{\pi}/kL$, Sec. 3.3
torsion modulus τ	1/50	content ratio, Sec. 4.3
left-handed count N_L	8	$N_L = N_g = N_c^2 - 1$, Sec. 4.1
right-handed count N_R	7	$N_R = N_g - 1$, Sec. 4.1
chiral asymmetry $N_L - N_R$	1	$\pi_1(RP^3) = \mathbb{Z}_2$, Sec. 4.1
hypercharge moment $\sum Y^2$	10/3	anomaly cancellation, Sec. 4.1

Table 8. The structural constants of the computation. They carry no reference value of their own. Every comparable quantity of Table 9 descends from them together with the single observed anchor.

tity, and each is computed in the derivation of Appendix C where it first appears rather than given a row of its own. The downstream quantities, namely the full electroweak precision block beyond its headline, the cosmological constant, the curvature amplitude, the tensor ratio, the acceleration scale, the string tension, and the deconfinement temperature, are computed in their derivations in Appendix C and in Section 10.6, but are not given rows here. The flavour- and cosmological-extension relations that are stated as structural closure rather than as precision comparison, namely the PMNS mixing angles, the CKM phase and the Jarlskog invariant, and the primordial abundances, are recorded as formal closure in Appendices E and F, where their closure forms are given without a decimal evaluation. The supplementary material lists the additional computed intermediates that are not displayed in this table.

Table 9: The comparable quantities of the computation, grouped by sector. Computed values are conditional on the closure prescriptions of Section 2; observed values are used only for comparison, never as inputs. The full derivation of every closed form is recorded in Appendix C.

Quantity	Computed	Observed	Deviation	Tier	Derivation
Internal scale					
generation count n_g	3	3	+0.000%	A	finite window capacity on the RP^3 spinor spectrum (App. C.1)
Gauge couplings					
weak coupling $g_2(M_G)$	0.508848	0.508845	+0.001%	B	Killing normalisation plus the colour and torsion screening law (App. C.4)

Table 9: The comparable quantities of the computation, grouped by sector. Computed values are conditional on the closure prescriptions of Section 2; observed values are used only for comparison, never as inputs. The full derivation of every closed form is recorded in Appendix C.

Quantity	Computed	Observed	Deviation	Tier	Derivation
hypercharge coupling $g_1(M_G)$	0.60499	0.605001	-0.002%	B	$g_1 = g_2\kappa(2\tau)$ with the squash normalisation and matching-scale running (App. C.4)
colour coupling $g_3(M_G)$	0.49776	0.497765	-0.001%	B	common-origin coupling with the long-root correction (App. C.4)
$\alpha^{-1}(M_Z)$	128.209	127.952	+0.200%	B	two-loop running from the geometric boundary condition (App. C.5)
$\alpha_s(M_Z)$	0.117986	0.1179	+0.073%	B	two-loop QCD running with the stated Yukawa-difference correction (App. C.5)
Fermion mass ratios					
top to charm m_t/m_c	135.242	136	-0.557%	B	$e^{2\alpha_{\text{up}}}$ from the up-sector Landau–Zener ladder (App. C.7)
bottom to strange m_b/m_s	44.8679	45	-0.293%	B	$e^{2\alpha_{\text{down}}}$ from the down-sector ladder (App. C.7)
top to up m_t/m_u	7.730364e+04	7.800000e+04	-0.893%	B	up-sector double step and first-generation window factor (App. C.7)
strange to down m_s/m_d	19.623	19.802	-0.904%	B	first-generation down-sector extrusion (App. C.7)
muon to electron m_μ/m_e	206.356	206.8	-0.215%	B	lepton ladder plus the $\sqrt{2\pi}$ endpoint factor (App. C.9)
Electroweak scale					

Table 9: The comparable quantities of the computation, grouped by sector. Computed values are conditional on the closure prescriptions of Section 2; observed values are used only for comparison, never as inputs. The full derivation of every closed form is recorded in Appendix C.

Quantity	Computed	Observed	Deviation	Tier	Derivation
Higgs vacuum expectation v (GeV)	246.19	246.22	-0.012%	B	window-squared electroweak line from M_G with the squash correction (App. C.11)
Electroweak precision					
Z mass M_Z (GeV)	91.1243	91.1876	-0.069%	B	self-consistent tree-level relation on the two-loop running couplings (Sec. 10.6)
W mass M_W (GeV)	80.3712	80.369	+0.003%	B	on-shell Sirlin relation with the one-loop t-b contribution to $\Delta\rho$ (Sec. 10.6)
W mass, leading-univ. M_W^{lu} (GeV)	80.3369	80.369	-0.040%	B	leading-universal on-shell completion with the two-loop ρ term (Sec. 10.6)
Γ_b (GeV)	0.37841	0.37705	+0.361%	C	Born width with QCD/QED radiators and the top-loop vertex correction (Sec. 10.6)
$\sin^2\theta^l_{\mathrm{eff}}$	0.230114	0.23153	-0.612%	C	improved-Born effective mixing angle (Sec. 10.6)
Higgs mass m_H (GeV)	124.983	125.2	-0.173%	B	$m_H = \sqrt{2\lambda_H}\,v$ at tree level (Sec. 10.6)
Neutrinos					
heaviest neutrino scale $m_{\nu 3}$ (eV)	0.0501797	0.0502	-0.040%	B	Weinberg two-period family; reference column is the atmospheric scale, not a direct mass measurement (App. C.10)
solar splitting Δm^2_{21} (eV^2)	7.412190e-05	7.410000e-05	+0.030%	B	tilt-dressed oscillation splitting from the absolute light-pair masses (App. C.10)

Table 9: The comparable quantities of the computation, grouped by sector. Computed values are conditional on the closure prescriptions of Section 2; observed values are used only for comparison, never as inputs. The full derivation of every closed form is recorded in Appendix C.

Quantity	Computed	Observed	Deviation	Tier	Derivation
atmospheric splitting Δm^2_{31} (eV^2)	0.00251121	0.002511	+0.008%	B	atmospheric splitting from the absolute mass eigenvalues (App. C.10)
lightest neutrino $m_{\nu 1}$ (eV)	0.00260742	–	–	B	absolute vacuum-floor mass; no direct oscillation central value is assigned (App. C.10)
Charged-fermion masses					
top mass m_t (GeV)	174.082	172.69	+0.806%	B	$m_t = y_0 v/\sqrt{2}$ with the zero-mode overlap $y_0 = 1$ (App. C.8)
bottom mass m_b (GeV)	4.23779	4.18	+1.383%	C	down-sector geometric mean tied to the scale-invariant top anchor (App. C.9)
electron mass m_e (MeV)	0.510354	0.511	-0.126%	B	$M_P e^{-20kL}$ with the squash correction (App. C.9)
muon mass m_μ (GeV)	0.105314	0.105658	-0.325%	B	electron mass times the lepton ladder factor (App. C.9)
tau mass m_τ (GeV)	1.77211	1.777	-0.275%	B	next lepton rung above the muon (App. C.9)
CP content					
baryon asymmetry η_B	6.090893e-10	6.100000e-10	-0.149%	D	Sakharov content relation from the Jarlskog invariant and weak coupling (App. C.13)
Cosmology					
scalar tilt $1 - n_s$	0.035	0.0351	-0.285%	D	$\tau(7/4)$ from the Weyl degree count and internal dimension (App. C.14)
Hubble rate H_0 (GeV)	1.438850e-42	1.440000e-42	-0.080%	D	infrared endpoint of the entropy integral (App. C.14)

Table 9: The comparable quantities of the computation, grouped by sector. Computed values are conditional on the closure prescriptions of Section 2; observed values are used only for comparison, never as inputs. The full derivation of every closed form is recorded in Appendix C.

Quantity	Computed	Observed	Deviation	Tier	Derivation
dark-energy fraction Ω_Λ	0.685044	0.6847	+0.050%	D	content ratio $\rho_\Lambda/(3H_0^2M_P^2)$ (App. C.14)
baryon fraction Ω_b	0.0492531	0.0493	-0.095%	D	raw photon floor, raw Sakharov η_B, proton mass, and critical density (App. C.14)
matter remainder Ω_Σ	0.265703	0.2645	+0.455%	D	maximum-entropy endpoint residual; reference column uses the conventional dark-matter fraction (App. C.14)
CMB temperature $T_{\rm CMB}$ (K)	2.72547	2.72548	-0.000%	D	corrected monopole $C_\gamma T_{\rm CMB}^{\rm raw}$; Ω_b uses the raw floor (App. C.14)
QCD					
QCD scale $\Lambda_{\rm QCD}$ (GeV)	0.20738	0.21	-1.248%	C	two-loop running and top-threshold matching from the colour boundary condition (App. C.15)
proton mass m_p (GeV)	0.938156	0.938272	-0.012%	D	constituent relation $(279/64)\Lambda_{\rm QCD}$ with the scheme correction (App. C.15)
lightest glueball m_G (GeV)	1.65904	1.7	-2.409%	D	lowest scalar spectral level $8\Lambda_{\rm QCD}$ (App. C.15)

B Reproducibility

Every quantity in the closure chain is returned by a closed-form derivation recorded in Appendix C, from the structural numbers and the single anchor alone. The derivations form an acyclic chain. Each quantity is defined in terms of quantities that precede it, and no quantity is defined in terms of a later one. Thus the algebra written in the text determines each internally computed quantity from the single anchor and the structural numbers.

Every quantity used to generate the numerical chain falls into one of two classes. It is either derived, computed from earlier quantities by a derivation recorded in this paper, or observed, the single Newton constant of Section 2.1. Apart from this anchor, observational reference values are kept outside the closure chain. External observational values and standard propagation tools enter only after the internal values have been computed, as reference comparisons.

The parameter table of Appendix A is produced from these closed forms rather than stated independently. The numerical constants used are the structural numbers 2, π, $3/2$, and $1/8$, together with the geometric factors $\sqrt{2\pi}$ and $\sqrt{3/\tau}$, and the single observed input G_{N}. The compressed CAMB, BAO, and SPARC entries, where quoted, are comparisons made with the closed outputs held fixed; they do not determine any closure parameter.

The supplementary materials give the algebraic chain in dependency order, together with the numerical values behind the tables. Observational reference values used for comparison are kept separate from the internally computed quantities and are not used as computation inputs. The weak-coupling comparison of Section C.1 inverts the conservation law of Section C.4 at the Standard Model reference value of $g_2(M_{\mathrm{G}})$ only after the spectral value of kL has been fixed, and returns the deviation $+0.0002\%$ recorded in the text.

Appendix C gives the algebraic derivations of the identities that carry the quantitative content of the paper, written in the standard mathematical language of the text.

The numerical precision of the window capacity is important because $kL = 2.493534\ldots$ is the central intermediate of the chain. It enters both the finite spectral capacity through $(kL)^2 = 6.2177$ of Section 3.5 and the exponential hierarchy through factors of the form $e^{-a\,kL}$, with $a = 4\pi$ for the electroweak scale and $a = 20$ for the electron mass. A small absolute perturbation δkL therefore induces the relative shift $\delta X/X \simeq -a\,\delta kL$ in any observable controlled by $e^{-a\,kL}$. With $a = 20$, a rounding of kL already at the third decimal moves the downstream value at the percent level. The value kL is consequently propagated at full numerical precision throughout the numerical chain, while the rounded display $kL = 2.49353$ quoted in the text is only a presentation convention. Its proximity to the Gaussian scale $\sqrt{2\pi}$ is likewise not used as an input but only noted after the closure. The same closed window parameter fixes the finite spectral capacity and the exponential hierarchy scale at once, and is the quantity whose numerical precision controls the stability of the full reconstruction.

Three reproducibility requirements are imposed. First, re-evaluating the chain returns the core parameters (kL, M_{G}, v, M_W, Γ_b, and Λ_{QCD}) unchanged within the displayed precision. Second, the observational reference values are held outside the closure chain and are used only after the computation has produced its outputs. Third, central closure values are propagated at full numerical precision, with rounding only at the display stage. Propagations using CAMB/DESI/SPARC enter only as comparisons with the closure parameters held fixed.

B.1 Scope of the formal verification

The optional Lean 4 formalisation supplied as supplementary material covers the discrete algebraic identities used in the construction before the numerical comparison stage. It covers integer counting relations, representation-content identities, normalisation factors, and closure equalities that enter the structural datum of the computation.

The formalisation verifies these identities as statements internal to the stated construction. It does not establish the physical adequacy of the construction, the uniqueness of its interpretation, or the empirical significance of the numerical agreement reported in the tables. Those questions are addressed separately by the derivations in Appendix C, the reproducibility statement, the separation of reference values from the closure chain, and the comparison summary.

C Key derivations

This appendix records the derivations that carry the quantitative content of the paper. Each subsection gives the algebra for one closure, so that every closed form in Sections 3–9 is specified within the present paper. The derivations use only the discrete spectrum of RP^3, the structural constants listed in Section 2, and the measured Newton constant as the single dimensional anchor.

The subsections follow the order of the main text and of the extensions, and each is the derivation behind the section or appendix of the same name. C.1–C.2 close the window capacity and the torsion modulus (Section 3, 4). C.3 records the geometric dynamics. C.4–C.6 derive the gauge couplings and their beta functions (Section 5). C.7–C.9 derive the flavour ladder, the top anchor, and the fermion masses (Section 6), and C.10 the neutrino masses (Appendix E). C.11 derives the electroweak order parameters (Section 7) and C.13 the CKM and CP content (Appendix E). C.14 derives the cosmological quantities (Section 8). C.15 derives the QCD scales (Section 9), and C.16 the primordial abundances (Appendix F).

C.1 The window-capacity closure

The window capacity kL is the fixed point of the spectral-pole condition of the spin-2 channel. The object is the transverse-traceless projection of the improved energy-momentum tensor, evaluated as a spectral sum over the discrete RP^3 spectrum. The amplitude at zero external momentum is

$$\Pi^2_{\mu\nu} = \frac{1}{V_3} \sum_{\text{fields}} \sum_{\text{modes}} d_n\, w(\text{field}) \int_0^\infty \frac{d\omega}{\pi}\, K_{\text{TT}}\big(\omega^2 + \lambda_n, m^2_{\text{eff}}\big)\, e^{-(\omega^2+\lambda_n)/\Lambda^2}, \tag{C.1}$$

where $V_3 = \pi^2 L^3$ is the internal volume, d_n the mode multiplicity, $w(\text{field})$ the channel weight, and Λ the running cutoff in Planck units. The transverse-traceless kernel is

$$K_{\text{TT}}(k^2, m^2) = \frac{k^4}{(k^2 + m^2)^2}. \tag{C.2}$$

It vanishes at $m^2 = 0$ by the Ward identity of the improved tensor, so the spin-2 channel is activated only by the curvature and torsion masses of the RP^3 modes. The channel weights are $w = 4$ for each gauge polarisation, $w = 4$ for each Weyl fermion, and $w = 1$ for each real scalar. The effective masses m^2_{eff} are the Einstein-Cartan shifts of each species. The scalar carries the conformal mass $m^2 = \xi R = 3/(4L^2)$ with $\xi = (d-2)/(4(d-1)) = 1/8$ and $R = 6/L^2$. The gauge fields carry the Camporesi curvature mass plus the torsion shift, $m^2 = C_2/(2L^2) + \tau^2/(6L^2)$, with $C_2 = 3, 2, 0$ for the SU(3), SU(2), U(1) generators. The Weyl fermions carry the torsion shift only, $m^2 = 3\tau^2/(8L^2)$, because the curvature is already inside the Dirac-squared spectrum through the Lichnerowicz identity $\not{D}^2 = \nabla^*\nabla + R/4$. The transverse-traceless tensor carries the Lichnerowicz shift $m^2 = 6/L^2$. All four shifts depend on L and τ, and since $L = kLM_{\mathrm{G}}/k$ along the self-similar flow, the effective masses depend on kL through L, so the pole condition is an implicit equation in kL. The exponential factor is the Gaussian window of the coarse-graining envelope.

The spectral-pole condition states that the graviton-like mode becomes massless at the emergence scale, which is the fixed point

$$\frac{V_3\, \Pi^2_{\mu\nu}(kL, k^2/M_{\mathrm{P}}^2)}{32\pi^2} = \frac{4}{27} \qquad \text{at} \qquad k = M_{\mathrm{G}}. \tag{C.3}$$

The threshold $4/27$ is the normalisation extremum of the massless-pole condition,

$$\frac{4}{27} = \max_{0 \le y \le 1} y(1-y)^2 \qquad \text{at} \qquad y = \frac{1}{3}, \tag{C.4}$$

where $y = m^2/(k^2 + m^2)$ is the mass fraction of the transverse-traceless kernel. The kernel is $K_{\mathrm{TT}} = k^4/(k^2+m^2)^2 = (1-y)^2$, and the mass-weighted spectral density is $y\, K_{\mathrm{TT}} = y(1-y)^2$, whose maximum over the mass fraction is $4/27$ at $y = 1/3$, the point $m^2 = k^2/2$ where the single mass fraction and the two kinetic fractions $(1/3)(2/3)^2$ balance. This extremum is the normalisation at which the spin-2 propagator develops its massless pole, not a fitted constant. The condition is the spectral-pole criterion of the construction. The massless pole appears when the mass-weighted spectral density $y(1-y)^2$ saturates its maximum, the point where the single mass fraction and the two kinetic fractions balance, the same extremal-normalisation structure as the dilaton pole condition $V\Pi^2 = 1$ of the geometric-series divergence. The criterion is the spectral statement of the emergence of a massless spin-2 mode, and $4/27$ is its fixed threshold. On the self-similar trajectory $L(k) = C/k$ with $C = M_{\mathrm{P}}\sqrt{\pi}$, the volume and the cutoff are functions of the single combination kL, so the condition is one equation in one unknown. It is solved by bisection on the interval $[0.3\, M_{\mathrm{G}}, 3\, M_{\mathrm{G}}]$ with the stopping criterion that the interval width fall below 10^{-15}. The self-consistency between $M_{\mathrm{G}} = C/kL$ and the pole scale $k_\star$ is then iterated, and the two-step iteration converges to the displayed precision. The fixed point is

$$kL = 2.49353. \tag{C.5}$$

The closure is cross-checked against the coupling conservation law of Section C.4, which returns the same kL to within 0.0002% when the window-capacity form $\alpha_W = 2/kL^5$ is inverted using the reference weak coupling $g_2(M_{\mathrm{G}}) = 0.508845$. This inversion is a comparison made after the spectral-pole closure has fixed kL from the F_{M_G} condition; it is not used as a computation input. The endpoint geometry $L_{\mathrm{Cg}} = \sqrt{\pi}$ fixes the critical curvature $R_{\mathrm{c}} = 6/\pi$, and the window capacity $(kL)^2 = 6.2177$ determines the generation count of Section 3.5.

C.2 The torsion window cancellation

The torsion modulus τ is fixed by three independent layers that meet in the Einstein-Cartan field equation.

The $\mathbb{Z}_2$ topology. The chiral content of Section 4 gives $N_L - N_R = 1$, which is odd. An odd chiral asymmetry is the non-trivial element of $\pi_1(RP^3) = \mathbb{Z}_2$, the non-trivial spin structure [21] of the internal space whose first cohomology $H^1(RP^3, \mathbb{Z}_2) = \mathbb{Z}_2$ has a single non-trivial class.

The hypercharge anomaly. The hypercharge assignment of Section 4 satisfies $\sum Y = 0$, the vanishing of the mixed gravitational anomaly, so the first non-zero hypercharge moment is $\sum Y^2 = 10/3$. This is the natural normalisation of the hypercharge polarisation.

The Einstein-Cartan field equation. For the totally antisymmetric torsion $T^a{}_{bc} = (\tau/L)\varepsilon^a{}_{bc}$, varying the Einstein-Cartan action with respect to the contorsion gives

$$\frac{\tau}{L} = \kappa^2 j_5, \qquad \kappa^2 = \frac{8\pi}{M_{\mathrm{P}}^2}, \tag{C.6}$$

where the chiral current is the intensive chiral drive screened by the hypercharge polarisation,

$$j_5 = \frac{\langle \chi \rangle}{\Pi_{\mathrm{ren}}} = \frac{N_L - N_R}{N_f \sum Y^2}. \tag{C.7}$$

The geometric factor $\kappa^2 L$ is evaluated exactly through the window-edge identity. The three-dimensional gravitational coupling κ^2 is reduced from the four-dimensional Planck mass by the Kaluza-Klein volume factor, and combined with the radius L the factor reads

$$\kappa^2 L = \frac{M_{\mathrm{G}}^2}{2\pi^2 M_{\mathrm{P}}^2 k L^2} = \frac{1}{2\pi k L^4}, \tag{C.8}$$

where the last step uses the window edge $kLM_{\mathrm{G}} = M_{\mathrm{P}}\sqrt{\pi}$, so $M_{\mathrm{G}}^2/(M_{\mathrm{P}}^2 kL^2) = \pi/kL^4$. The polarisation requires one further layer, stated here explicitly. The bare one-loop hypercharge polarisation is small,

$$\Pi_{\mathrm{bare}} = 0.0014 \times \sum Y^2, \tag{C.9}$$

with 0.0014 the one-loop coefficient of the earlier loop-normalisation record (the V2 legacy value). The polarisation is renormalised at the emergence scale by the condition

$$\Pi_{\mathrm{ren}}(M_{\mathrm{G}}) = \sum Y^2 = \frac{10}{3}, \tag{C.10}$$

the coupling convention of the construction, the analogue of a fixed renormalisation point in a standard scheme. The counter-term

$$\Delta\Pi = \Pi_{\mathrm{ren}} - \Pi_{\mathrm{bare}} = \sum Y^2 \,(1 - 0.0014) \tag{C.11}$$

absorbs the bare loop. The torsion modulus is then the chiral drive over the renormalised polarisation,

$$\tau = \frac{\langle\chi\rangle}{\Pi_{\rm ren}} = \frac{N_L - N_R}{N_f \sum Y^2} = \frac{1}{50}. \tag{C.12}$$

The window capacity $2\pi kL^4$ of the bare field equation is a notation common to the bare factor $1/(2\pi kL^4)$ and the screening factor $2\pi kL^4$. It carries no independent spectral sum (the three-dimensional RP^3 spectral sums close to $(kL)^3$, not $(kL)^4$), it cancels identically between the two factors, and τ depends only on the content ratio. The torsion enters the spectrum only through τ^2, so its effect is of order 10^{-4}.

C.3 Geometric dynamics

The torsion enters the physical content through the geometric quantities of the transverse-traceless sector. The $J = 2$ Lichnerowicz eigenvalue of the Einstein-Cartan sector is

$$\lambda_{\rm EC} = N_g\Big(1 + \frac{\tau}{2}\Big)^2 + 6 = 14 + 8\tau + 2\tau^2, \tag{C.13}$$

the $\mathfrak{su}(2)_L$ spin connection $8(1 + \tau/2)^2$ dressed by the torsion plus the Lichnerowicz shift 6, and the transverse-traceless eigenvalue is

$$\lambda_{\rm TT} = 14 = 2N_R, \tag{C.14}$$

the content double of the seven right-handed singlets. The squash amplitude is the broken-generator content,

$$s_0 = n_{\rm broken}\,\tau = 2\tau, \tag{C.15}$$

with $n_{\rm broken} = \dim {\rm SU}(2)_R - \dim {\rm U}(1)_R = 3 - 1 = 2$ the two broken generators absorbed as the Goldstone directions of the W_R bosons, each contributing the torsion modulus. Its content ratio is

$$\frac{s_0}{N_R} = \frac{n_{\rm broken}}{N_f \sum Y^2 N_R} = \frac{2}{15 \cdot (10/3) \cdot 7} = \frac{1}{175}, \tag{C.16}$$

half the $J = 2$ first-order torsion shift $N_g\tau/14 = 8\tau/14$. The 2π Euclidean period appears throughout the construction as a common factor of the ultraviolet and infrared closures, in the amplitude $e^{1/(2\pi)}$, the window width $2L = \sqrt{2\pi}$, and the tensor ratio $r = (1/2\pi)^2$.

C.4 The gauge-coupling conservation law

The weak coupling at the emergence scale is the Killing normalisation of the isometry zero modes. The raw coupling is $g_2^2 = 16\pi^2/I_{kv}$ with the Killing integral $I_{kv} = 2\pi^2 L^3$, and the four-dimensional reduction inserts the normalisation factor $(M_{\rm G}/M_{\rm P})^2$. At the emergence scale $L = kL$ this gives

$$g_2^2 = \frac{16\pi^2 (M_{\rm G}/M_{\rm P})^2}{2\pi^2 kL^3} = 8\Big(\frac{M_{\rm G}}{M_{\rm P}}\Big)^2 kL^{-3}. \tag{C.17}$$

The window-edge identity $kLM_{\rm G} = M_{\rm P}\sqrt{\pi}$ collapses this to

$$g_2^2 = \frac{8\pi}{kL^5}, \qquad \alpha_W = \frac{g_2^2}{4\pi} = \frac{2}{kL^5}. \tag{C.18}$$

The bare Killing value sits +0.345% above the Standard Model value. The correction is the geometric-dynamics conservation law

$$\frac{1}{\alpha_{\rm SM}} = \frac{1}{\alpha_W} + \frac{1}{N_c} - \frac{\tau^2\pi}{2}, \tag{C.19}$$

equivalently

$$N_c\Big(\frac{1}{\alpha_{\rm SM}} - \frac{1}{\alpha_W} + \frac{\tau^2\pi}{2}\Big) = 1. \tag{C.20}$$

Here $\alpha_{\rm SM}$ is the physical weak coupling returned by the law, the screened value computed from α_W and the two correction terms, not an observed input. The law reads the window-capacity coupling α_W and returns the physical coupling $g_2(M_{\rm G}) = \sqrt{4\pi\alpha_{\rm SM}}$. The two correction terms have independent physical origins. The colour-number term is the fermion screening of the gauge coupling, the ratio of the adjoint Casimir to the total fermion Dynkin index,

$$\frac{1}{N_c} = \frac{C_2(\mathrm{SU}(2))}{\sum_f T(R_f)} = \frac{2}{6} = \frac{1}{3}, \tag{C.21}$$

with $C_2(\mathrm{SU}(2)) = 2$ and $\sum_f T(R_f) = 6$ the Dynkin sum of the three generations of doublets, so the screening is the content ratio one over the colour number. The torsion term $-\tau^2\pi/2$ is the Einstein-Cartan correction of the geometric dynamics, the square of the torsion modulus $\tau^2 = 1/2500$ times the half-period $\pi/2$, the Euclidean causal-horizon factor $T_{\rm eff} = k/(2\pi)$ of the window. The torsion enters the couplings only through its square, because the totally antisymmetric torsion contributes to the kinetic term of the geometric dynamics at order τ^2, so the correction is the torsion content ratio times the period factor. The bare coupling α_W is the unscreened Killing value, and the physical coupling $\alpha_{\rm SM}$ is the screened value, which is why the physical coupling, not the bare one, stands on the left of the law. The colour-number term $1/N_c$ and the torsion term $\tau^2\pi/2$ remove the overshoot of the bare Killing value, and the resulting coupling is

$$g_2 = \sqrt{\frac{4\pi}{1/\alpha_W + 1/N_c - \tau^2\pi/2}} = 0.508848. \tag{C.22}$$

The two correction terms are both content ratios, the colour number and the torsion squared, so the law has no free coefficient. The algebraic core of the law is the colour self-inverse $N_c \cdot (1/N_c) = 1$ together with the cancellation of the torsion term, and the colour-number term is exactly the Casimir-over-Dynkin ratio $C_2(\mathrm{SU}(2))/\sum_f T(R_f)$ established above. The screening is the non-perturbative spectral-sum polarisation of Section C.2, the discrete RP^3 spectral sum in closed form (the window capacity), not a standard one-loop vacuum-polarisation shift. The torsion factor carries a spectral-sum origin. Its explicit form is the product of three spectral factors,

$$\frac{\tau^2\pi}{2} = (2\tau^2)\,(2\pi)\,\xi = \frac{4\pi\tau^2}{N_g}, \tag{C.23}$$

each traced to the spectrum. The factor $2\tau^2$ is the second-order torsion term of the Einstein-Cartan curvature $\lambda_{\rm EC} = N_g(1+\tau/2)^2 + 6 = 14 + 8\tau + 2\tau^2$ of Section C.2. The totally antisymmetric torsion $T^a{}_{bc} = (\tau/L)\varepsilon^a{}_{bc}$ raises the scalar curvature by the six-component square $2\tau^2$. The factor 2π is the Euclidean period, the Matsubara zero-frequency of the spectral sum, the same 2π family as $\varepsilon = e^{1/2\pi}$, the window capacity $2\pi k L^4$ and $g_A = 4/\pi$. The factor $\xi = (d-2)/(4(d-1)) = 1/8 = 1/N_g$ is the conformal coupling, the scalar-field curvature coupling of the frame whose duality $N_g\xi = 1$ closes the law. The torsion mode's Euclidean spectral sum, $2\tau^2 \cdot 2\pi$, normalised by the generator count through $\xi = 1/N_g$, is the physical origin of the term, not an assigned constant. The two terms of the law therefore carry the same conformal coupling. The colour term $1/N_c = N_g\xi/N_c$ and the torsion term $\tau^2\pi/2 = 4\pi\tau^2\xi$ are both proportional to ξ, so the law is the conformal-gauge duality $N_g\xi = 1$ written in the coupling-inverse basis, not an independent assumption. The hypercharge coupling follows from the same geometry through the squash mixing $g_1 = g_2\kappa(2\tau)$ with $\kappa^2(s) = (1+s)/(1-2s)^{5/2}$ at $s = 2\tau$, applied at the isometry-breaking scale. The mixing is the metric normalisation of the squashed internal space, not a fitted form. The isometry breaking $\mathrm{SU}(2)_R \to \mathrm{U}(1)_R$ squashes the internal S^3 along the σ_3 axis, the $\mathrm{U}(1)_Y$ direction, giving the metric

$$ds^2 = (1+s)\,\sigma_1^2 + (1+s)\,\sigma_2^2 + (1-2s)\,\sigma_3^2, \tag{C.24}$$

with eigenvalues $\lambda_1 = \lambda_2 = 1+s$ and $\lambda_3 = 1-2s$. The volume ratio is $V_3(s)/V_3(0) = \sqrt{\det g} = (1+s)\sqrt{1-2s}$, and the σ_3-axis inverse metric is $g^{33} = 1/(1-2s)$. The $\mathrm{U}(1)_Y$ normalisation is the volume ratio times the third moment of the axis inverse metric,

$$\kappa^2(s) = \frac{V_3(s)}{V_3(0)}\,(g^{33})^3 = (1+s)\sqrt{1-2s}\,(1-2s)^{-3} = \frac{1+s}{(1-2s)^{5/2}}. \tag{C.25}$$

The third moment is the Killing normalisation of the $\mathrm{U}(1)_Y$ gauge field on the squashed internal space. The F^2 contraction of the σ_3-axis zero mode carries three inverse-metric factors, two from the field strength and one from the polarisation, so the normalisation integral is $\int \sqrt{g}\,(g^{33})^3\,d^3x$ normalised to its round value. At $s = 0$ (round S^3) $\kappa^2 = 1$, so $g_1 = g_2$, the unification. At $s = 2\tau$ the mixing is $\kappa(2\tau) = 1.13183$. The mixing coefficient is corrected by the content ratio

$$\delta_{g_1} = -\tau\, r_{23} \sum Y^2 \Delta_f \xi, \tag{C.26}$$

with $r_{23} = 3/(10\sqrt{3})$ the hypercharge-trace hierarchy ratio and the product $\sum Y^2\Delta_f\xi = (10/3)(3/2)(1/8) = 5/8$ the hypercharge capacity times the fermion conformal weight times the conformal coupling, the same conformal-gauge content that closes the $1/N_c$ term above. The colour coupling carries the long-root bifurcation $g_3 = g_2(1 + \alpha_{\mathrm{GUT}}^2/K)$ with $K = J(J+2)/d = 8/3$, where $J = 2$ is the squash angular momentum and $d = 3$ the internal dimension, the kinetic eigenvalue $J(J+2) = 8$ of the $J = 2$ squash over the internal dimension, the long-root condensation diluting the two-loop self-coupling of the colour factor.

C.5 Two-loop gauge coefficients

The renormalisation group running requires the gauge beta functions, and their coefficients are derived from the group-theoretic content rather than imported from a table.

The five fermion representations are the minimal combination of the construction, not a standard-model input. The chiral clause of the coarse-graining principle is carried by the two semispinor representations of Spin(4), the left-handed $(1/2, 0)$ being an $\mathrm{SU}(2)_L$ doublet. The colour algebra is selected as the minimal non-trivial one ($\mathfrak{su}(3)$, with $8 < 10 < 14$), and the fermion sits in its smallest irreducible representation, the fundamental 3. Colour vectoriality (the $\mathrm{SU}(3)^3$ anomaly cancellation) puts the left-handed fermion in 3 and the right-handed in $\bar{3}$. The five representations are then chirality times colour assignment. The left doublet carries colour (Q_L) or not (L_L), and the right singlets mirror the doublet components, $u_L, d_L \to u_R, d_R$ (colour $\bar{3}$) and $e_L \to e_R$ (colour 1), with ν_L left Majorana. Hence two left doublets and three right singlets, the content $N_L = 6 + 2 = 8 = N_g$ of Section 4.1. The uniqueness of these five follows from the same minimal combination. The chiral clause fixes the left sector as the single semispinor $(1/2, 0)$, an $\mathrm{SU}(2)_L$ doublet, and the colour assignment has exactly two options for this doublet. The minimal non-trivial 3 (giving Q_L) or the trivial 1 (giving L_L). The right sector is the Yukawa mirror of the left. Each charged left component acquires exactly one right singlet, $u_L, d_L \to u_R, d_R$ and $e_L \to e_R$, while the neutral ν_L acquires none (Majorana). The counting is therefore two left doublets and three right singlets, exactly five, and no other representation is produced by the minimal combination. The scalar content is the colour-singlet zero mode of the construction, whose electroweak part is the doublet $(1, 2)_{1/2}$, whose hypercharge is fixed by the same charge convention, not imported.

The hypercharge assignment is derived next. Given the fermion representations and the Higgs hypercharge $Y_H = 1/2$, the charge convention $Q = T_3 + Y$ gives the Yukawa relations

$$Y_u = Y_Q + \frac{1}{2}, \qquad Y_d = Y_Q - \frac{1}{2}, \qquad Y_e = Y_L - \frac{1}{2}. \tag{C.27}$$

The mixed $\mathrm{SU}(2)^2\mathrm{U}(1)$ anomaly vanishes when $3Y_Q + Y_L = 0$, and the gravitational anomaly when $6Y_Q - 3Y_u - 3Y_d + 2Y_L - Y_e = 0$ (the right-handed singlets enter with the opposite chirality sign). Substituting the Yukawa relations into these two equations gives $3Y_Q - 1/2 = 0$, hence $Y_Q = 1/6$ and $Y = (1/6, 2/3, -1/3, -1/2, -1)$. The per-generation hypercharge moment is

$$\sum Y^2 = 6 \cdot \frac{1}{6^2} + 3 \cdot \frac{4}{9} + 3 \cdot \frac{1}{9} + 2 \cdot \frac{1}{4} + 1 \cdot 1 = \frac{10}{3}. \tag{C.28}$$

The one-loop gauge coefficients follow from the group-theoretic content,

$$b_1 = \frac{2}{5}\left(3 \sum Y^2 + Y_H^2\right) = \frac{41}{10}, \tag{C.29}$$

$$b_2 = -\frac{11}{3} C_2(\mathrm{SU}(2)) + \frac{2}{3} T_f(\mathrm{SU}(2)) + \frac{1}{3} T_H = -\frac{19}{6}, \tag{C.30}$$

$$b_3 = -\frac{11}{3} C_2(\mathrm{SU}(3)) + \frac{2}{3} T_f(\mathrm{SU}(3)) = -7, \tag{C.31}$$

with $C_2(\mathrm{SU}(2)) = 2$ and $C_2(\mathrm{SU}(3)) = N_c = 3$ the adjoint Casimirs, $T_f(\mathrm{SU}(2)) = T_f(\mathrm{SU}(3)) = 6$ the three-generation Dynkin sums, and $T_H = 1/2$ the Higgs doublet index. The two-loop matrix is derived from the same content. The diagonal entries are

$$B_{ii} = -\frac{34}{3} C_2(G_i)^2 + \left(2C_2^i(F) + \frac{10}{3} C_2(G_i)\right) S_2^i(F) + \left(2C_2^i(S) + \frac{1}{3} C_2(G_i)\right) S_2^i(S), \tag{C.32}$$

and the off-diagonal entries are

$$B_{ij} = 2C_2^j(F)S_2^i(F) + 2C_2^j(S)S_2^i(S), \tag{C.33}$$

with C_2 the Casimir and S_2 the Dynkin index of each representation, summed over the five Weyl states and the Higgs scalar. The Casimir and Dynkin index are defined as follows. The adjoint Casimir of SU(N) is $C_2(\mathrm{SU}(N)) = N$, and the Dynkin index of a representation R is fixed by $\mathrm{Tr}(T_R^aT_R^b) = S_2(R)\delta^{ab}$. The fundamental doublet of SU(2) has $C_2 = 3/4$ and $S_2 = 1/2$, and the fundamental triplet of SU(3) has $C_2 = 4/3$ and $S_2 = 1/2$. For the U(1) factor, in the GUT normalisation $g_1 = \sqrt{5/3}\,g_Y$, the Casimir and Dynkin index are $C_2 = (3/5)Y^2$ and $S_2 = (3/5)\sum Y^2$.

The per-generation Weyl content, whose six entries are the Casimirs and Dynkin indices $(C_2^{\mathrm{U}(1)}, C_2^{\mathrm{SU}(2)}, C_2^{\mathrm{SU}(3)}, S_2^{\mathrm{U}(1)}, S_2^{\mathrm{SU}(2)}, S_2^{\mathrm{SU}(3)})$, is

$$\begin{aligned}
Q_L(3,2)_{1/6} &: \Big(\frac{3}{5}\frac{1}{36}, \frac{3}{4}, \frac{4}{3}, \frac{3}{5}\frac{1}{6}, \frac{3}{2}, 1\Big),\\
u_R(3,1)_{2/3} &: \Big(\frac{3}{5}\frac{4}{9}, 0, \frac{4}{3}, \frac{3}{5}\frac{4}{3}, 0, \frac{1}{2}\Big),\\
d_R(3,1)_{-1/3} &: \Big(\frac{3}{5}\frac{1}{9}, 0, \frac{4}{3}, \frac{3}{5}\frac{1}{3}, 0, \frac{1}{2}\Big),\\
L_L(1,2)_{-1/2} &: \Big(\frac{3}{5}\frac{1}{4}, \frac{3}{4}, 0, \frac{3}{5}\frac{1}{2}, \frac{1}{2}, 0\Big),\\
e_R(1,1)_{-1} &: \Big(\frac{3}{5}, 0, 0, \frac{3}{5}, 0, 0\Big),
\end{aligned} \tag{C.34}$$

and the Higgs scalar $(1,2)_{1/2}$ contributes the real-scalar content $\big(\frac{3}{5}\frac{1}{4}, \frac{3}{4}, 0, \frac{3}{5}, 1, 0\big)$. The diagonal entry B_{11} is evaluated term by term. With $C_2(G_1) = 0$ for the abelian factor,

$$B_{11} = 2\sum_f C_2^{\mathrm{U}(1)}(f)S_2^{\mathrm{U}(1)}(f)\,n_g + 2C_2^{\mathrm{U}(1)}(H)S_2^{\mathrm{U}(1)}(H), \tag{C.35}$$

and inserting the hypercharge content, with $\sum_f C_2^{\mathrm{U}(1)}S_2^{\mathrm{U}(1)} = (3/5)^2\sum_f Y^4 = (3/5)^2\big(\frac{1}{216} + \frac{16}{27} + \frac{1}{27} + \frac{1}{8} + 1\big)$ and $n_g = 3$, gives $B_{11} = 199/50$. The remaining entries follow by the same insertion. The full matrix is

$$B = \begin{pmatrix} 199/50 & 27/10 & 44/5 \\ 9/10 & 35/6 & 12 \\ 11/10 & 9/2 & -26 \end{pmatrix}. \tag{C.36}$$

No entry of this matrix is taken from a table. The coefficients are the quadratic content ratios of the fermion and scalar representations, each carried to its Casimir and Dynkin index.

The Yukawa-gauge mixing coefficients A_i are derived from the same content. The gauge beta carries the top-Yukawa term through the invariant $Y_4(F) = (1/d(G_i))\,\mathrm{Tr}[C_2^{(i)}(F)Y^aY^{+a}]$ with $A_i = 2\kappa Y_4/y_t^2$ ($\kappa = 1/2$ for Weyl fermions). The trace is the explicit matrix element

$$\mathrm{Tr}[C_2^{(i)}(F)Y^aY^{+a}] = 6\,y_t^2\,\big[C_2^{(i)}(Q_L) + C_2^{(i)}(u_R)\big], \tag{C.37}$$

where the factor $6 = 2 \times 3$ is the weak doublet times the colour triplet of Q_L. Each fermion block contributes the same 6. The Yukawa contraction $Q_L^b H^a u_R \varepsilon_{ab}$ closes the SU(2) indices through $\sum_a \varepsilon_{ab}\varepsilon_{ab'} = \delta_{bb'}$, so the u_R trace carries the weak-contraction factor $2 = \dim \mathrm{SU}(2)$. Hence

$$A_i = \frac{6}{d(G_i)}\big[C_2^{(i)}(Q_L) + C_2^{(i)}(u_R)\big], \tag{C.38}$$

with $d(G) = (1, 3, 8)$ and the Casimirs $C_2^{(1)}(Q_L) = \frac{3}{5}\frac{1}{36}$, $C_2^{(1)}(u_R) = \frac{3}{5}\frac{4}{9}$, $C_2^{(2)}(Q_L) = \frac{3}{4}$, $C_2^{(2)}(u_R) = 0$, $C_2^{(3)}(Q_L) = C_2^{(3)}(u_R) = \frac{4}{3}$. The three entries are

$$A = \Big(\frac{17}{10},\ \frac{3}{2},\ 2\Big), \tag{C.39}$$

reproducing the standard two-loop values from the SM content alone, with no external table.

C.6 The top-Yukawa and Higgs-quartic beta functions

The scalar-sector running uses the top-Yukawa and Higgs-quartic beta functions, evaluated on the same content by the general two-loop formulas of [22] specialised to the standard-model content in [23]. The conventions here are $V = -m^2|H|^2 + \lambda|H|^4$ and the GUT normalisation $g_1 = \sqrt{5/3}\, g_Y$. [23] writes the potential as $m^2\varphi^\dagger\varphi + (\lambda_{LX}/2)(\varphi^\dagger\varphi)^2$, so the rescaling $\lambda = \lambda_{LX}/2$, $\beta_\lambda = \beta_{\lambda,LX}/2$ (a pure coupling rescaling) converts their results to the convention used here. The content enters through the trace invariants

$$\begin{gathered} Y_2(S) = \mathrm{Tr}[3H^\dagger H] = N_c\, y_t^2, \qquad H(S) = \mathrm{Tr}[3(H^\dagger H)^2] = N_c\, y_t^4, \\ \chi_4(S) = \frac{9}{4}\,\mathrm{Tr}[3(H^\dagger H)^2] = \frac{9}{4} N_c\, y_t^4, \end{gathered} \tag{C.40}$$

and

$$Y_4(S) = \Big(\frac{17}{20}g_1^2 + \frac{9}{4}g_2^2 + 8g_3^2\Big)y_t^2, \tag{C.41}$$

where the gauge coefficients are themselves the one-loop content of Section C.5, the anti-commutator $-3g^2\{C_2(F), Y\}$ evaluated per gauge group,

$$\begin{gathered} \frac{17}{20} = 3\big[C_2^{\mathrm{U}(1)}(Q_L) + C_2^{\mathrm{U}(1)}(u_R)\big] = 3\Big(\frac{3}{5}\frac{1}{36} + \frac{3}{5}\frac{4}{9}\Big), \qquad \frac{9}{4} = 3\, C_2^{\mathrm{SU}(2)}(Q_L) = 3\cdot\frac{3}{4}, \\ 8 = 3\big[C_2^{\mathrm{SU}(3)}(Q_L) + C_2^{\mathrm{SU}(3)}(u_R)\big] = 3\Big(\frac{4}{3} + \frac{4}{3}\Big). \end{gathered} \tag{C.42}$$

The family count $n_g = 3$ entering below is the window-capacity generation number of Section 4.1.

The one-loop top Yukawa is

$$\frac{\beta_{y_t}^{(1)}}{y_t} = \Big(\frac{3}{2} + N_c\Big)y_t^2 - \frac{17}{20}g_1^2 - \frac{9}{4}g_2^2 - 8g_3^2 = \frac{9}{2}y_t^2 - \frac{17}{20}g_1^2 - \frac{9}{4}g_2^2 - 8g_3^2, \tag{C.43}$$

the Yukawa self-coupling 3/2 plus the colour trace $Y_2(S) = N_c y_t^2$. The U(1) coefficient 17/20 is in the GUT normalisation, i.e. $-(17/12)g_Y^2$ in the hypercharge coupling. The two-loop top Yukawa is

$$\begin{aligned}\frac{\beta_{y_t}^{(2)}}{y_t} = & -12\,y_t^4 + \Big(36\,g_3^2 + \frac{225}{16}g_2^2 + \frac{393}{80}g_1^2\Big)y_t^2 - 108\,g_3^4 - \frac{23}{4}g_2^4 + \frac{1187}{600}g_1^4 \\ & + 9\,g_2^2g_3^2 + \frac{19}{15}g_1^2g_3^2 - \frac{9}{20}g_1^2g_2^2 + 6\,\lambda^2 - 12\,\lambda y_t^2,\end{aligned} \tag{C.44}$$

with the content structure

$$-12 = \frac{3}{2} - \frac{9}{4}N_c - \frac{9}{4}N_c, \qquad 36 = \frac{5}{2}\cdot 8 + 16, \qquad \frac{225}{16} = \frac{5}{2}\cdot\frac{9}{4} + \frac{135}{16}, \qquad \frac{393}{80} = \frac{5}{2}\cdot\frac{17}{20} + \frac{223}{80}, \tag{C.45}$$

$$\frac{1187}{600} = \frac{9}{200} + \frac{29}{45}n_g, \qquad -\frac{23}{4} = -\Big(\frac{35}{4} - n_g\Big), \qquad -108 = -\Big(\frac{404}{3} - \frac{80}{9}n_g\Big), \tag{C.46}$$

where the first line is the χ_4- and Y_4-content of the trace invariants above, and the remaining coefficients (16, 135/16, 223/80, 9/200, 29/45, 35/4, 404/3, 80/9, 9, 19/15, 9/20, 6, 12) are the universal two-loop numbers of the general evaluation. The one-loop Higgs quartic is

$$\beta_\lambda^{(1)} = 24\,\lambda^2 - 3\lambda\Big(3g_2^2 + \frac{3}{5}g_1^2\Big) + \frac{3}{8}\Big(2g_2^4 + \big(g_2^2 + \frac{3}{5}g_1^2\big)^2\Big) - 6\,y_t^4 + 12\,\lambda y_t^2, \tag{C.47}$$

whose Yukawa part is the content combination $4Y_2(S)\lambda - 4H(S) = 12\lambda y_t^2 - 6y_t^4$. The full two-loop Higgs quartic is

$$\begin{aligned}\beta_\lambda^{(2)} = & -312\,\lambda^3 + 36\lambda^2\Big(3g_2^2 + \frac{3}{5}g_1^2\Big) - 144\,\lambda^2 y_t^2 - \frac{73}{8}\lambda g_2^4 + \frac{117}{20}\lambda g_2^2 g_1^2 + \frac{1887}{200}\lambda g_1^4 \\ & + \Big(\frac{17}{2}g_1^2 + \frac{45}{2}g_2^2 + 80\,g_3^2\Big)\lambda y_t^2 - 3\,\lambda y_t^4 + 30\,y_t^6 \\ & - 32\,g_3^2 y_t^4 - \frac{8}{5}g_1^2 y_t^4 - \frac{9}{4}g_2^4 y_t^2 + \frac{63}{10}g_2^2 g_1^2 y_t^2 - \frac{171}{100}g_1^4 y_t^2 \\ & + \frac{305}{16}g_2^6 - \frac{289}{80}g_2^4 g_1^2 - \frac{1677}{400}g_2^2 g_1^4 - \frac{3411}{2000}g_1^6,\end{aligned} \tag{C.48}$$

with the content structure

$$\begin{aligned}&-312 = -78\cdot 4, \qquad 108 = 54\cdot 2, \qquad -144 = -24\cdot Y_2(S)/y_t^2\cdot 4, \\ &\frac{17}{2} = 10\cdot\frac{17}{20}, \qquad \frac{45}{2} = 10\cdot\frac{9}{4}, \qquad 80 = 10\cdot 8,\end{aligned} \tag{C.49}$$

$$\begin{aligned}&30 = \frac{60}{2}, \quad -32 = -\frac{64}{2}, \quad -\frac{8}{5} = -\frac{16}{5}\Big/2, \\ &-\frac{9}{4} = -\frac{9}{2}\Big/2, \quad \frac{63}{10} = \frac{63}{5}\Big/2, \quad -\frac{171}{100} = -\frac{171}{50}\Big/2,\end{aligned} \tag{C.50}$$

$$\begin{aligned}&\frac{305}{16} = \Big(\frac{497}{8} - 8n_g\Big)\Big/2, \quad -\frac{289}{80} = -\Big(\frac{97}{40} + \frac{8}{5}n_g\Big)\Big/2, \\ &-\frac{1677}{400} = -\Big(\frac{717}{200} + \frac{8}{5}n_g\Big)\Big/2, \quad -\frac{3411}{2000} = -\Big(\frac{531}{1000} + \frac{24}{25}n_g\Big)\Big/2,\end{aligned} \tag{C.51}$$

$$-\frac{73}{8} = -\Big(\frac{313}{8} - 10n_g\Big), \qquad \frac{1887}{200} = \frac{687}{200} + 2n_g, \tag{C.52}$$

where the $\lambda y_t^2 g^2$ terms are the $10\,\lambda Y_4(S)$ content, the y_t^6, $g^2 y_t^4$, $g^4 y_t^2$ terms are the fermionic traces evaluated on the top content, and the n_g-carrying coefficients are the fermionic Dynkin sums over the three generations. Every coefficient above is displayed as a content combination (or a documented universal two-loop number). None is imported from a table, and each rational identity is elementary (the displayed equalities are the decompositions).

C.7 The Landau-Zener ladder

The generation masses are suppressed exponentially by the extrusion of the modes $n = \{0, 2, 4\}$ under the non-adiabatic squeezing of the scale flow. The extrusion is a Landau-Zener two-level crossing in the scale flow $L(k) = C/k$, with the avoided-crossing Hamiltonian

$$H(t) = \begin{pmatrix} vt & \Delta \\ \Delta & -vt \end{pmatrix}, \qquad P = e^{-2\pi\Delta^2/v}, \tag{C.53}$$

and the LZ index $\gamma = 2\pi\Delta^2/v$ decomposes into the adiabatic window width and the non-adiabatic torsion. The mass ratios are $m_i \propto e^{-\alpha n_i}$, and the sector index is fixed by the internal ladder. The up-sector index is the adiabatic window width minus the non-adiabatic torsion correction,

$$\alpha_{\rm up} = kL - 2\tau = 2.4535, \tag{C.54}$$

where kL is the adiabatic part (the extrusion of a mode across the full coarse-graining window) and $2\tau = n_{\rm broken}\tau$ is the non-adiabatic deficit (the Einstein-Cartan torsion of the two broken generators, each carrying τ), so the top-to-charm ratio is $m_t/m_c = e^{2\alpha_{\rm up}}$, and the difference $kL - \alpha_{\rm up} = 2\tau$ is exact. The down and lepton sectors descend by the step

$$\Delta = 6(1 - n_s)kL_{\rm CMB}, \tag{C.55}$$

the product of the six generators of the $\mathfrak{so}(4)$ isometry, the spectral tilt $1 - n_s$, and the microwave-pivot window $kL_{\rm CMB}$. The down and lepton steps are split by the 9/8 hypercharge identity

$$\frac{9}{8} = \frac{1}{1 - (Y_d/Y_l)^2}, \tag{C.56}$$

with $Y_d = 1/3$ and $Y_l = 1$, so $(Y_l^2 - Y_d^2)/Y_l^2 = 8/9$, the lepton step carrying 8/9 of the down step. Writing the lepton step as s and the down step as $(9/8)s$, and using that the lepton rung spans two steps, $(9/8)s + s = 2\Delta$, one obtains $s = 16\Delta/17$ and the down step $(9/8)(16\Delta/17) = 18\Delta/17$. The indices are

$$\alpha_{\rm dn} = \alpha_{\rm up} - \frac{18}{17}\Delta, \qquad \alpha_{\rm lp} = \alpha_{\rm up} - 2\Delta. \tag{C.57}$$

The first-generation step of the down sector carries the colour dilution

$$\alpha_{\rm sd} = \alpha_{\rm dn} - \frac{kL_{\rm CMB}}{6}, \tag{C.58}$$

the microwave window divided by the six generators of $\mathfrak{so}(4)$, the colour dilution of the first-generation extrusion. The symmetry form of this step is $\alpha_{\rm sd} = \Delta_f(1 - s_0/N_R) = (3/2)(1-2\tau/7)$, the fermion conformal weight $\Delta_f = d/2 = 3/2$ times the gravity correction $1 - s_0/N_R$ with $s_0/N_R = 1/175$.

The four coefficients share one physical origin, the non-adiabatic extrusion of the generation modes. The up index is the window width with the torsion subtracted, $kL - \alpha_{\rm up} = 2\tau$ exactly, because the Einstein-Cartan torsion is the non-adiabatic part of the scale flow, and the LZ exponent is the adiabatic window width minus the non-adiabatic torsion. The factor 2 is the number of broken generators of the squash, $n_{\rm broken} = 2$. The step Δ is the extrusion rate of the sector ladder, the product of the six $\mathfrak{so}(4)$ generators (the extrusion coupling, the isometry of the internal space), the spectral tilt $1 - n_s$ (the closed window-evolution rate, itself $\tau \cdot 7/4$ with 7 the RP^3 Weyl degree-of-freedom count), and the microwave window $kL_{\rm CMB}$ (the scale at which the extrusion is observed). The 9/8 split is the hypercharge weight in the covariant derivative on RP^3 shifting the LZ exponent. The down and lepton doublets carry different hypercharge weights $Y_d = 1/3$ and $Y_l = 1$, so the lepton rung carries 8/9 of the down rung, the exact identity $9/8 = 1/(1 - (Y_d/Y_l)^2)$. The colour dilution $kL_{\rm CMB}/6$ is the first-generation extrusion divided by the six generators, the content of the first-generation mode spread over the full $\mathfrak{so}(4)$ isometry. None of the four coefficients is a fitted rule. Each is a content ratio of the non-adiabatic extrusion.

C.8 The top-quark anchor

The top Yukawa is the $(0,0)$ diagonal overlap of the $n = 0$ spinor mode with the scalar condensate. The overlap is normalised to unity,

$$y_0 = 1, \tag{C.59}$$

because the mass operator is the $(0,0)$ scalar of SO(4), the torsion singlet $T_{abc}\gamma^{abc}$, and its matrix element is evaluated explicitly on the generation modes. A generation mode n carries the SO(4) content $(j_L, j_R) = ((n+1)/2, n/2)$, so the top $n = 0$ is $(j_L, j_R) = (1/2, 0)$. The matrix element of the $(0,0)$ scalar between a mode and itself is the product of the two SU(2) scalar-channel Clebsch-Gordan coefficients,

$$y_0 = \langle (j_L, j_R)|(0,0)|(j_L, j_R)\rangle = C(j_L, 0, j_L)\, C(j_R, 0, j_R), \tag{C.60}$$

where $C(j, 0, j) = 1$ is the SU(2) Clebsch-Gordan coefficient of the scalar channel. The identity representation couples $j \otimes 0 = j$ with the unique coefficient one. Hence

$$y_0 = C(\tfrac{1}{2}, 0, \tfrac{1}{2})\, C(0, 0, 0) = 1 \cdot 1 = 1. \tag{C.61}$$

This is the explicit wave-function matrix element, evaluated on the $n = 0$ generation mode. The same element is 1 for every generation mode, $C(j_L, 0, j_L)\, C(j_R, 0, j_R) = 1$ (the scalar channel couples each mode to itself with unit coefficient), so the generations differ only through the ladder of Section C.7, not through this element. It is also scale-invariant, in the sense that the geometric renormalisation group leaves it unchanged, so y_0 does not run. The top mass is then fixed by the vacuum expectation value of Section 7.3, which is itself derived from the emergence scale through the window-squared line, so $m_t = y_0 v/\sqrt{2}$ carries no observed mass and no free Yukawa coupling. The lower generations descend by the ladder of Section C.7. Writing $n_s^{\rm tilt} \equiv 1-n_s$, the bottom absolute mass is the geometric mean of the strange and the top dressed by the window evolution,

$$\frac{y_b}{y_t} = e^{-(2\alpha_{\rm dn} - n_s^{\rm tilt}(kL_{\rm CMB}+2\tau))}, \tag{C.62}$$

so $m_b^2 = m_s m_t e^{n_s^{\rm tilt}(kL_{\rm CMB}+2\tau)}$.

C.9 Charged-fermion masses

The exponent 20 can be understood systematically as a heat-kernel coefficient. For the Laplacian on RP^3, the Seeley–deWitt expansion [24] gives $\operatorname{Tr} e^{-t\Delta_{RP^3}} \sim a_0\, t^{-3/2} + a_1\, t^{-1} + \cdots$, where the first non-trivial coefficient is $a_1 = (d+1)(\sum Y^2\Delta_f) = 4\times 5 = 20$. This is the same coefficient structure that appears in the spectral action principle of Chamseddine and Connes [4], specialised to the present spectral datum. The closed form $m_e = M_{\rm P} e^{-20kL}(1-s_0\kappa)$ is the evaluation of this heat-kernel coefficient on the scale trajectory $L(k) = C/k$, Wick-rotated to the physical mass. The exponent is therefore not an ad hoc product but the first non-trivial heat-kernel invariant of the spectral datum.

The electron mass is the Planck-anchored exponential chain with the low-scale squash correction,

$$m_e = M_{\rm P}\, e^{-20kL}(1 - s_0\kappa), \tag{C.63}$$

where the exponent $20 = 4\times 5$ is the structural counting of the fourfold and fivefold content, and $s_0\kappa$ is the squash correction of Section C.4. The two factors have independent structural origins. The factor $4 = d+1$ is the internal dimension $d = 3$ plus one, the four levels of the spectral cascade. The factor $5 = \sum Y^2\Delta_f$ is the hypercharge capacity $\sum Y^2 = 10/3$ times the fermion conformal weight $\Delta_f = d/2 = 3/2$, which equals the five fermion representations of one generation, Q_L, u_R, d_R, L_L, e_R, the content of Section 4.1. The product, not the sum, appears because each of the four cascade levels acts on the complete content of one generation, the five species, so the exponent is $20 = (d+1)(\sum Y^2\Delta_f) = 4\times 5$, a pure content ratio. The cascade form of the electron mass is

$$m_e = y_0\, O_e\, \frac{v_{\rm dil}}{\sqrt{2}}, \qquad O_e = 1 - \frac{1}{1+kL^2} e^{-1/kL}, \tag{C.64}$$

the universal Yukawa seed $y_0 = 1$ times the $(0,0)$ overlap O_e of the electron $l = 0$ mode with the dilaton scalar, times the dilaton vacuum expectation value $v_{\rm dil}$ at the electron scale. The overlap is the explicit constant-mode integral,

$$O_e = \int_{RP^3} d^3y \sqrt{g}\, \bar\chi_0(y)\, \varphi_{\rm dil}^{(0)}(y)\, \chi_0(y), \tag{C.65}$$

where $\chi_0 = 1/\sqrt{2\pi^2 a^3}$ is the $l = 0$ constant spinor and $\varphi_{\rm dil}^{(0)} = v_{\rm dil}/\sqrt{2\pi^2 a^3}$ the $l = 0$ constant dilaton, both spatially uniform, so the volume integral cancels the normalisation factors, giving $O_e = 2\pi^2 a^3 (1/\sqrt{2\pi^2 a^3})(v_{\rm dil}/\sqrt{2\pi^2 a^3})(1/\sqrt{2\pi^2 a^3}) = v_{\rm dil}/\sqrt{2\pi^2 a^3}$, i.e. $O_e = 1$ after normalising by $v_{\rm dil}/\sqrt{2\pi^2 a^3}$ (the two constant modes overlap exactly, the unit $(0,0)$

Clebsch-Gordan weight). The small deficit $\delta(kL) = \frac{1}{1+kL^2} e^{-1/kL}$ is the finite-size back-reaction. The dilaton profile has a weak kL-dependent gradient on RP^3, and $\delta(kL)$ is its first correction, so $O_e = 1 - \delta(kL)$.

The exponential chain $m_e = M_{\rm P} e^{-20kL}$ is the compressed statement of this cascade, and the equivalence is made explicit as follows. The electron-scale dilaton VEV descends from the Planck scale through the spectral cascade,

$$v_{\rm dil} = M_{\rm P}\, e^{-20kL}\, \frac{\sqrt{2}\,(1 - s_0\kappa)}{y_0\, O_e}, \tag{C.66}$$

so that the cascade form reproduces the exponential chain exactly,

$$m_e = y_0\, O_e\, \frac{v_{\rm dil}}{\sqrt{2}} = M_{\rm P}\, e^{-20kL}\,(1 - s_0\kappa). \tag{C.67}$$

The descent exponent $20 = (d+1)(\sum Y^2 \Delta_f) = 4 \cdot 5$ is the KK cascade. The dilaton (the scalar zero mode) descends through $d + 1 = 4$ levels (the internal dimension 3 plus the scale-flow direction), each level acting on the complete one-generation content $\sum Y^2 \Delta_f = (10/3)(3/2) = 5$ species. Each (level $\times$ species) unit crosses the coarse-graining window once and is suppressed by the LZ survival factor e^{-kL} (one window width kL, the same window whose circumference $4\pi kL$ sets the electroweak hierarchy $\ln(M_G/v)$ of Section C.11). The four steps therefore give $(e^{-kL})^{4\cdot 5} = e^{-20kL}$. The $O(1)$ prefactor $\sqrt{2}(1 - s_0\kappa)/(y_0 O_e)$ carries the normalisation (the Yukawa factor $\sqrt{2}$ and the seed $y_0 = 1$) and the two level-corrections, namely the finite-size back-reaction δ inside O_e and the $J = 2$ squash $s_0\kappa$, both $O(1)$ corrections to the dominant exponential, in the same two-correction pattern as the electroweak hierarchy $\ln(M_G/v) = 4\pi kL - \ln(3\alpha/\pi) + s_0\kappa$. The compression is thus explicit. The $4 \times 5 = 20$ window crossings collapse into the single exponent $20kL$, and the $O(1)$ factor is the residual normalisation plus the two corrections. The muon-to-electron ratio carries the Euclidean period,

$$\frac{m_\mu}{m_e} = e^{2\alpha_{\rm lp} + \sqrt{2\pi}}, \tag{C.68}$$

the lepton LZ index plus the entropy-minimum distance $2L = \sqrt{2\pi}$.

C.10 Neutrino masses and the PMNS matrix

The heaviest neutrino mass is set by the Weinberg dimension-five operator with the two-period family scale,

$$m_{\nu_3} = \frac{v^2 (2\pi)^2}{k_{\rm GUT}} (1 + s_0\kappa) = 0.0502 \text{ eV}, \tag{C.69}$$

the electroweak scale squared times the Euclidean-period factor divided by the breaking scale, with the squash correction $s_0\kappa$ of Section C.4. The form is the dimension-five operator $\mathcal{L} = c\,(LH)(LH)/\Lambda$ with the seesaw family scale $\Lambda = k_{\rm GUT}/(2\pi)^2$, giving $m_{\nu_3} = c\, v^2/\Lambda$ with $c = (2\pi)^2$. The $(2\pi)^2$ is the Euclidean period squared, in the same 2π family as the amplitude $e^{1/(2\pi)}$ and the window width $2L = \sqrt{2\pi}$. The lighter masses descend by the hypercharge-trace hierarchy ratios,

$$\frac{m_{\nu_1}}{m_{\nu_2}} = \frac{1}{\mathrm{Tr}(Y^2)} = \frac{3}{10}, \qquad \frac{m_{\nu_2}}{m_{\nu_3}} = \frac{1}{\sqrt{3}\,\mathrm{Tr}(Y^2)} = \frac{3}{10\sqrt{3}}, \tag{C.70}$$

so $m_{\nu_1} = m_{\nu_3} r_{12} r_{23}$ with $r_{12} = 3/10$ and $r_{23} = 3/(10\sqrt{3})$. The ratios are not assigned by hand. They are the eigenvalue ratios of an explicitly constructed mass matrix. The effective matrix is the spectral assembly

$$M_\nu = U\,\mathrm{diag}(\lambda_3, \lambda_2, \lambda_1)\,U^\dagger\,\frac{v^2}{\Lambda}, \qquad \lambda = \Big\{1,\ \tfrac{1}{\sqrt{3}\,\mathrm{Tr}(Y^2)},\ \tfrac{1}{\sqrt{3}\,\mathrm{Tr}(Y^2)^2}\Big\}, \tag{C.71}$$

where U is the PMNS matrix of the content-ratio angles (Section C.10 below) and the eigenvalues are the inverse powers of the first non-zero hypercharge moment. Diagonalising M_ν returns the hierarchy as the eigenvalue ratios, $m_1/m_2 = \lambda_1/\lambda_2 = 1/\,\mathrm{Tr}(Y^2) = 3/10$ and $m_2/m_3 = \lambda_2/\lambda_3 = 1/(\sqrt{3}\,\mathrm{Tr}(Y^2))$, to the displayed precision. The hierarchy is therefore the eigenvalue structure of the hypercharge-trace flavour matrix, the content ratio of the same trace that closes the gauge sector, and no mass is inserted by hand. The $\sqrt{3}$ is the internal-space geometry factor $\sin(\pi/3)\cdot 2$ that also enters the reactor angle.

The absolute masses and the oscillation comparison are kept distinct. The raw rest-mass split of the light pair is

$$\Delta m^2_{21,\mathrm{raw}} = m^2_{\nu_2} - m^2_{\nu_1} = 6.874 \times 10^{-5}\ \mathrm{eV}^2. \tag{C.72}$$

Since m_{ν_1} is a nonzero vacuum-floor mass in the present construction, comparing m_{ν_2} directly with $\sqrt{\Delta m^2_{21}}$ would be only a massless-floor proxy. The solar oscillation channel is therefore evaluated as a propagation observable. The second light state carries the finite window-tilt dressing already fixed by the cosmological tilt,

$$m^{\mathrm{osc}}_{\nu_2} = m_{\nu_2}\big(1 + n_s^{\mathrm{tilt}}\big), \qquad n_s^{\mathrm{tilt}} \equiv 1 - n_s = \tau\frac{7}{4}, \tag{C.73}$$

while m_{ν_1} remains the absolute floor used in the dark-energy and microwave-temperature closures. This gives

$$\Delta m^2_{21} = \big(m^{\mathrm{osc}}_{\nu_2}\big)^2 - m^2_{\nu_1} = 7.412 \times 10^{-5}\ \mathrm{eV}^2, \qquad \Delta m^2_{31} = m^2_{\nu_3} - m^2_{\nu_1} = 2.511 \times 10^{-3}\ \mathrm{eV}^2. \tag{C.74}$$

The tilt factor is not fitted in the neutrino sector; it is the previously closed RP^3 Weyl-window ratio of Section C.14. It also leaves the absolute mass sum and the cosmological floor unchanged.

The PMNS mixing angles carry the two-period imprint. The solar angle is the neutrino mass ratio,

$$\sin^2\theta_{12} = \frac{m_{\nu_1}}{m_{\nu_2}} = \frac{3}{10} = 0.30, \tag{C.75}$$

the light-pair mixing of the two-state texture. In the near-tribimaximal texture the two lightest states decouple from the heavy state, and the mixing of the light pair is the ratio of their masses, $\sin^2\theta_{12} = m_{\nu_1}/m_{\nu_2}$, the standard two-state seesaw relation. The reactor angle is the Euclidean-period imprint times the internal-space geometry factor,

$$\sin^2\theta_{13} = \left(\frac{1}{2\pi}\right)^2 \frac{\sqrt{3}}{2} = 0.02194, \tag{C.76}$$

and the atmospheric angle is

$$\sin^2\theta_{23} = \frac{1}{2} + \frac{\mathrm{Tr}(T_3^2)}{(2\pi)^2} = 0.5507. \tag{C.77}$$

All three angles are fixed by the same content ratios that fix the charged sector, so the PMNS matrix is closed within the stated prescription. The solar angle is the neutrino mass ratio, the hypercharge-trace reciprocal of Section C.10. The atmospheric angle is the maximum-mixing value 1/2 corrected by the isospin trace over the Euclidean period, $\mathrm{Tr}(T_3^2) = 2$. The reactor angle is the period imprint $1/(2\pi)^2$ times the internal-space geometry factor $\sqrt{3}/2 = \sin(\pi/3)$, the $S^3 \to RP^3$ quotient projection. The structure is the near-tribimaximal form dressed by the two-period imprint, and no angle is fitted.

C.11 The electroweak order parameter

The isometry breaking is driven by the $J = 2$ squash mode, the order parameter φ of the internal geometry. Its dynamics is the Landau potential on the curvature axis,

$$V(\varphi) = \frac{1}{2}\xi(R - R_\mathrm{c})\varphi^2 + \frac{\lambda}{4}\varphi^4, \tag{C.78}$$

with $\xi = 1/8$ the conformal coupling and $R_\mathrm{c} = 6/\pi$ the critical curvature. The effective mass squared is $m^2 = \xi(R - R_\mathrm{c})$, tachyonic for $R < R_\mathrm{c}$, so the condensation is triggered by the curvature crossing, not by a free-spectrum instability. The quartic coupling is fixed by the stationarity of the potential at the squash vacuum $\varphi_0 = 2\tau$,

$$\frac{\partial V}{\partial\varphi} = 0 \quad \Rightarrow \quad \lambda = \frac{\xi(R_\mathrm{c} - R_\mathrm{GUT})}{(2\tau)^2} = 149, \tag{C.79}$$

with $R_\mathrm{GUT} = 6/L_\mathrm{GUT}^2$ and $L_\mathrm{GUT} = \sqrt{3}/\tau$ the isometry-breaking scale. The Einstein-Cartan torsion algebra fixes the three coefficients of the parity-preserving torsion Lagrangian

$$L_\mathrm{torsion} = a\,T^2 + b\,T^{bac}T_{abc} + c\,(T^a{}_{ab})^2. \tag{C.80}$$

The algebraic-torsion condition, the requirement that the torsion field equation be algebraic rather than dynamical, fixes $b = 4a$, the Holst-Immirzi condition, and the trace term fixes $c = -(a + b/3) = -(7/3)a$. The remaining coefficient $a = M_\mathrm{G}^3/4$ is fixed by the canonical normalisation of the torsion kinetic term at the emergence scale, and the stationarity $\partial V/\partial\varphi = 0$ at $\varphi_0 = 2\tau$ then selects the quartic coupling. The vacuum expectation value is the emergence scale times the window-squared line,

$$v = M_\mathrm{G}\frac{3\alpha}{\pi}e^{-4\pi kL}(1 - s_0\kappa), \tag{C.81}$$

the exponential $e^{-4\pi kL}$ carrying the hierarchy $v/M_\mathrm{G} \sim 10^{-16}$, with the squash correction $s_0\kappa$ of Section C.4. The two factors of the line have independent origins. The prefactor $3\alpha/\pi$ is the product of three pieces, the $J = 2$ squash bifurcation $3\alpha/2$ with $\alpha = 1/(16\pi^2)$ the one-loop weight, the Fourier prefactor $1/\pi$ of the window, and the factor 2 from the double crossing, $(3\alpha/2)(1/\pi)(2) = 3\alpha/\pi$. The exponential $e^{-4\pi kL}$ is the Landau-Zener survival of the $J = 2$ mode crossing the coarse-graining window twice, once at creation and once at stabilisation, each crossing suppressing by $e^{-2\pi kL}$, the same exponential that governs the curvature amplitude of Section C.14.

C.12 The Higgs quartic

The Higgs quartic λ_H is the self-coupling of the pseudo-dilaton, the Higgs identified as the pseudo-dilaton of the scale flow. Its closed form is the dilaton quartic reduced by the one-loop factor,

$$\lambda_H = \frac{\lambda_{\rm dil} + \sigma_{\rm SM}}{32\pi^2} = \frac{12\pi + 3}{32\pi^2} = 0.129, \tag{C.82}$$

where $\lambda_{\rm dil} = 3 \times 4\pi = 12\pi$ is the dilaton self-coupling, the trace anomaly coupling to all three generations, three times the single-generation strong-coupling bound 4π of the Nambu–Jona-Lasinio/Bethe-Salpeter normalisation, and $\sigma_{\rm SM} = 3$ is the Standard-Model loop contribution, one unit per generation. The tree-level Higgs mass is then

$$m_H = \sqrt{2\lambda_H}\, v = 124.98 \text{ GeV}, \tag{C.83}$$

against the observed 125.2 GeV, a deviation of -0.17%. The trace-anomaly coefficient that feeds the pseudo-dilaton mass is

$$\beta_{\rm eff} = \frac{3g_2^2 + g_1^2 + 4y_t^2 + 2\lambda_H}{16\pi^2} + \frac{\lambda_{\rm dil}}{16\pi^2}, \tag{C.84}$$

the pure-loop Standard-Model part plus the dilaton strong-coupling part. This λ_H is the Higgs quartic, distinct from the colour-singlet order parameter $\lambda = 149$ of Section C.11; the two couplings belong to different sectors.

C.13 CKM mixing and CP

The quark-mixing elements follow from the mass ladder through the Gatto relation. The first-generation ratio carries the hypercharge first-generation factor $|Y_d|/|Y_u| = 1/2$,

$$\frac{m_d}{m_s} = e^{-2\alpha_{\rm dn}}\left(1 + \frac{|Y_d|}{|Y_u|}\right)^2 = e^{-2\alpha_{\rm dn}}\left(\frac{3}{2}\right)^2 = 0.05015, \tag{C.85}$$

and the Cabibbo element is the Gatto dominant term,

$$|V_{us}| = \sqrt{\frac{m_d}{m_s}} = 0.2239. \tag{C.86}$$

The Gatto ratio here differs from the absolute-mass ladder ratio $m_s/m_d = e^{2\alpha_{\rm sd}} = 19.62$ of Section C.7 by 1.6%. The two ratios carry the first-generation colour-dilution step $\alpha_{\rm sd} = \alpha_{\rm dn} - kL_{\rm CMB}/6$ in different powers, the absolute ladder carrying $e^{2\alpha_{\rm sd}}$ and the Gatto element carrying $e^{-2\alpha_{\rm dn}}(3/2)^2$. They are therefore kept as separate closure relations. The cross-generation elements are

$$|V_{cb}| = \frac{1}{2}e^{-\alpha_{\rm up}}(1 - s_0\kappa), \qquad |V_{ub}| = \frac{1}{2}e^{-2\alpha_{\rm up}}(1 + \tau\kappa), \tag{C.87}$$

the adjacent-generation element carrying the squash correction $s_0\kappa$ and the cross-generation element the chiral correction $\tau\kappa$. The three-element product is numerically close to the weak cube,

$$|V_{us}||V_{cb}||V_{ub}| \approx \alpha_W(v)^3, \tag{C.88}$$

with $\alpha_W(v) = g_2(v)^2/(4\pi)$ the electroweak-scale weak coupling of the geometric renormalisation group (not the window-capacity coupling $2/kL^5$ of Section 5.1). The product matches the weak cube to -2.45%, and the near-equality is recorded as a numerical coincidence of the ladder and the running coupling, not as an exact identity.

The CKM phase is the colour-number dilution of the left-right content ratio. The left-right ratio is

$$\frac{N_L}{N_R} = \frac{8}{7}, \tag{C.89}$$

eight left-handed doublets against seven right-handed singlets per generation. The lepton PMNS phase carries the undiluted ratio $\delta_{\text{PMNS}} \sim (8/7)\pi$, and the quark phase is diluted by the colour number,

$$\delta_{\text{CKM}} = \frac{8}{7}\frac{\pi}{N_c} = \frac{8\pi}{21} = 68.57^\circ. \tag{C.90}$$

The Jarlskog invariant is the exact formula with the closed elements and the derived phase,

$$J = |V_{us}||V_{cb}||V_{ub}|\, c_{12}c_{23} \sin\delta = 3.15 \times 10^{-5}. \tag{C.91}$$

The baryon asymmetry is the Sakharov content, written here as a model relation,

$$\eta_B = \frac{J\alpha_W(v)^2}{56} = 6.09 \times 10^{-10}, \tag{C.92}$$

where J is the Jarlskog invariant above and $1/56 = 1/(N_L N_R)$ the content count. The factor $56 = N_L N_R = 8 \times 7$ is the product of the left and right content counts of one generation, and it decomposes as $1/56 = \xi/n_R$, the conformal-gauge duality $\xi = 1/N_g = 1/8$ times the inverse right-handed count $1/n_R = 1/7$, the same content-ratio structure that closes the $1/N_c$ term of the gauge conservation law of Section C.4. The two explicit factors $\alpha_W(v)^2$ stand for the two weak sphaleron vertices, and the Jarlskog invariant itself carries a weak-cube factor through the three-element product. But because that product matches the weak cube only to -2.45%, the total power is only approximately the five-weak-factor sphaleron form, and the relation is stated here as a model relation built from the content ratios rather than as an exact sphaleron-rate identity. The Sakharov out-of-equilibrium condition is the content-count normalisation $1/(N_L N_R)$, not a fitted rate.

C.14 Cosmology

The dark-energy density is the lightest-neutrino mass floor,

$$\rho_\Lambda = Y_u\, m_{\nu_1}^4\,(1 - 4s_0\kappa) = \frac{2}{3} m_{\nu_1}^4 (1 - 4s_0\kappa), \tag{C.93}$$

where $Y_u = 2/3$ is the up-quark hypercharge and the lightest neutrino mass is derived from the Weinberg operator

$$m_{\nu_3} = \frac{v^2(2\pi)^2}{k_{\text{GUT}}}(1 + s_0\kappa), \qquad m_{\nu_1} = m_{\nu_3} \cdot \frac{m_1}{m_2} \cdot \frac{m_2}{m_3}, \tag{C.94}$$

with the hierarchy ratios $m_1/m_2 = 1/\operatorname{Tr}(Y^2) = 3/10$ and $m_2/m_3 = 1/(\sqrt{3}\operatorname{Tr}(Y^2))$. The weight $(1 - 4s_0\kappa)$ is the dark-energy-weight symmetry correction. The seesaw mass carries

$+s_0\kappa$ (the level transfer from the electroweak scale, whose v carries $-s_0\kappa$), so $m_{\nu_1}^4$ carries $+4s_0\kappa$, and the weight cancels it to order $s_0^2\kappa^2$, keeping the dark-energy density conserved under the $J = 2$ squash level transfer. The physical content of this relation is that the lightest fermion sets the vacuum-energy floor. The neutrino is the lightest fermion of the spectrum, and its mass is the smallest nonzero energy scale the spectrum carries, so the vacuum energy is anchored at that floor rather than at an arbitrary scale. The weight $Y_u = 2/3$ is the up-quark hypercharge, the hypercharge of the neutral seesaw partner of the up quark, so the floor is the lightest-neutrino mass raised to the fourth power, weighted by the hypercharge of the partner it pairs with through the seesaw.

The entropy integral is the logarithm of the ratio of two Gaussian widths, the Planck width $M_{\rm P}^2$ and the vacuum floor $\sqrt{\rho_\Lambda}$,

$$\int \gamma_M = \ln \frac{M_{\rm P}^2\sqrt{2\pi + r_{23}}}{\sqrt{\rho_\Lambda}} = 139.2537, \tag{C.95}$$

with $r_{23} = 3/(10\sqrt{3})$ the hypercharge-trace hierarchy ratio. The statistical-mechanical origin is the association entropy of two Gaussians. The Planck anchor is a Gaussian of variance $M_{\rm P}^2$ and the vacuum floor a Gaussian of variance $\sqrt{\rho_\Lambda}$. A Gaussian of variance σ^2 has differential entropy $H(\sigma) = \frac{1}{2}\ln(2\pi e\,\sigma^2)$, and the association of the two ends, twice their entropy difference plus the entropy-minimum distance $\ln\sqrt{2\pi + r_{23}}$, gives

$$\int \gamma_M = 2\big[H(M_{\rm P}) - H(\sqrt{\rho_\Lambda})\big] + \ln\sqrt{2\pi + r_{23}} = \ln \frac{M_{\rm P}^2\sqrt{2\pi + r_{23}}}{\sqrt{\rho_\Lambda}}. \tag{C.96}$$

The physical content is the Boltzmann entropy of the scale flow. The scale flow carries two Gaussian measures, one at the Planck anchor of width $M_{\rm P}^2$ and one at the vacuum floor of width $\sqrt{\rho_\Lambda}$, and the number of available states between the two ends of the emergence window is the ratio of the two widths, $W = M_{\rm P}^2\sqrt{2\pi + r_{23}}/\sqrt{\rho_\Lambda}$. The factor $\sqrt{2\pi + r_{23}}$ is the width of the correlation window between the two ends, the Euclidean period 2π dressed by the hypercharge-trace hierarchy ratio r_{23}, so the entropy is the Boltzmann logarithm $S = \ln W$, and W is the window span over which the flow states are counted. The self-similar branch has $\gamma_M = 0$ (no entropy production, the scale-invariant spectrum), and the infrared end is the maximum-entropy state, $S = \ln W = 139.25$, the Jaynes maximum-entropy endpoint of the duality emergence. This single number is the bridge between the ultraviolet Planck anchor and the infrared vacuum floor. The Hubble rate is the infrared endpoint of the emergence window,

$$H_0 = M_{\rm P}\sqrt{\pi}\, e^{-\int \gamma_M}. \tag{C.97}$$

The dark-energy fraction is

$$\Omega_\Lambda = \frac{\rho_\Lambda}{3H_0^2 M_{\rm P}^2} = \frac{2}{3} + \frac{r_{23}}{3\pi} = 0.68504. \tag{C.98}$$

The algebra is exact. With $H_0^2 = M_{\rm P}^2\pi\, e^{-2\int\gamma_M}$ and $e^{-2\int\gamma_M} = \rho_\Lambda/(M_{\rm P}^4(2\pi + r_{23}))$, the ratio is $\rho_\Lambda/(3H_0^2 M_{\rm P}^2) = (2\pi + r_{23})/(3\pi) = 2/3 + r_{23}/(3\pi)$, the maximum-entropy closure

2/3 plus the hypercharge-trace correction $r_{23}/(3\pi)$. The endpoint residual is the remaining Hamiltonian fraction,

$$\Omega_\Sigma = 1 - \Omega_\Lambda - \Omega_b = 0.26570. \tag{C.99}$$

In the linear cosmological propagation this residual is used as a conserved cold source, $\Omega_{\rm cdm} := \Omega_\Sigma$. The reference column is the conventional cold-dark-matter fraction used for comparison, not an input to the closure. The microwave temperature is the photon floor set by the lightest neutrino,

$$T^{\rm raw}_{\rm CMB} = \frac{m_{\nu_1} r_{12}}{\pi}(1 - s_0\kappa)(1 - \tau\Delta_s) = 2.7310\ \mathrm{K}, \tag{C.100}$$

with $r_{12} = 3/10$ and $\Delta_s = 1/2$ the scalar conformal weight. The squash correction $(1-s_0\kappa)$ is the same level transfer as the electroweak scale. The photon floor carries $-s_0\kappa$ while the seesaw mass carries $+s_0\kappa$. The physical content is that the lightest fermion sets the photon temperature, the neutrino mass scaled by the hypercharge ratio $r_{12} = m_1/m_2 = (N_L - N_R)/\sum Y^2$, the pure content ratio of the chiral difference over the hypercharge capacity, the geometric factor $1/\pi$, and the chiral correction $1-\tau\Delta_s$ with $\Delta_s = (d-2)/2 = 1/2$ the scalar conformal weight, the same Δ_s that enters the proton-mass correction of Section C.15. The redshift equals the spectrum, in the sense that the photon temperature is the lightest neutrino mass scaled by the content ratio and the period factor. The finite endpoint photon zero mode multiplies the raw floor by

$$C_\gamma = 1 - \frac{\tau}{\pi^2}, \qquad T_{\rm CMB} = C_\gamma T^{\rm raw}_{\rm CMB} = 2.72547\ \mathrm{K}. \tag{C.101}$$

This correction is used for the CMB-monopole comparison and for the linear CMB comparison. The internal baryon-density closure remains evaluated with the raw photon floor and the raw Sakharov baryon asymmetry,

$$\Omega_b = \frac{\eta^{\rm raw}_B n^{\rm raw}_\gamma m_p}{\rho_{\rm crit}}, \qquad n^{\rm raw}_\gamma = \frac{2\zeta(3)}{\pi^2}\left(T^{\rm raw}_{\rm CMB}\right)^3,$$

so no factor C_γ^{-3} is multiplied into η_B in the computation of Ω_b.

The primordial spectral tilt is the torsion modulus times the RP^3 Weyl window-weight ratio,

$$1 - n_s = \tau \cdot \frac{7}{4} = 0.035, \tag{C.102}$$

where the numerator is the Weyl-law degree-of-freedom count of the internal fields and the denominator is the four-level window normalisation,

$$\frac{7}{4} = \frac{1+2+1+3}{d+1}, \qquad d = 3.$$

The conformal-curvature expression gives the same rational value as a cross-check,

$$\frac{7}{4} = 1 + \xi R_{\rm LC} L^2 = 1 + \frac{1}{8}6.$$

Here $1, 2, 1, 3$ are the scalar, transverse-vector, spinor-projected and TT Weyl weights. The primordial curvature amplitude is the spin-$\frac{1}{2}$ Gaussian zero-point suppressed by the window period,

$$\Delta_R^2 = \frac{1}{2}\Big(\frac{1}{2\pi}\Big)^2 e^{-2\pi k L_{\rm CMB}}(1-\tau\kappa) = 2.10 \times 10^{-9}, \tag{C.103}$$

with $\Delta_0^2 = (1/2)(1/2\pi)^2$ the spin-$\frac{1}{2}$ zero-point, $e^{-2\pi k L_{\rm CMB}}$ the window suppression, and $(1-\tau\kappa)$ the chiral-squash correction, with $kL_{\rm CMB} = kL(1-\tau/4)$ the window at the microwave pivot, so the amplitude closes without inflation. The tensor-to-scalar ratio is the Euclidean zero-point of the tensor sector, the same period factor squared without the spin-$\frac{1}{2}$ half that enters Δ_0^2,

$$r = \Big(\frac{1}{2\pi}\Big)^2 = 0.02533, \tag{C.104}$$

so the tensor amplitude is $\Delta_t^2 = r\,\Delta_s^2 = 5.32 \times 10^{-11}$. The value sits below the current combined upper bound $r < 0.036$ [20], so it is consistent with observation rather than a signed deviation. It is therefore not entered as a signed-deviation row in Appendix A; it is stated here and in Section 10.4 as a bound comparison.

C.15 QCD and the mass gap

The QCD sector closes in three steps. First the condensate energy of the long-root is

$$\Delta E = \frac{1}{8} M_{\rm G}, \tag{C.105}$$

the curvature gap $K = 8/3$ times the conformal coupling $\xi = 1/8$. The generator mass $m_{\rm gen} = g_2(2\tau)M_{\rm G}/\sqrt{2}$ sets the running initial condition, and the two-loop QCD running with the top matching gives $\Lambda_{\rm QCD}$. The running uses the two-loop gauge beta functions of Section C.5, whose coefficients are derived there from the group-theoretic content, the Casimir and Dynkin indices of the representations, so no coefficient is imported from a table. The running equation itself, $\beta_g = g^3/(16\pi^2)(b_i + B_{ij}g_j^2/(16\pi^2) - A_i y_t^2/(16\pi^2))$, is the standard form of the renormalisation group, and it enters at this single point as the statement that the gauge coupling evolves logarithmically with the scale. The independent content is the initial condition $g_3(M_{\rm G})$, the geometric Yukawa $y_0 = 1$, and the coefficients b_i and B_{ij} of Section C.5. The two-loop terms shift $\Lambda_{\rm QCD}$ at the level of a few percent, and the one-loop content of the running is fixed entirely by these coefficients. Second the topological gap is the $l = 2$ scalar mode of RP^3, whose eigenvalue is $\lambda_{\rm glue} = 8/L^2 > 0$, the lowest colour-singlet level. Third the glueball tower is the two-gluon bound-state spectrum,

$$\lambda = 2\lambda_{\rm gluon} + C_2(J) + n(N_g\xi), \qquad N_g\xi = 1, \tag{C.106}$$

with $\lambda_{\rm gluon} = 4/L^2$ the Killing eigenvalue. The lightest glueball is

$$m_G = 8\Lambda_{\rm QCD}, \tag{C.107}$$

and the tower ratios are $2^{++}/0^{++} = \sqrt{2}$, $0^{++*}/0^{++} = 3/2$, $0^{-+}/0^{++} = \sqrt{17/8}$. The proton mass is the constituent-quark content with the conformal-duality correction,

$$m_p = N_c\Delta_f\Big(1 - \frac{1}{N_g^2\Delta_s}\Big)\Lambda_{\rm QCD}\Big(1 + \tau\kappa\sum Y^2\Delta_s\Big) = \frac{279}{64}\Lambda_{\rm QCD}\Big(1 + \frac{5\tau\kappa}{3}\Big), \tag{C.108}$$

where $1/(N_g^2\Delta_s) = 1/32 = \xi/4$ follows from the two identities $N_g\xi = 1$ and $N_g\Delta_s = 2(d-1)$. The coefficient is the product of the three constituent quarks, $N_c = 3$, the fermion conformal weight $\Delta_f = d/2 = 3/2$, and the conformal-duality factor $31/32$, so $(9/2)(31/32) = 279/64$. The scheme bracket $(1+\tau\kappa\sum Y^2\Delta_s) = (1+5\tau\kappa/3)$ is the correction between the constituent-quark scale and the $\overline{\mathrm{MS}}$ scale, with $\sum Y^2\Delta_s = (10/3)(1/2) = 5/3$ the hypercharge capacity times the scalar conformal weight. The string tension and the deconfinement temperature are

$$\sigma = \frac{\lambda_{\mathrm{TT}}}{\pi}\Lambda_{\mathrm{QCD}}^2 = \frac{14}{\pi}\Lambda_{\mathrm{QCD}}^2, \qquad T_d = \frac{\lambda_{\mathrm{vector}}}{N_c}\Lambda_{\mathrm{QCD}}(1-\tau\kappa) = \frac{4}{3}\Lambda_{\mathrm{QCD}}(1-\tau\kappa), \tag{C.109}$$

with the chiral-squash factor $(1-\tau\kappa)$ on the deconfinement scale (deconfinement is accompanied by chiral restoration, so the scale inherits the chiral-squash content), and the self-consistent ratio

$$\frac{\sigma}{T_d^2} = \frac{14}{\pi}\frac{9}{16}(1-\tau\kappa)^{-2} = \frac{126}{16\pi(1-\tau\kappa)^2} = 2.6242, \tag{C.110}$$

the content ratio $126/(16\pi) = 2.5068$ softened by the factor $(1-\tau\kappa)^2$. The QCD scale itself carries the correction $(1-s_0\kappa/N_g)$ for the Yukawa difference between the scale-invariant geometric Yukawa $y_0 = 1$ and the running Standard Model top Yukawa in the two-loop gauge beta function, applied at the extraction point. This factor is part of the displayed scale definition.

C.16 Primordial nucleosynthesis

The six constants of the weak-rate freeze-out are derived from the closed content, not taken from observation. The Fermi constant is $G_F = 1/(\sqrt{2}v^2)$ from the closed electroweak scale. The axial coupling is the conformal-weight form

$$g_A = \frac{N_g\Delta_s}{\pi} = \frac{2(d-1)}{\pi} = \frac{4}{\pi}. \tag{C.111}$$

This is the axial coupling of the closed weak-rate model, the generator count $N_g = 2(d-1) = 8$ times the scalar conformal weight $\Delta_s = (d-2)/2 = 1/2$ over the Euclidean period π, the same conformal-gauge content that closes the gauge sector of Section C.4. It is not the nucleon axial charge of the standard model. It is the conformal-weight axial coupling used in the internal weak-rate freeze-out, not substituted into an external nucleon matrix element. The electromagnetic self-energy is $\Delta_{\mathrm{EM}} = (1-1/(2\pi))\alpha_{\mathrm{em}}(0)\Lambda_{\mathrm{QCD}}$, where the low-energy fine-structure constant is itself derived from the closed content. The fine-structure constant at the internal Z mass is $\alpha^{-1}(M_Z) = 128.208$ (Section 5.2, the two-loop geometric renormalisation group, the same running as $\alpha_s(M_Z)$), and the one-loop QED vacuum polarisation of the same content runs it down to zero momentum,

$$\frac{1}{\alpha_{\mathrm{em}}(0)} = \frac{1}{\alpha(M_Z)} + \frac{2}{3\pi}\sum_f Q_f^2 N_c^f \ln\frac{M_Z}{m_f}, \tag{C.112}$$

with the leptons on the internal ladder masses $(m_e,\ m_\mu,\ m_\tau)$, the c and b quarks on the internal masses, and the light quarks $(u,\ d,\ s)$ frozen at the internal confinement scale

$\Lambda_{\rm QCD}$. This gives $\alpha_{\rm em}^{-1}(0) = 137.049$ against the observed 137.036, a deviation of $+0.010\%$. No external fine-structure constant enters. The radiative correction is $\delta_R = 1+(1-\tau)/(8\pi)$. The neutron-proton mass difference is the quark mass difference minus the self-energy,

$$\Delta m_{np} = (m_d - m_u) - \Delta_{\rm EM}, \tag{C.113}$$

with the quark masses from the ladder of Section C.7. The neutron lifetime is

$$\tau_n = \frac{2\pi^3}{G_F^2 |V_{ud}|^2 (1+3g_A^2) m_e^5 f\, \delta_R}, \tag{C.114}$$

with $|V_{ud}| = \sqrt{1-|V_{us}|^2}$ and f the phase-space integral. The phase-space factor $f = \int F(Z,W)\, W\, p\, (W_0 - W)^2\, dW$ is the spectral integral of the weak-decay kinematics, the Fermi Coulomb function $F(Z,W)$ (with $Z = 1$ for the proton) times the phase-space weight $W\, p\, (W_0 - W)^2$ integrated over the electron energy, computed numerically to full precision, with $W_0 = \Delta m/m_e$ the endpoint. It is a pure-kinematics spectral sum with no continuous free parameter and no imported decay constant. The recoil and weak-magnetism corrections of the standard extraction are absent and shift f by only -0.2%. The dominant sensitivity is the endpoint weight. The internal Δm_{np} (observed 1.2933 MeV) enters $(W_0 - W)^2$ with a roughly four-fold amplification. The freeze-out temperature solves $\Gamma_{\rm weak}(T) = H(T)$, and the helium yield is

$$Y_p = \frac{2(n/p)_{\rm BBN}}{1+(n/p)_{\rm BBN}}, \qquad (n/p)_{\rm BBN} = e^{-\Delta m/T_f}\, e^{-t/\tau_n} = 0.2514. \tag{C.115}$$

The effective neutrino species is $N_{\rm eff} = 3 + \sqrt{3}/(2\pi)^2 = 3.0439$. The five nuclear constants close through the conformal-weight form, the 2π Euclidean period, and the torsion content ratio, with no QCD loop and no relativistic correction. The single perturbative ingredient is the one-loop QED running inside $\alpha_{\rm em}(0)$, evaluated on the closed masses and the closed $\alpha(M_Z)$.

C.17 Correction-factor inventory

Several displayed formulas of the preceding subsections carry correction factors of the schematic form $(1 \pm c\,\tau\kappa)$ or $(1 \pm c\, s_0\kappa)$, with $s_0 = 2\tau$ the squash amplitude and κ the isometry-breaking normalisation of Section C.4. These factors are of order a few percent, and they are collected here in one place, together with the physical quantity each one modifies. The factors are:

- $(1 - 4s_0\kappa)$, the dark-energy weight of Section C.14. The seesaw mass m_{ν_1} carries $+s_0\kappa$ through the level transfer, so $m_{\nu_1}^4$ carries $+4s_0\kappa$, and the weight cancels it, keeping $\rho_\Lambda = Y_u m_{\nu_1}^4 (1 - 4s_0\kappa)$ conserved under the $J = 2$ squash transfer.

- $(1 - s_0\kappa)$, the squash level-transfer factor on the electroweak scale v of Section C.11 and on the raw microwave floor $T_{\rm CMB}^{\rm raw}$ of Section C.14. Both carry $-s_0\kappa$ while the seesaw mass carries $+s_0\kappa$.

- $(1+s_0\kappa)$, the seesaw level-transfer factor on the heaviest neutrino scale m_{ν_3} of Section C.10.

- $(1-\tau\kappa)$, the chiral-squash factor on the deconfinement temperature T_d of Section C.15 (deconfinement is accompanied by chiral restoration) and on the curvature amplitude Δ_R^2 of Section C.14.

- $(1+\tau\kappa\sum Y^2\Delta_s) = (1+5\tau\kappa/3)$, the scheme correction of the proton mass m_p of Section C.15, between the constituent-quark scale and the $\overline{\mathrm{MS}}$ scale, with $\sum Y^2 = 10/3$ the hypercharge capacity and $\Delta_s = 1/2$ the scalar conformal weight.

- $(1-s_0\kappa/N_g)$, the Yukawa-difference factor applied at the extraction of Λ_{QCD} in Section C.15, between the scale-invariant geometric Yukawa $y_0 = 1$ and the running Standard Model top Yukawa inside the two-loop gauge beta function.

- $C_\gamma = 1-\tau/\pi^2$, the finite endpoint photon zero-mode factor multiplying $T_{\mathrm{CMB}}^{\mathrm{raw}}$ in Section C.14. This factor is used for the microwave-monopole comparison and for the linear CMB comparison. The internal Ω_b closure uses $T_{\mathrm{CMB}}^{\mathrm{raw}}$ and the raw Sakharov η_B, so no C_γ^{-3} rescaling is applied inside the computation.

- $f = \int F(Z,W)\,W\,p\,(W_0-W)^2\,dW$, the phase-space integral of the neutron lifetime of Section C.16, a pure-kinematics spectral sum rather than a factor of the $1\pm c\,\tau\kappa$ type.

The first six factors are built on the derived isometry-breaking normalisation κ and the squash amplitude $s_0 = 2\tau$ (the geometric and torsion content of Section C.4). The photon zero-mode factor is instead an endpoint finite-size correction. The dark-energy weight $(1-4s_0\kappa)$ and the squash factors on m_e and $T_{\mathrm{CMB}}^{\mathrm{raw}}$ are inherited algebraically from the base factor once its sign is fixed. The coefficients of v, m_{ν_3}, T_d, Δ_R^2, m_p, and α_s reduce to the following sequence of field-equation and geometric-moment steps:

1. *Einstein–Cartan action* (explicit). The torsion Lagrangian $L_{\mathrm{tors}} = a\,T^2 + b\,T^{bac}T_{abc} + c\,(T^a{}_{ab})^2$ with $b = 4a$ (the Holst/Immirzi algebraic-torsion condition), the order-parameter Landau potential $V(\varphi) = \frac{1}{2}\xi(R-R_c)\varphi^2 + \frac{\lambda}{4}\varphi^4$ ($\xi = 1/8$, $R_c = 6/\pi$), and the surviving $\mathrm{U}(1)_Y$ coupling whose Killing normalisation is the geometric integral $\kappa^2(s) = [V_3(s)/V_3(0)](g^{33})^3 = (1+s)/(1-2s)^{5/2} = \int\sqrt{g}\,(g^{33})^3 d^3x/\int\sqrt{g}\,d^3x$ (the third moment, namely two inverse-metric factors from the field strength and one from the polarisation).

2. *Torsion field equation.* $\delta S/\delta K = 0$ gives $\tau/L = \kappa^2 j_5$ with $j_5 = \langle\chi\rangle/\Pi_{\mathrm{ren}} = (N_L - N_R)/(N_f\sum Y^2)$, so $\tau = 1/50$ (Section C.2).

3. *Squash field equation.* $\delta S/\delta\varphi = 0$ gives $s_0 = n_{\mathrm{broken}}\tau = 2\tau$ (Section C.4).

4. *Geometric moment $c_Q = a_Q\,r_Q$.* Each quantity responds to the traceless shear $ds^2 = (1+s)\sigma_1^2 + (1+s)\sigma_2^2 + (1-2s)\sigma_3^2$ by the first-order integral $c_Q = d\ln Q/ds|_{s=0} = 2n_3 - n_{\mathrm{eq}}$ (the weighted count of the squash deformation across the content), factorised

as an amplitude fraction a_Q (geometric = 1 coupling to both broken generators $n_{\text{broken}} = 2$, chiral = 1/2 coupling to the single chiral asymmetry $N_L - N_R = 1$) times a content ratio r_Q.

Each coefficient is then $Q \to Q\,(1 + c_Q\, s_0\kappa)$ with $s_0\kappa = N_g\tau\kappa/(d+1)$:

- v, V_{cb} carry $-s_0\kappa$ ($a_Q = 1$, $r_Q = -1$). A geometric quantity on the full geometry responds to the full amplitude s_0 with the compression sign $-$. The dilaton-stop lengthens by $+s_0\kappa$, so $v = M_G e^{-\varphi}$ shrinks by $(1 - s_0\kappa)$.
- m_{ν_3} carries $+s_0\kappa$ ($a_Q = 1$, $r_Q = +1$). The seesaw relation $m_{\nu_3} = v^2/M_R$ ($M_R = k_{\text{GUT}}/(2\pi)^2$) combines with the level-transfer conservation $v\, m_{\nu_3} = v^3/M_R = \text{const}$. The shifts $v \to (1 - s_0\kappa)$ and $M_R \to (1 - 3s_0\kappa)$ force $m_{\nu_3} \to (1 + s_0\kappa)$.
- T_d and Δ_R^2 carry $-\tau\kappa$, and V_{ub} carries $+\tau\kappa$ ($a_Q = 1/2$, $r_Q = \mp 1$). The chiral level couples to the chiral asymmetry $\tau = s_0/2$ (the amplitude fraction $a_Q = (N_L - N_R)/n_{\text{broken}} = 1/2$). The sign is $-$ for chiral restoration (T_d, Δ_R^2) and $+$ for the cross-generation element (V_{ub}).
- m_p carries $+\tau\kappa \sum Y^2\Delta_s = +5\tau\kappa/3$ ($a_Q = 1/2$, $r_Q = \sum Y^2\Delta_s = 5/3$). The constituent-vs-MSBAR scheme couples the chiral asymmetry to the hypercharge capacity $\sum Y^2 = 10/3$ and the scalar conformal weight $\Delta_s = 1/2$.
- α_s carries $-s_0\kappa/N_g = -s_0\kappa\,\xi$ ($a_Q = 1$, $r_Q = -1/N_g$). The Yukawa difference $y_0 = 1$ vs the running top Yukawa is a conformal effect normalised by the conformal coupling $\xi = 1/N_g$ of the duality $N_g\xi = 1$.

The amplitude fraction a_Q and the power $r_Q = 4$ are derived from the broken-generator / chiral-asymmetry / $m_{\nu_1}^4$ content, and the seesaw $r_Q = +1$ from the level-transfer conservation. The two remaining content ratios are each reduced to an independent Einstein–Cartan field equation, the direct analogue of the torsion equation $\tau/L = \kappa^2 j_5$ of Section C.2:

1. *Constituent scheme* (m_p, $r_Q = \sum Y^2\Delta_s = 5/3$). The constituent-quark action of the χSB sector,

$$S_q = \int \sqrt{g}\, d^3x \left[\bar\psi\,(i\gamma^a e_a{}^\mu D_\mu)\psi + G_s\,(\bar\psi\psi)^2\right], \qquad D_\mu = \partial_\mu + ig_1 Y A_\mu, \tag{C.116}$$

with the $\mathrm{U}(1)_Y$ covariant derivative, has the gap equation $\delta\Gamma/\delta\langle\bar\psi\psi\rangle = 0$ giving $m_q = -2G_s\langle\bar\psi\psi\rangle$. Its first-order squash response is

$$\frac{\delta m_p}{m_p} = \tau\,\kappa\,\Delta_s \sum_c c\,Y^2 = \frac{5\tau\kappa}{3}, \qquad \sum_c c\,Y^2 = 6\,(\tfrac{1}{6})^2 + 3\,(\tfrac{2}{3})^2 + 3\,(-\tfrac{1}{3})^2 + 2\,(-\tfrac{1}{2})^2 + 1 = \frac{10}{3}, \tag{C.117}$$

the chiral asymmetry τ times the Killing normalisation κ times the scalar conformal weight $\Delta_s = (d-2)/2 = 1/2$ of the condensate times the hypercharge spectral sum $\sum Y^2 = 10/3$ (the first non-zero hypercharge moment).

2. *Yukawa difference* (α_s, $r_Q = -1/N_g = -\xi$). The two-loop gauge beta carries the top-Yukawa term through the invariant $Y_4(F) = (1/d(G_a))\,\mathrm{Tr}[C_2^{(a)}(F)Y^aY^{+a}]$ with $A_i = 2\kappa Y_4/y_t^2$ (Luo–Wang–Xiao, Eq. 31, with $\kappa = 1/2$ for Weyl fermions), the explicit trace $\mathrm{Tr}[C_2(F)YY^+] = 6y_t^2[C_2(Q_L) + C_2(u_R)]$ with $6 = 2 \times 3$ the weak-doublet $\times$ colour-triplet factor. The geometric Yukawa $y_0 = 1$ (the SO(4) diagonal overlap) exceeds the running y_t, and the difference is a conformal effect normalised by the Yamabe coupling

$$\delta \ln \alpha_s = -s_0\,\kappa\,\xi = -\frac{s_0\kappa}{N_g}, \qquad \xi = \frac{d-2}{4(d-1)} = \frac{1}{N_g} = \frac{1}{8}, \tag{C.118}$$

the squash amplitude s_0 times the Killing normalisation κ times the conformal coupling ξ of the duality $N_g\xi = 1$ (the same ξ that closes g_1's $5/8 = \sum Y^2\Delta_f\xi$).

Each charge is a *computed* moment from an explicit field equation, no coefficient a fitted input. The factors are disclosed here as displayed formulas so that the correction inventory and the numerical table carry the same content.

D Theoretical sensitivity

D.1 Sensitivity of the hierarchy

The central values of this paper are fixed within the adopted closure prescription in the sense of Section 2.3. No continuous free parameter and no adjusted phenomenological coefficient enter the chain once the structural conventions are fixed. Exactness of the central values, however, does not by itself quantify how the outputs respond to the inputs and conventions that carry the chain. This appendix records that response in full. The inputs separate into four layers. The single observed anchor G_N carries the experimental relative uncertainty of order 2×10^{-5}, which propagates only into the dimensionful quantities (each with elasticity $+1$) and leaves every dimensionless quantity unchanged, the M_P-rescaling invariance. The structural numbers 2, π, 3/2, 1/8 and the geometric factors are exact. The remaining two layers, namely the window-capacity fixed point kL and the torsion modulus τ, are the derived intermediates whose defining conventions carry the genuine theoretical sensitivity, and it is these that the analysis quantifies. These central values are therefore fixed, but the interpretation of that fixedness is limited. The sensitivity analysis of this appendix is the quantitative statement of that limitation.

Table 10 gives the elasticity matrix $d\ln O/d\ln p$ for the principal outputs and the three inputs $p = kL, \tau, M_\mathrm{P}$, evaluated by central finite difference on the closed forms of Appendix C. The dominant sensitivities are those with respect to kL. The electroweak scale v, the electron mass m_e, the heaviest neutrino mass m_{ν_3}, the dark-energy density ρ_Λ and the Hubble rate H_0 all carry large negative elasticities, because they are built from exponentials of kL. The relations are $v \propto e^{-4\pi kL}$, $m_e \propto e^{-20kL}$, $m_{\nu_3} \propto v^2$, $\rho_\Lambda \propto m_{\nu_1}^4$, $H_0 \propto m_{\nu_1}^2$. A one-percent shift of kL therefore moves v by -32%, m_e by -50%, and ρ_Λ by -262%. The mass ratios carry only order-unity elasticities, and the dark-energy fraction $\Omega_\Lambda = 2/3 + r_{23}/(3\pi)$ is a pure content ratio whose elasticity is exactly zero. The hierarchy lives in the exponentials of kL, not in Ω_Λ.

	Elasticity $d\ln O/d\ln p$		
Quantity O	$p = kL$	$p = \tau$	$p = M_{\mathrm{P}}$
M_{G}	-1.0	0.0	$+1.0$
$g_2(M_{\mathrm{G}})$	-2.5	0.0	0
m_t/m_c	$+5.0$	-0.1	0
v	-32.3	-0.1	$+1.0$
m_e	-49.9	-0.1	$+1.0$
$m_{\nu 3}$	-64.7	-1.1	$+1.0$
ρ_Λ	-261.6	-4.5	$+4.0$
H_0	-129.7	-2.2	$+1.0$
Ω_Λ	0.0	0.0	0

Table 10. The elasticity matrix, which gives the percentage change of each output per one-percent change of kL, τ, or M_{P}. The large negative entries in the kL column reflect the exponential hierarchy $e^{-4\pi kL}$, e^{-20kL}, and their powers.

Convention (1% shift)	$\to kL$	$\to v$	$\to m_e$	$\to \rho_\Lambda$
Kernel exponent $(1-y)^p$, $p = 2$	$+0.8\%$	-26%	-40%	-212%
Threshold $4/27$	-1.0%	$+32\%$	$+50\%$	$+262\%$
Window kL	—	-32%	-50%	-262%

Table 11. The convention chain, showing how a one-percent shift of each upstream convention propagates into the fixed point kL and then into the hierarchy.

The elasticity into kL traces back to the three conventions that fix the fixed point, through the spectral-pole condition $V_3\,\Pi^2/(32\pi^2) = 4/27$ of Section C.1. Table 11 records this chain. The kernel exponent is the transverse-traceless kernel $K_{\mathrm{TT}} = (1-y)^2$ regarded as $p = 2$ of the one-parameter family $(1-y)^p$. The threshold $4/27$ is the maximum of the mass-weighted kernel $y(1-y)^2$. A one-percent shift of the kernel exponent moves the threshold by $p\ln(p/(p+1)) = -0.81\%$, the fixed point by $+0.81\%$ (the elasticity $d\ln kL/d\ln(4/27)$ is exactly -1), and v by -26%. The three conventions therefore sit at the top of a steep amplification chain.

The status of the three conventions is not uniform. The kernel $K_{\mathrm{TT}} = (1-y)^2$ is the standard spectral density of the transverse-traceless projection of the improved energy-momentum tensor, $K_{\mathrm{TT}} = k^4/(k^2+m^2)^2$, and the substitution $y = m^2/(k^2+m^2)$ turns it into $(1-y)^2$. This is an established consequence of the spin-2 Ward identity. The kernel vanishes at $m^2 = 0$, as a massless graviton is transverse-traceless, and is inherited from the standard spin-2 tensor structure. The threshold $4/27$ is the exact maximum of $y(1-y)^2$ at $y = 1/3$, a calculus identity. The two choices that precede it, namely the mass weight y, which measures how far the torsion has activated the TT channel, and the selection of the maximum as the normalisation, are physically motivated conventions, not unique deductions. The window capacity $2\pi kL^4$ is a notation rather than an independent spectral sum. It enters the bare field equation as $1/(2\pi kL^4)$ and the hypercharge screening as $2\pi kL^4$, so it cancels exactly and drops out of τ. The torsion modulus is therefore fixed by the content ratio alone, $\tau = (N_L - N_R)/(N_f \sum Y^2) = 1/50$, an integer identity, and does not depend on the specific value of $2\pi kL^4$. This is the precise sense in which the hierarchy is closed within the prescription but convention-sensitive. The central values carry no free parameter, yet they rest on three upstream conventions. One of them, the transverse-traceless kernel, is fixed by the spin-2 tensor structure. The second, the threshold 4/27, is an exact extremum with a conventional selection. The third, the window capacity $2\pi kL^4$, is a notation that cancels out of the physical content.

The selection of the maximum can be stated explicitly. The mass-weighted density $y(1-y)^2$ has a unique interior stationary point, where $[y(1-y)^2]' = (1-y)(1-3y)$ vanishes on $(0,1)$ only at $y = 1/3$, where the second derivative is -2, so 4/27 is the unique interior extremum, the endpoints $y = 0, 1$ giving zero. The maximum is the only non-trivial scale-invariant datum carried by the kernel. It is also the point of maximal activation of the transverse-traceless channel, the mass fraction $y = 1/3$ balancing the two kinetic fractions $(2/3)^2$, which is the point at which the massless pole appears. The unbiasedness clause of the coarse-graining principle, together with the maximum-entropy theorem of the companion paper [1], selects the configuration of maximal spectral weight. A complementary content reading gives the same number: the stationary condition $y = 1/3$ balances one mass fraction against two kinetic fractions, $(1/3)(2/3)^2$, the content of a spin-2 mode. Thus the longitudinal activation by the torsion mass is balanced against the two transverse polarisations carried by the two powers of k^2 in the kernel. The value 4/27 is therefore treated as the spin-2 content datum in the present prescription, alongside the integer content identities $N_L = N_g = 8$ and $\sum Y^2 = 10/3$ that fix the other structural numbers.

The kernel is likewise fixed by the stated spin-2 requirement. The transverse-traceless projection of a spin-2 mode is fixed by the Ward identity (transversality) together with tracelessness, and its spectral density is $K_{\mathrm{TT}} = k^4/(k^2 + m^2)^2$. The two powers of k^2 are the two transverse polarisations and the denominator is the propagator squared, so with $y = m^2/(k^2 + m^2)$ the kernel is exactly $(1-y)^2$. Within this one-variable TT kernel, no further functional coefficient is available.

The flatness of the extremum softens only the first link of the chain, and the remainder stays first-order. Since 4/27 is a stationary point, a shift of the mass fraction by Δy moves the threshold only at second order, where $y(1-y)^2 = 4/27 - (y - 1/3)^2 + \mathcal{O}((y - 1/3)^3)$ locally, the curvature being $\left|[y(1-y)^2]''_{1/3}\right| = 2$. But the threshold acts on the fixed point kL at first order (the elasticity $d \ln kL/d \ln(4/27) = -1$), and kL acts on the hierarchy at first order through the exponentials of Table 10. The two links downstream of the threshold are therefore not softened by the flatness of the extremum. Table 12 records the response of the principal outputs to representative variations of the upstream quantities.

The conclusion is therefore a sensitivity statement, not an additional fit. The threshold is flat, so small displacements of y enter the threshold only quadratically. The hierarchy downstream is exponential, however, so variations of the upstream prescription are amplified into the dimensionful outputs. The central values are fixed within the adopted closure

Convention variation	$\to kL$	$\to v$	$\to m_e$	$\to \rho_\Lambda$
$y \to 1.10\,y$	$+0.73\%$	-23%	-36%	-190%
$y \to 1.30\,y$	$+6.1\%$	-196%	-303%	-1590%
$p \to 1.01\,p$	$+0.81\%$	-26%	-41%	-212%

Table 12. Response of the hierarchy to representative variations of the mass fraction y and of the kernel exponent p. The flatness of the extremum softens only the first link ($y \to$ threshold). The threshold-to-kL and kL-to-output links remain first-order, so a 10% variation of y moves v by about 23%.

prescription, while Table 12 gives representative numerical responses to variations around that prescription.

The representative variations of Table 12 are supplemented by a scheme comparison. The scheme used here is the smooth Gaussian window, the coarse-graining envelope fixed by the unbiasedness clause of the coarse-graining principle. The clause requires the scale flow to carry no preferred spectral weight, and the Gaussian is the maximum-entropy envelope of the spectral datum (the maximum-entropy theorem of the companion paper [1]). A sharp Litim step [25] gives a useful comparison because it sharply separates retained and discarded modes. Re-solving the fixed point with that step gives $kL = 1.09$ against the Gaussian $kL = 2.49$, a spread of -56%. Under the Litim step the window capacity is $(1.09)^2 = 1.19 < 1.5$, so no spinor mode fits the window and the window-capacity theorem of Section C.1 would return a generation count of zero, in conflict with the chiral content that the companion paper derives from the same principle. The sharp-step comparison is therefore recorded as a scheme-sensitivity check, while the Gaussian window remains the prescription selected by the unbiasedness clause.

D.2 The spectral-scale flow and the renormalisation group

The spectral-scale flow is the evolution of the spectral datum with the coarse-graining scale, parametrised by the trajectory $L(k) = C/k$ of Section C.1. Its relation to the standard renormalisation group is as follows. Below the emergence scale M_{G} the spectral sums are dominated by the zero modes and the lowest Kaluza–Klein excitations, the regime in which the spectral-sum definition and the standard measure-theoretic formulation coincide, as proved in Appendix C of the companion paper [1]. In this regime the scale flow of the effective couplings reduces to the standard two-loop renormalisation group equations of Section C.5, whose coefficients are the group-theoretic content of the construction rather than imported values. The spectral-scale flow therefore matches the renormalisation group equations in the precise sense that both describe the same low-energy running of the effective couplings in the shared regime. The flow supplies the spectral datum and the boundary condition at M_{G}, while the standard equations reproduce the running below M_{G} with coefficients fixed by the same content. This is an identification in the shared low-energy regime, not a claim of global equivalence between the two formulations.

E Flavour and CP structures (formal closure)

The relations recorded in this appendix close the flavour sector at the level of the spectral data, and two kinds of content are distinguished explicitly. (i) The neutrino masses and the baryon asymmetry are evaluated numerically, because they are the two values that the cosmological closure of Section 8 consumes downstream and that must be carried forward for the chain to close. Their numbers are computed from the closure forms below, not taken from observation. (ii) The remaining relations, namely the PMNS mixing angles, the CKM phase, and the Jarlskog invariant, are stated as exact closure forms without a decimal evaluation and without comparison against any measured value. Their evaluation depends on the effective matching prescriptions of Appendix C and does not affect the core closure of Sections 5–9. No relation in this appendix is a precision comparison against a measured value.

E.1 Neutrinos

The neutrino sector is closed by the Weinberg dimension-five operator. The heaviest mass is

$$m_{\nu_3} = \frac{v^2(2\pi)^2}{k_{\rm GUT}}(1 + s_0\kappa) = 0.0502 \text{ eV},$$

the electroweak scale squared times the two-period factor divided by the breaking scale, with the squash correction $s_0\kappa$ of Section C.4. The lighter masses descend by the hypercharge-trace hierarchy ratios

$$\frac{m_{\nu_1}}{m_{\nu_2}} = \frac{1}{\mathrm{Tr}(Y^2)} = \frac{3}{10}, \qquad \frac{m_{\nu_2}}{m_{\nu_3}} = \frac{1}{\sqrt{3}\ \mathrm{Tr}(Y^2)} = \frac{3}{10\sqrt{3}},$$

so that

$$m_{\nu_2} = m_{\nu_3}\, r_{23} = 0.0087 \text{ eV}, \qquad m_{\nu_1} = m_{\nu_3}\, r_{12} r_{23} = 0.0026 \text{ eV},$$

with $r_{12} = 3/10 = (N_L - N_R)/\sum Y^2$ and $r_{23} = 3/(10\sqrt{3})$. The tower is fixed by the two hypercharge-trace ratios, and its lightest member m_{ν_1} is the seed that sets the dark-energy density of Section 8.1.

For oscillation comparisons the mass-squared splittings, not the absolute masses themselves, are the directly measured quantities. The raw light-pair rest split is $m_{\nu_2}^2 - m_{\nu_1}^2 = 6.874 \times 10^{-5}$ eV^2. The solar propagation channel uses the already closed spectral-tilt dressing of the second light state,

$$m_{\nu_2}^{\rm osc} = m_{\nu_2}(1 + n_s^{\rm tilt}), \qquad n_s^{\rm tilt} \equiv 1 - n_s = \tau\frac{7}{4},$$

which gives

$$\Delta m_{21}^2 = (m_{\nu_2}^{\rm osc})^2 - m_{\nu_1}^2 = 7.412 \times 10^{-5} \text{ eV}^2, \qquad \Delta m_{31}^2 = m_{\nu_3}^2 - m_{\nu_1}^2 = 2.511 \times 10^{-3} \text{ eV}^2.$$

The tilt dressing is a comparison-layer propagation factor and does not replace the absolute masses used by the cosmological closure.

The mixing angles, in the near-tribimaximal form, are encoded by the same content ratios. Their numerical evaluation is not performed here:

$$\sin^2\theta_{12} = \frac{m_{\nu_1}}{m_{\nu_2}}, \qquad \sin^2\theta_{13} = \frac{\sqrt{3}}{2\,(2\pi)^2}, \qquad \sin^2\theta_{23} = \frac{1}{2} + \frac{\mathrm{Tr}(T_3^2)}{(2\pi)^2}.$$

The solar angle is the light-pair mass ratio, the reactor angle carries the two-period imprint times the internal-space factor $\sqrt{3}/2$, and the atmospheric angle carries the isospin trace over the Euclidean period. They are proposed by the same content ratios that fix the charged sector. These relations are treated as a phenomenological ansatz, not as precision comparisons.

E.2 CKM and CP

The quark-mixing closure is expressed by the Gatto relation and the left-right content ratio. The cross-generation elements follow from the mass ladder of Appendix C.7:

$$|V_{us}| = \sqrt{\frac{m_d}{m_s}}, \qquad |V_{cb}| = \frac{1}{2}\,e^{-\alpha_{\mathrm{up}}}(1 - s_0\kappa), \qquad |V_{ub}| = \frac{1}{2}\,e^{-2\alpha_{\mathrm{up}}}(1 + \tau\kappa),$$

the Cabibbo element being the Gatto dominant term and the two cross-generation elements the adjacent- and next-generation ladder suppressions. The CKM phase is the colour-number dilution of the left-right content ratio,

$$\delta_{\mathrm{CKM}} = \frac{8}{7}\frac{\pi}{N_c} = \frac{8\pi}{21},$$

and the Jarlskog invariant is the product

$$J = |V_{us}|\,|V_{cb}|\,|V_{ub}|\,c_{12}c_{23}\sin\delta.$$

The baryon asymmetry closes the same content through the Sakharov count,

$$\eta_B = \frac{J\,\alpha_W(v)^2}{56},$$

where $\alpha_W(v) = g_2(v)^2/(4\pi)$ is the electroweak-scale weak coupling of the geometric renormalisation group and $1/56 = 1/(N_L N_R)$ is the content count. This one quantity is evaluated, because it is the single member of this block that the baryon fraction of Section 8.4 consumes. With the ladder masses of Appendix C.7 the closure above gives

$$\eta_B = 6.09 \times 10^{-10}.$$

No external value enters. The quantity η_B is the evaluation of $J\,\alpha_W(v)^2/56$ on the closure above, and the Jarlskog invariant and the CKM phase are stated as structural relations rather than as precision comparisons.

These expressions complete the formal correspondence between the spectral datum and the flavour sector. The complete derivation of each closed form is recorded in Appendix C, Sections C.10 and C.13.

F Cosmological extension and BBN (formal closure)

The primordial nucleosynthesis calculation recorded here depends on the core outputs of Sections 5, 8, and 9, and on the effective relations of Appendix E (through the CKM unitarity). The closure forms of the weak-rate freeze-out are stated below, without a decimal evaluation and without comparison against any measured value. Their evaluation requires the effective scale choices of Appendix C and does not affect the core closure of Sections 5–9. They are stated as structural relations rather than as precision comparisons.

F.1 Primordial helium

The weak-rate freeze-out closes through six internal constants, none taken from observation. The axial coupling is the conformal-weight form, a closure relation of the content

$$g_A = \frac{N_g \Delta_s}{\pi} = \frac{4}{\pi},$$

the electromagnetic self-energy is

$$\Delta_{\mathrm{EM}} = \left(1 - \frac{1}{2\pi}\right)\alpha_{\mathrm{em}}(0)\Lambda_{\mathrm{QCD}},$$

where the low-energy fine-structure constant $\alpha_{\mathrm{em}}(0)$ is itself internal: it is obtained by running the closed $\alpha^{-1}(M_Z)$ of Section 5.2 down through the one-loop QED vacuum polarisation of the same content (Appendix C.16). The radiative correction is

$$\delta_R = 1 + \frac{1-\tau}{8\pi},$$

the neutrino-species correction is

$$\delta_N = \frac{\sqrt{3}}{3(2\pi)^2},$$

and the remaining two inputs are $|V_{ud}| = \sqrt{1 - |V_{us}|^2}$ from the CKM unitarity of Appendix E, and the phase-space integral f. The neutron-proton mass difference is

$$\Delta m_{np} = (m_d - m_u) - \Delta_{\mathrm{EM}},$$

and the neutron lifetime is the closed form

$$\tau_n = \frac{2\pi^3}{G_F^2 |V_{ud}|^2 (1 + 3g_A^2) m_e^5 \, f \, \delta_R},$$

with $G_F = 1/(\sqrt{2}\, v^2)$ from the closed electroweak scale. The freeze-out temperature solves $\Gamma_{\mathrm{weak}}(T_f) = H(T_f)$, and the helium yield and the effective neutrino number are the closed forms

$$Y_p = \frac{2(n/p)_{\mathrm{BBN}}}{1 + (n/p)_{\mathrm{BBN}}}, \qquad (n/p)_{\mathrm{BBN}} = e^{-\Delta m/T_f} e^{-t/\tau_n},$$

$$N_{\mathrm{eff}} = 3 + \frac{\sqrt{3}}{(2\pi)^2}.$$

The six constants close through the conformal-weight form, the two-period Euclidean imprint, and the torsion content ratio, with the single perturbative ingredient the one-loop QED running inside $\alpha_{\mathrm{em}}(0)$. The complete derivation of the closed forms is recorded in Appendix C, Section C.16.

G The acceleration scale and the endpoint residual

The maximum-entropy endpoint has two projections in the present construction. Its normal Hamiltonian projection is the conserved residual density Ω_Σ used as the cold source in the linear cosmological propagation of Section 8.5. Its local spatial projection defines the low-acceleration scale and response described below. The two projections are kept separate to avoid counting the same endpoint contribution twice.

G.1 The acceleration scale

The acceleration scale is

$$a_0 = \frac{cH_0}{2\pi}\sqrt{\frac{4}{3}},$$

the endpoint Hubble rate divided by the period, times the three-ball coefficient $\sqrt{4/3} = 2/\sqrt{3}$ of the spectral zero-mode infrared behaviour. This gives $a_0 = 1.204 \times 10^{-10}\ \mathrm{m\,s^{-2}}$ against the empirical Milgrom scale $1.2 \times 10^{-10}\ \mathrm{m\,s^{-2}}$, a deviation of $+0.36\%$. The relation $a_0 \sim cH_0/(2\pi)$ is the same order relation that the MOND phenomenology encodes empirically, and the coefficient $\sqrt{4/3}/(2\pi)$ is fixed by the spectral content rather than fitted. In the closed branch used here, a_0 is the endpoint value and is not replaced by the instantaneous Friedmann rate $H(z)$.

G.2 The local acceleration response

The local spatial projection is written in terms of

$$y = \frac{|\nabla\Phi|}{a_0}, \qquad \mu(y) = \frac{y}{\sqrt{1+y^2}}.$$

The quasi-static scalar equation is

$$\nabla\cdot\left[\mu\left(\frac{|\nabla\Phi|}{a_0}\right)\nabla\Phi\right] = 4\pi G\left(\rho_b + \rho_\Sigma^{\mathrm{free}}\right).$$

In the isolated relaxed deep-infrared limit, where the free endpoint residual density is negligible inside the baryonic system, this gives

$$a(r) = \frac{\sqrt{GM_b a_0}}{r}, \qquad v^4 = GM_b a_0.$$

This is used as a fixed-response branch. The present paper does not perform a full nonlinear structure-formation calculation or a joint galaxy-cluster-cosmology likelihood.

G.3 No double counting

The observed effective dark source is decomposed as

$$\rho_{\mathrm{dark,eff}} = \rho_\Sigma^{\mathrm{free}} + \rho_{\mathrm{pol}}, \qquad \rho_{\mathrm{pol}} = \frac{1}{4\pi G}\nabla\cdot[(1-\mu)\nabla\Phi].$$

Thus the linear cosmological propagation uses ρ_Σ as the conserved cold source with $\rho_{\mathrm{pol}} = 0$, while isolated low-acceleration galaxies may be dominated by the polarization term. In clusters and mergers the free residual component $\rho_\Sigma^{\text{free}}$ can separate from collisional baryonic gas and contribute to lensing offsets. Public DESI and SPARC data are used only for comparison with the closed parameters held fixed; full weak-lensing map likelihoods and nonlinear simulations are left to separate work.

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Declarations

Conflict of interest. The author declares no competing interests.

Data and code availability. No new observational dataset is introduced in this work. The numerical outputs, comparison tables, reproduction scripts, and formal verification files are available at `https://github.com/bookwormseagull1980/coarse-graining-genesis`. The computed quantities and observational reference values are listed separately. The reproduction procedure and the scope of the formal checks are described in Appendix B.

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